KEIO RESEARCH PROGRAM

Arithmetic Alternatives for Euler's Constant

Approximation, binomial arithmetic,
functional transcendence and mixed specialization

An integrated research manuscript

From the Two-World review to the rank-four
and cross-field investigations

KEIO

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Status. This manuscript consolidates proof-bearing deductions, conditional criteria, counterexamples and the limits of the investigated methods. It does not establish irrationality or transcendence of Euler's constant or Gompertz's constant, or linear independence of $1, e, \delta, e\,\mathrm{Ein}(1)$ over $\overline{\mathbb{Q}}$. Literature-wide novelty and journal-level significance are not asserted.

Abstract

We integrate a sequence of investigations prompted by the evaluation of a two-world approach to Euler's constant. Algebraicity and transcendence are treated as alternative assumptions inside the same mathematics, while the analytic definitions of the constants remain fixed. The quantitative review gives the sharp fixed-framing exponent, a rational-assumption floor, and a compressed degree-dependent norm bound. Binomial closures, prime certificates, denominator growth, rational Borel kernels, Fourier positivity and moment reconstruction provide exact auxiliary structures. Independent Euler-integral values yield functional differential transcendence, while a separate arithmetic-cost analysis records the height and denominator losses in polynomial constructions. The designated four-value problem is reduced to a precise rank-three alternative, and its exceptional points are constrained by affine multiplicative independence. A final set of imports from spectral theory, difference Galois theory and p-adic differential equations supplies explicit counterexamples and exact local diagnostics: a positive integer Jacobi operator has a differentially transcendental Weyl function with a rational value at the designated point, and the Euler initial-value family has an exact p-adic radius dichotomy. Full proofs, corrected hypotheses and a source-to-section record are retained. The missing arithmetic specialization step is stated separately from what is proved.

Keywords. Euler's constant; Gompertz's constant; E-functions; arithmetic Gevrey series; binomial rings; differential transcendence; Jacobi operators; p-adic differential equations.

한국어 요약과 읽기 안내

이 논문은 "이거 평가해볼래? 혹시 이거 가지고 너가 더 추가로 뭔가 감마 닫을 수 있거나 초월성 증명할 수 있다면 도전"이라는 질문에 대한 검토 답변부터, 그 뒤의 A/B 세계 확장, Moneyball, Snowball, 핵심 재평가, 네 값 랭크 연구, 마지막 교차분야 시도까지를 통합한 연구 원고다. 기존 원고의 영문 정리와 증명을 본문으로 유지하되, 반복되는 입력을 공유하고 표기와 참고문헌을 통일했다. 대화의 모든 문장을 옮긴 기록이 아니라, 그 구간의 수학적 내용을 추적할 수 있게 정리한 논문이다.

현재의 수학적 결론.

$$\gamma, \delta\text{의 개별 무리성·초월성 및} \quad \operatorname{rank}_{\overline{\mathbb{Q}}}\{1, e, \delta, e\,\mathrm{Ein}(1)\} = 4$$

는 이 논문에서 증명되지 않는다. 함수의 미분초월성, 보조값들의 대수적 독립성, 특정 값의 초월성을 구별한다. 완결된 보조 정리가 있다는 것과 감마 문제의 미증명 단계를 해결했다는 것은 별개의 주장이다.

연구 단계	이 통합본에서 다루는 내용
초기 원고 검토	고정 구간의 정확한 지수 $1/2$, 유리수 가정 아래의 하한, 대수차수에 따른 노름 하한, 기존 표현의 수정.
A/B 세계 확장	이항계수 환, 소수별 역원 증명서, 분모와 밀도, Borel 나눗셈, 감마 곱셈공식과 Fourier 양성 조건.
누적 구성과 전이	소수 분해, 양의 모멘트、Hankel 근사, 미분초월 함수, 역모멘트 문제, 유한 Fourier 재구성.
핵심 재평가	차수、높이、분모의 총비용, 고정 변형의 범위, 유한 실험과 무한족의 차이, 연구 가치에 대한 제한.
네 값 랭크	랭크 3 세계의 정확한 분류, 예외점의 곱셈 관계 제한, 혼합 영점 제거의 미증명 단계.
교차분야 감사	정수 Jacobi 반례, 차분 갈루아 이론, 정확한 p진 반경, p곡률, 값 수송의 오류 점검.

이 원고의 “세계 A”는 실제 정의로 정해진 γ가 대수적이라는 가정이고, “세계 B”는 초월적이라는 가정이다. 두 세계의 결과를 만든다는 것만으로 어느 쪽의 모순이 보장되지는 않는다. 소거할 수 있는 세계가 되려면 동일한 가정 아래에서 어떤 명제와 그 부정을 함께 증명해야 한다. 축소된 항등식 체계의 반모형을 실제 상수의 정의 전체에 대한 반모형으로 해석하지 않는다.

원고 001、002와 그 이전 통합본은 연구의 배경이다. 이 논문에는 지정한 시작 질문 이후에 사용되는 입력과 증명을 재수록하며, 그 이전 문서들의 모든 절을 다시 싣지는 않는다. 마지막의 출처 대응표는 어느 연구 묶음이 어느 부분으로 들어왔는지를 명시한다.

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1 Scope, notation and the logic of arithmetic alternatives

This is a thematic, proof-bearing consolidation of the investigations listed in the source record at the end. Repeated source front matter and repeated statements of the research objective have been removed. Established external theorems are cited; internal deductions are proved; conditional implications keep their hypotheses. A computational example is not used in place of an asymptotic quantifier.

The common normalization is

$$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \log n\right), \tag{1.1}$$

$$\mathrm{Ein}(z) = \sum_{n\geq 1}\frac{(-1)^{n-1}z^n}{n\,n!}, \qquad E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\,dt \quad (x>0), \tag{1.2}$$

$$D(x) = e^x E_1(x) = \int_0^\infty \frac{e^{-t}}{x+t}\,dt, \qquad \delta = D(1), \qquad \eta = \mathrm{Ein}(1). \tag{1.3}$$

Integration by parts also gives $\delta = \int_0^\infty \log(1+t)e^{-t}\,dt$. The connection identities are

$$\mathrm{Ein}(x) = \gamma + \log x + E_1(x), \qquad \delta = e\eta - e\gamma. \tag{1.4}$$

All logarithms of positive real numbers are real logarithms. Analytic continuation elsewhere uses the indicated branch.

Throughout, $\overline{\mathbb{Q}}$ denotes the field of algebraic numbers. The synonymous macros $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}}$ in the source notation have this meaning. The letter K is defined locally: it may denote a number field, an exponential-value field, or a Jacobi operator, and is never silently transported between these meanings. A denominator-clearing integer for Taylor coefficients is not the same object as the reduced denominator of a value at a point. Likewise, integer coefficients, a nonzero coefficient vector and a nonzero evaluated form are distinct conditions.

1.1 A, B and C are a proof organization, not separate axioms

Let C denote the ordinary background theory together with the analytic definitions above, and write

$$A = C + \{\gamma \in \overline{\mathbb{Q}}\}, \qquad B = C + \{\gamma \notin \overline{\mathbb{Q}}\}.$$

These assumptions are exhaustive. To reject A one needs a finite valid derivation of a contradiction from A; to reject B one needs such a derivation from B. Producing many compatible consequences, even from unrelated subjects, does not ensure that either derivation will be found. An additional finite partition can refine the search: if $A \Rightarrow C_1 \vee \cdots \vee C_s$ and every $A \wedge C_i$ is contradictory, then A is contradictory. Both exhaustiveness and each exclusion must be proved. For an infinite partition a uniform argument or another theorem covering all cases is required.

The "overlapping worlds" viewpoint is therefore used as a search discipline. It does not replace the proof of a separating arithmetic statement. The rank-three classification and the spectral counterexample later in the manuscript illustrate why strong-looking consequences can remain compatible with the undesired arithmetic alternative.

1.2 Four distinct levels of assertion

1. **Established input:** an external theorem with its stated coefficient field, singularity, regularity and integrality assumptions.

2. **Deduction proved here:** a statement with a complete argument from those inputs. This label does not itself assert historical novelty.
3. **Conditional criterion:** an implication whose additional hypothesis has not been established for the designated constants.
4. **Finite diagnostic:** an exact finite calculation or a numerical observation. Neither proves the existence of a required infinite family.

The familiar tests for small integer forms belong to a specific Diophantine route: an explicit construction, a fixed output space, adequate error relative to height, and nonvanishing of the evaluated form. They are not necessary conditions for every possible proof of transcendence. A relation-lifting theorem, a structural algebraization theorem or an entirely different argument could in principle bypass that route, provided its own hypotheses are proved.

1.3 The designated target and its strength

Define

$$\rho(1) = \operatorname{rank}_{\overline{\mathbb{Q}}}\{1, e, \delta, e\eta\}.$$

The established independence of e and η gives $\rho(1) \geq 3$. If $\rho(1) = 4$, then both γ and δ are transcendental: algebraicity of γ would turn (1.4) into a nontrivial algebraic-coefficient relation, and algebraicity of δ would give a relation with 1. The converse is not established by this argument; indeed the rank-three classification below allows generic coefficient cases in which γ and δ are algebraically independent. The target is therefore kept separate from the weaker individual transcendence questions.

1.4 Editorial and verification policy

The publication-style exposition is assembled from the recovered source manuscripts and proof notes. Mathematical assistance and internal cross-checking used ChatGPT. Internal checking is not external peer review or proof-assistant verification. Where an earlier wording required correction, the corrected quantifier or scope is used here and recorded in the integration notes. Source-reported computations that were not rerun are not presented as new computations. All mathematical source text and bibliography needed to typeset this edition are contained in the accompanying single TeX file.

Part I
Foundations

2 Shared E-function foundations

The following dependency proofs supply the inputs needed by the subsequent deductions. They are reused results, not additional claims of novelty. Only the order-one Ein theorem is used; no higher-polyexponential assertion is required.

2.1 Order-one multipoint independence

We use

$$\mathrm{Ein}(z)=\sum_{n\ge 1}\frac{(-1)^{n-1}z^n}{n\,n!},\qquad E_1(u)=\int_u^\infty\frac{e^{-t}}{t}\,dt\quad(u>0),\qquad K_{\exp}=\overline{\mathbb{Q}}(e^{\alpha}:\alpha\in\overline{\mathbb{Q}}).$$

The connection formula and derivative are

$$E_1(u)=\mathrm{Ein}(u)-\gamma-\log u,\qquad \mathrm{Ein}'(z)=\frac{1-e^{-z}}{z}. \tag{2.1}$$

These normalizations agree with [6, §6.2]. The entire function Ein is an E-function: its normalized Taylor coefficients are $(-1)^{n-1}/n$, whose common denominators grow at most exponentially, and the displayed differential identity places it in a finite linear differential system. In contrast, E_1 is not an E-function at zero.

Lemma 2.1 (Laurent derivative obstruction)**.** *Let $\nu_1,\dots,\nu_d\in\overline{\mathbb{Q}}$ be linearly independent over $\mathbb{Q}$. Endow*

$$R=\overline{\mathbb{Q}}(z)[X_1^{\pm1},\dots,X_d^{\pm1}],\qquad F=\mathrm{Frac}(R),\qquad DX_j=-\nu_jX_j.$$

If $g\in F$ satisfies $Dg=z^{-1}\sum_u c_uX^u$ with finitely supported $c_u\in\overline{\mathbb{Q}}$, then all c_u vanish. The constant field of F is $\overline{\mathbb{Q}}$.

Proof. An irreducible nonunit $P\in R$ cannot divide DP. Indeed, from $DP=AP$, comparison of Newton polytopes gives $A\in\overline{\mathbb{Q}}(z)$; the case $DP=0$ is included in the coefficient argument. For any two nonzero Laurent coefficients a_u,a_v of P one obtains

$$(a_u/a_v)'/(a_u/a_v)=(u-v)\cdot\nu.$$

A rational function has no nonzero constant logarithmic derivative. Independence of the ν_j therefore forces $u=v$, making P a Laurent monomial times a unit, a contradiction.

Any nonunit denominator factor of a reduced g would consequently have its pole order increased by differentiation. The proposed right side has no such pole, so $g\in R$. Writing $g=\sum a_u(z)X^u$ gives

$$a_u'-(u\cdot\nu)a_u=c_u/z.$$

For $u=0$, rational derivatives have zero residue. For $u\neq0$, a pole of a_u at zero produces a higher-order pole, while regularity of a_u at zero makes the left side regular there. Thus $c_u=0$ in all cases. With zero right side, $u\neq0$ admits no nonzero rational solution, and a_0 is constant. This proves the final assertion. □

Theorem 2.2 (Multipoint Ein input)**.** *For distinct nonzero algebraic $\lambda_1, \dots, \lambda_s$, put*

$$d = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\lambda_1, \dots, \lambda_s\}.$$

Then

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{\lambda_i}, \operatorname{Ein}(\lambda_i) : 1 \le i \le s) = d + s.$$

In particular all the $\operatorname{Ein}(\lambda_i)$ *are algebraically independent over $K_{\exp}$, in the finite-subset sense.*

Proof. Choose a $\mathbb{Z}$-basis $\nu_1, \dots, \nu_d$ of the additive group generated by the λ_i, with $\lambda_i = m_i \cdot \nu$. Distinct exponential characters are linearly independent over $\overline{\mathbb{Q}}(z)$, so $X_j = e^{-\nu_j z}$ realizes the field in Lemma 2.1. Set $Y_i(z) = \operatorname{Ein}(\lambda_i z)$; then $DY_i = (1 - X^{m_i})/z$. The generated field embeds in $\overline{\mathbb{Q}}((z))$, and has no new differential constants. The primitive Kolchin–Ostrowski theorem [22] turns any algebraic dependence of the Y_i over F into

$$Dg = z^{-1}\left(\sum_i c_i - \sum_i c_i X^{m_i}\right), \qquad g \in F, \quad c_i \in \overline{\mathbb{Q}},$$

with some $c_i \neq 0$. This contradicts Lemma 2.1, since the characters m_i are distinct and nonzero.

Extending constants from $\overline{\mathbb{Q}}$ to $\mathbb{C}$ does not change this functional transcendence degree: any proposed relation uses finitely many monomials with algebraic Taylor coefficients, whose coefficient-matrix rank is unchanged by scalar extension. Include 1, the required exponential characters, and the Y_i in a finite linear system of E-functions. Its coefficients have no finite pole except possibly zero. Siegel–Shidlovskii specialization at $z = 1$, in the refined form of Beukers [4], gives the asserted transcendence degree. The exponential coordinates have exactly their toric relations, so the s remaining coordinates are independent over their field. Finally every coefficient in a hypothetical relation over $K_{\exp}$ uses only finitely many algebraic exponential arguments. Add those arguments to the finite system; the same result excludes the relation. □

A full-row-rank matrix over $\overline{\mathbb{Q}}$ therefore sends a finite independent Ein tuple to an independent tuple over $K_{\exp}$. This follows by extending its rows to an invertible matrix and applying a linear change of indeterminates.

2.2 The integer-point rank bound

Write $U_n = \operatorname{Ein}(n)$, $y_n = E_1(n)$, $I_n = e^n y_n$, and $\Omega(n) = \sum_p v_p(n)$. For every composite integer n, the previously constructed residual is

$$\begin{aligned} R_n &= y_n - \sum_{p \mid n} v_p(n) y_p + (\Omega(n) - 1) y_1 \\ &= U_n - \sum_{p \mid n} v_p(n) U_p + (\Omega(n) - 1) U_1. \end{aligned}$$

Both the logarithms and the common γ term cancel by (2.1). Each composite column U_n is a unique pivot in its own row, so the residuals are algebraically independent over $K_{\exp}$. There are $N - \pi(N) - 1$ such columns through N. Consequently

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(I_1, \dots, I_N) \ge N - \pi(N) - 1, \tag{2.2}$$

since multiplication by $e^n \in K_{\exp}^{\times}$ does not alter the generated field. These residuals are also positive: with $h(s) = E_1(e^s)$ one has $h''(s) = e^s e^{-e^s} > 0$, whence $h(a+b) + h(0) > h(a) + h(b)$ for $a, b > 0$. Iterate over the logarithms of the prime factors of n, counted with multiplicity.

Part II

Quantitative review and arithmetic alternatives

3 Logical scope and the review of the Two-World argument

The supplied review addendum audits the 52-page manuscript [1]. The main logical and quantitative corrections below have also been compared with the recovered source text. The larger lattice searches are not independently reproduced here; exact lattice reduction alone does not certify the real input constants. The degree-twelve example preserved in the review archive has been checked again by exact rational arithmetic.

The arithmetic alternatives concern the same fixed real number

$$\gamma = \lim_{n\to\infty}\left(\sum_{j=1}^{n}\frac{1}{j} - \log n\right) = -\Gamma'(1), \qquad 0 < \gamma < 1.$$

Write A for the hypothesis $\gamma \in \overline{\mathbb{Q}}$, B for $\gamma \notin \overline{\mathbb{Q}}$, and C for the usual mathematical background including this definition. A contradiction to A requires both implications

$$\mathrm{C} + \mathrm{A} \Longrightarrow \mathcal{O} \text{ has property } P, \qquad \mathrm{C} \Longrightarrow \mathcal{O} \text{ cannot have property } P.$$

Rejecting an additional auxiliary property does not reject either arithmetic alternative unless that property has first been derived from it. The constructions below are classifications and conditional criteria; none constructs the required infinite approximation family.

Write

$$\mathrm{Ein}(z) = \sum_{n\geq 1}\frac{(-1)^{n-1}z^n}{n\,n!}, \qquad T = e\,\mathrm{Ein}(1), \qquad \delta = eE_1(1) = \int_0^\infty \frac{e^{-t}}{1+t}\,dt = T - e\gamma.$$

The scalar used to vary the framing will be x, so a fixed integer form becomes

$$F_{a,b,c}(x) = a + bx + c(T - ex) = (a + cT) + (b - ce)x. \tag{3.1}$$

At $x = \gamma$ it is the actual form $a + b\gamma + c\delta$.

Restricted models are not models of the full definition. The two-model theorem applies to the restricted theory which omits the identifications of its new symbols with the defining limits, integrals and canonical sums. It does not prove independence of the arithmetic nature of the actual, analytically defined γ, or impossibility for every available mathematical method. The manuscript itself partly recognizes this distinction; its abstract and concluding generalizations should retain it throughout.

Uniformity is essential. Every true strict inequality $|F_{a,b,c}(\gamma)| < \eta$ remains true in some neighborhood of γ, by continuity. Hence, with the manuscript's definition allowing an arbitrary finite enclosure, a single finite exceptional cancellation is locally shear-invariant. What the tie argument restricts is an entire named family with *one fixed common interval*. The statement in Remark 18.3 that a finite cancellation is itself not shear-invariant needs this qualification.

Numerical exponents are not asymptotic exponents. The finite statistic $300/\log_{10} H_d$ in the proposition entitled "Approximation exponents of γ" does not establish $w_d^*(\gamma) = d$. The latter requires asymptotic quantifiers, nontrivial approximants and the actual approximation errors. Report the finite ratios only. Exact verification of LLL reduction on rounded input does not by itself certify enclosures of the original constants or root locations. Such claims need rational or ball intervals in addition to exact lattice calculations.

Other specific repairs and scope qualifications.

1. In the finite-observation theorem, $S = \varnothing$ gives a false assertion of a degree-one algebraic integer in every neighborhood of γ. There is no such integer in $(0, 1)$. Allow degree-one algebraic numbers (rationals), or require degree at least two for the algebraic-integer assertion in this case. The neighborhood proof also requires the interval enclosures to contain γ in their interiors.
2. The "one-sidedness" proposition cannot infer that the program can only terminate on the algebraic-assumption side from finitely many excluded relations. A relation of larger height or degree has not thereby been ruled out. Under transcendence, existence of a derivation taking γ to 1 is also a positive existential consequence, contrary to its universal-negative description of every consequence.
3. The last item of the linear-hierarchy theorem needs the declared subspace V restricted, for example, to the listed countable subspaces of **E**. For arbitrary $V \subset \mathbb{C}$ with $e \in V$, take $V = \mathbb{C}$: its claimed exclusion is identically false.
4. The collision identity's citation to Fischler–Rivoal uses $a = 1, s = r$, not $s = r - 1$. With their notation, $\omega_{1,r} = (-1)^{r-1} f_{1,r+1;0}(1)/r!$.
5. The shadow proposition's title is broader than its proved conclusion: it establishes an E-value realization for δ_c, while its body already notes logarithmic terms for general λ; higher entries also contain zeta values. Coefficient conjugation of a power series should not be identified with arbitrary Galois conjugation of its value. The dilation identity for the connection polynomials P_r does not rescale the entire ω-tower.
6. The mixed-value proposition establishes **SE** membership; it does not assert canonical Borel membership in **D**, and that stronger conclusion must not be inferred. A calculation of poles in the parameter a is not a proof about all Stokes data in the variable z.
7. The E-membership decision theorem already requires an explicit certified E-function representation in its algorithmic branch. Its broader claim of termination on deciding membership must retain that requirement in the affirmative direction: bare nonconstructive membership does not supply the input. The m-fold multiplication twist factor is $e^{(m-1)\tau/2}$; $e^{\tau/2}$ is the duplication case.

The central connection identities and the fixed-interval affine tie estimate survive these corrections.

4 The fixed-interval exponent is exactly one half

We use Shidlovskii's rational-coefficient linear independence measure, as stated in [2, Theorem A]: an independent N-component rational E-system, evaluated at a nonsingular nonzero rational point, has a lower bound $C_\varepsilon H^{-N+1-\varepsilon}$ for its nonzero integer linear forms.

Lemma 4.1. *For each $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that*

$$|h_0 + h_1 e + h_2 T| \geq C_\varepsilon \max(1, |h_0|, |h_1|, |h_2|)^{-2-\varepsilon}$$

for every nonzero integer triple. Moreover $1, e, T$ are linearly independent over $\overline{\mathbb{Q}}$.

Proof. The vector $(1, e^z, F(z))$, $F(z) = e^z \operatorname{Ein}(z)$, satisfies

$$(1)' = 0, \quad (e^z)' = e^z, \quad F' = F + (e^z - 1)/z.$$

It consists of E-functions with rational coefficients, with only 0 a possible finite singularity. Its functions are linearly independent over $\mathbb{C}(z)$: otherwise, after division by e^z, $\operatorname{Ein}(z)$ would be a rational function plus a rational multiple of e^{-z}. On the positive real axis this is incompatible with $\operatorname{Ein}(z) = \log z + \gamma + O(e^{-z}/z)$. The remaining two functions are plainly independent. Apply the rational measure at $z = 1$, and Beukers' linear independence theorem [4] for the algebraic-coefficient assertion. □

Lemma 4.2 (Elementary simultaneous transference)**.** *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$ such that, for all $q \geq 1$ and integers p, b,*

$$\max(|qe - b|, |qT - p|) \geq C_\varepsilon q^{-1/2-\varepsilon}. \tag{4.1}$$

Proof. Put $E_q = \max(|qe - b|, |qT - p|)$. The case $E_q \geq 1$ is immediate. Otherwise b/q and p/q are bounded by absolute constants. Put $M = \lfloor \sqrt{q} \rfloor$. The $(M+1)^2 > q$ residues $ub + vp \pmod q$, $0 \leq u, v \leq M$, yield a nonzero pair (h_1, h_2) with $|h_i| \leq M$ and $q \mid h_1 b + h_2 p$. Set $h_0 = -(h_1 b + h_2 p)/q$. Then $\max |h_i| \ll \sqrt{q}$ and

$$0 < |h_0 + h_1 e + h_2 T| = \left| \frac{h_1(qe - b) + h_2(qT - p)}{q} \right| \leq \frac{2E_q}{\sqrt{q}}.$$

Lemma 4.1, with exponent parameter 2ε, bounds the left side below by $Cq^{-1-\varepsilon}$. Rearrangement proves the claim. □

Theorem 4.3 (Sharp fixed framing interval)**.** *Let $I = [u, v]$ be a fixed bounded interval with $u < v$. For all $\varepsilon > 0$ there exists $C_{I,\varepsilon} > 0$ such that every nonzero integer triple, with $H = \max(1, |a|, |b|, |c|)$, satisfies*

$$\sup_{x \in I} |a + bx + c(T - ex)| \geq C_{I,\varepsilon} H^{-1/2-\varepsilon}. \tag{4.2}$$

Conversely, infinitely many triples with unbounded H satisfy an upper bound $C_I H^{-1/2}$. Thus the supremum of the uniform approximation exponents for this particular pair is exactly $1/2$.

Proof. Write the supremum as η. The endpoint differences give

$$|b - ce| \leq \frac{2\eta}{v - u}, \qquad |a + cT| \leq \eta \left(1 + \frac{2\max(|u|, |v|)}{v - u}\right).$$

If $c \neq 0$, apply Lemma 4.2 with $q = |c|$, $b' = \operatorname{sgn}(c) b$, $p' = -\operatorname{sgn}(c) a$. It follows that $\eta \geq Cq^{-1/2-\varepsilon} \geq CH^{-1/2-\varepsilon}$. If $c = 0$, either the integer b has modulus at least one, giving $\eta \geq (v - u)/2$, or a is a nonzero integer, giving $\eta \geq 1$.

For the upper bound, simultaneous Dirichlet approximation gives arbitrarily large Q and $1 \leq q \leq Q$ with both errors at most $CQ^{-1/2}$. Take $(a, b, c) = (-p, b, q)$. Then $H \ll q \leq Q$ and the uniform error is $O_I(Q^{-1/2}) = O_I(H^{-1/2})$. The denominators must become unbounded because e is irrational. □

Remark 4.4. This sharpens the manuscript's ceiling 1 and its merely metric observation about 1/2. It restricts the proved uniform bound, not the true error at the single point $x = \gamma$. It gives no universal bound on all approximations to γ.

Corollary 4.5 (Moving intervals)**.** *Suppose $I_n \subset [-M, M]$ has length $0 < \ell_n \le L$, and H_n is the height of a nonzero integer triple. For every $\varepsilon > 0$,*

$$\eta_n := \sup_{x \in I_n} |F_{a_n,b_n,c_n}(x)| \ge C_{M,L,\varepsilon} \ell_n H_n^{-1/2-\varepsilon}.$$

Consequently, if $H_n \to \infty$, $\ell_n \ge H_n^{-\sigma+o(1)}$ and $\eta_n \le H_n^{-\kappa+o(1)}$, then $\kappa \le \sigma + 1/2$.

Proof. The endpoint argument gives both engine errors at most $C_{M,L}\eta_n/\ell_n$. Use (4.1); the $c_n = 0$ case follows directly from integrality as above. Take logarithms and then let $\varepsilon \downarrow 0$. □

This is a genuine allowed way around the fixed-interval theorem. It spends increasing information about the position of γ, so it is not by itself an arithmetic construction or a proof of nonvanishing.

5 A lower irrationality threshold without growth regularity

Theorem 5.1 (Rational-assumption floor)**.** *Assume $\gamma \in \mathbb{Q}$. For every $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that every integer triple with $L = a + b\gamma + c\delta \ne 0$ satisfies*

$$|L| \ge C_\varepsilon \max(1, |a|, |b|, |c|)^{-1-\varepsilon}. \tag{5.1}$$

This uses the irrationality-exponent theorem for rational-coefficient E-functions in [3, Theorem 1] (arXiv version 2, 2025).

Proof. Write $\gamma = r/s$ with $s > 0$. The function $e^z(\mathrm{Ein}(z) - r/s)$ is an E-function with rational coefficients, and its value at 1 is δ. This value is irrational: otherwise $T - (r/s)e - \delta = 0$ contradicts Lemma 4.1. Therefore [3] gives $\mu(\delta) = 2$.

If $c \ne 0$, put $m = sa + rb$ and $k = sc$. Then $sL = m + k\delta$. The assertion $\mu(\delta) = 2$ implies $|m + k\delta| \ge C_\varepsilon |k|^{-1-\varepsilon}$ for all integer m and nonzero integer k (absorb the finitely many exceptional denominators into the positive constant). Since $|k| \le sH$, (5.1) follows. If $c = 0$, nonzero L is a nonzero rational with denominator dividing s, so $|L| \ge 1/s$. □

Corollary 5.2 (Irregular-sequence irrationality test)**.** *If for some fixed $\kappa > 1$ there are integer triples with $H_n \to \infty$ and*

$$0 < |a_n + b_n\gamma + c_n\delta| \le H_n^{-\kappa},$$

then γ is irrational. No regular growth of H_n and no matching lower asymptotic for the forms are required.

Proof. Under rationality choose $0 < \varepsilon < \kappa - 1$ in Theorem 5.1; its lower bound contradicts the displayed upper bound for large n. □

Remark 5.3. The conclusion is irrationality, not transcendence: algebraic irrational values of γ have not been excluded by this threshold. The nonzero condition cannot be dropped. Under rationality, infinitely many coefficient triples have exactly zero value with $c = 0$.

6 Compressing the degree-dependent algebraic floor

Theorem 6.1 (Norm and symmetric powers)**.** *Assume γ is algebraic of degree $d \geq 1$. For every $\varepsilon > 0$, all integer triples with $L = a + b\gamma + c\delta \neq 0$ satisfy*

$$|L| \geq C_{\gamma,\varepsilon} H^{-K_d - \varepsilon}, \qquad K_d = d\binom{d+2}{2} - 1. \tag{6.1}$$

For $d = 1$, Theorem 5.1 improves $K_1 = 2$ to 1.

Proof. Let $f(X) = A\prod_{j=1}^{d}(X - \gamma_j) \in \mathbb{Z}[X]$ be the primitive minimal polynomial, with $\gamma_1 = \gamma$. Define $U = a + cY$, $V = b - cX$ and

$$P(X,Y) = (-1)^d V^d f(-U/V) = A\prod_{j=1}^{d}\big(a + b\gamma_j + cY - c\gamma_j X\big).$$

The first expression is understood as its polynomial expansion even at $V = 0$. Its coefficients are integers, its total degree is at most d, and $\mathrm{H}(P) \ll_\gamma H^d$.

At $(X,Y) = (e,T)$ the first factor is L. If $c \neq 0$, each factor is nonzero by the algebraic linear independence in Lemma 4.1. If $c = 0$, a zero factor would imply that a conjugate γ_j is rational (unless $a = b = 0$); hence $d = 1$ and $L = 0$, contrary to the hypothesis. Thus $P(e,T) \neq 0$. Each of the other $d - 1$ factors has modulus $O_\gamma(H)$.

Set $F(z) = e^z \operatorname{Ein}(z)$ and consider

$$F_{ij}(z) = e^{iz}F(z)^j, \qquad i,j \geq 0, \quad i + j \leq d.$$

These $m_d = \binom{d+2}{2}$ rational-coefficient E-functions form a closed differential system:

$$F'_{ij} = (i+j)F_{ij} + \frac{j}{z}F_{i+1,j-1} - \frac{j}{z}F_{i,j-1},$$

with the last two terms omitted for $j = 0$. Use [2, Theorem 1] with coefficient field $\mathbb{Q}$ and $N = m_d$. This version only needs the evaluated form to be nonzero, already checked; no unproved high-order independence statement is needed. It gives

$$|P(e,T)| \geq C\,\mathrm{H}(P)^{-m_d+1-\varepsilon/d} \geq C'H^{-d(m_d-1)-\varepsilon}.$$

Divide by the other factors, bounded by $O_\gamma(H^{d-1})$, and the fixed A. The resulting exponent is $dm_d - 1 + \varepsilon$. □

Degree d	1	2	3	4	10
Manuscript $d3^d - 1$	2	17	80	323	590489
Norm bound $d\binom{d+2}{2} - 1$	2	11	29	59	659
Best rational-assumption bound	1	–	–	–	–

The constants depend on the alleged algebraic value and on ε. These are asymptotic bounds, not numerical certificates obtained by inserting a large height. This is a quantitative strengthening relative to the reviewed manuscript, by a standard norm construction.

Corollary 6.2 (What an unbounded exponent would imply)**.** *Suppose for every $\kappa > 0$ there is an integer triple of height $H \geq 2$ with $0 < |a + b\gamma + c\delta| < H^{-\kappa}$. Then γ is transcendental.*

Proof. If γ had degree d, (6.1) would bound all sufficiently large-height exponents by K_d+1. The finitely many remaining triples of bounded height with nonzero values cannot realize unbounded exponents either. □

No such family is constructed here. The improved floor does not convert the manuscript's finite or approximately 1/2-exponent forms into a transcendence proof.

7 Coefficient rank and conditional contradiction criteria

Put $T = e\,\mathrm{Ein}(1)$ and $\delta = T - e\gamma$. For an integer row $v = (a, b, c)$ write

$$L(v) = a + b\gamma + c\delta, \qquad H(v) = \max(1, |a|, |b|, |c|).$$

The results below replace some value-nonvanishing requirements by exact coefficient-rank tests. They do not construct the required infinite families.

Arithmetic inputs. Lemma 4.1 supplies both algebraic linear independence of $1, e, T$ and its integer-form lower bound. Theorem 5.1 supplies the rational-assumption floor. In the notation of this section, these previously proved bounds read

$$|h_0 + h_1 e + h_2 T| \geq C_\epsilon \max(1, |h_0|, |h_1|, |h_2|)^{-2-\epsilon} \quad (0 \neq h \in \mathbb{Z}^3), \tag{7.1}$$

and, under $\gamma \in \mathbb{Q}$,

$$L(v) \neq 0 \Longrightarrow |L(v)| \geq C_{\gamma,\epsilon} H(v)^{-1-\epsilon}. \tag{7.2}$$

Finally, if γ is algebraic irrational, Roth's theorem [8] gives

$$|A + B\gamma| \geq C_{\gamma,\epsilon} \max(1, |A|, |B|)^{-1-\epsilon} \quad ((A, B) \in \mathbb{Z}^2 \setminus \{(0, 0)\}). \tag{7.3}$$

All constants may depend on the alleged algebraic value of γ; these bounds are asymptotic statements, not effective numerical certificates.

Theorem 7.1 (Rank replaces individual nonvanishing)**.** *Suppose $H_n \to \infty$ and, for a fixed $\rho > 0$, there are $\mathbb{Q}$-linearly independent integer rows v_n, w_n of heights at most H_n such that*

$$|L(v_n)|,\ |L(w_n)| \leq H_n^{-1-\rho}.$$

Then γ is irrational. Neither value is separately assumed nonzero.

Proof. Assume $\gamma = r/s \in \mathbb{Q}$ in lowest terms. Choose $0 < \epsilon < \rho$. By (7.2), both displayed values must be exactly zero for sufficiently large n. Since δ is irrational under this assumption, $L(a, b, c) = 0$ implies $c = 0$ and $sa + rb = 0$. Thus both rows lie on the single rational line $\mathbb{Q}(-r, s, 0)$, contradicting their independence. □

Theorem 7.2 (A shared auxiliary coefficient)**.** *Suppose $H_n \to \infty$, $c_n \in \mathbb{Z} \setminus \{0\}$, and for a fixed $\rho > 0$ there are distinct pairs $(a_n, b_n), (a'_n, b'_n) \in \mathbb{Z}^2$ such that*

$$H(a_n, b_n, c_n),\ H(a'_n, b'_n, c_n) \leq H_n$$

and

$$|a_n + b_n\gamma + c_n\delta|,\ |a'_n + b'_n\gamma + c_n\delta| \leq H_n^{-1-\rho}.$$

Then γ is transcendental. No regularity of the growth of H_n or separate certification of value nonvanishing is required.

Proof. The rows are independent: proportional rows with the same nonzero last coordinate would be identical. Theorem 7.1 excludes rational γ. If γ were algebraic irrational, subtracting the two forms would give

$$0 < |A_n + B_n\gamma| \le 2H_n^{-1-\rho}, \qquad \max(|A_n|, |B_n|) \le 2H_n,$$

where $A_n = a_n - a'_n$, $B_n = b_n - b'_n$. The pair is nonzero, and irrationality makes its value nonzero. Equation (7.3), with $0 < \epsilon < \rho$, gives the incompatible lower bound $C(2H_n)^{-1-\epsilon}$. □

Corollary 7.3 (Conditional occupancy of a fixed coefficient)**.** *If γ is algebraic, then for each fixed $\rho > 0$ and all sufficiently large H, every fixed nonzero integer c admits at most one pair (a, b) with $H(a, b, c) \le H$ and $|a + b\gamma + c\delta| \le H^{-1-\rho}$. Under rationality of γ there are no such pairs.*

Proof. In the rational case $c \neq 0$ makes the value nonzero, so use (7.2). In the algebraic irrational case, subtract two candidates and use (7.3) as above. The constants are independent of c. □

Theorem 7.4 (Primitive content in auxiliary elimination)**.** *Suppose $v_n = (a_1, b_1, c_1)$ and $w_n = (a_2, b_2, c_2)$ are independent integer rows of heights at most $H_n \to \infty$, and*

$$|L(v_n)|,\ |L(w_n)| \le H_n^{-\kappa} \qquad \text{for a fixed } \kappa > 1.$$

Set

$$A_n = c_2a_1 - c_1a_2, \qquad B_n = c_2b_1 - c_1b_2.$$

Suppose $(A_n, B_n) \neq (0, 0)$ and, for some fixed $0 \le \sigma < 2$,

$$g_n := \gcd(|A_n|, |B_n|) \ge H_n^{\sigma}.$$

If $\kappa + 2\sigma > 3$, then γ is transcendental.

Proof. Theorem 7.1 excludes rationality. Under algebraic irrationality, divide both coefficients by g_n. The resulting primitive pair has nonzero value, height $h_n \le 2H_n^{2-\sigma}$, and

$$0 < \left|\frac{A_n + B_n\gamma}{g_n}\right| = \frac{|c_2L(v_n) - c_1L(w_n)|}{g_n} \le 2H_n^{1-\kappa-\sigma}.$$

Choose $\epsilon > 0$ with $\kappa + \sigma - 1 > (2 - \sigma)(1 + \epsilon)$. Roth gives a lower bound $Ch_n^{-1-\epsilon}$, which is at least $C'H_n^{-(2-\sigma)(1+\epsilon)}$ and contradicts this upper bound. □

For $\sigma = 0$ the sufficient threshold is $\kappa > 3$; for $\sigma = 1$ it is $\kappa > 1$. The actual gcd is essential. A large raw determinant alone supplies no primitive gain.

Proposition 7.5 (A power-gap consequence in the algebraic world)**.** *Assume γ is algebraic irrational. Let v, w be independent integer rows with nonzero last coordinates, heights $H \le K$, and $|L(v)| \le H^{-\kappa}$, $|L(w)| \le K^{-\kappa}$ for $\kappa > 0$. For every $\epsilon > 0$ one has*

$$K^{2+\epsilon} \ge C_{\gamma,\epsilon}H^{\kappa-1-\epsilon}.$$

Proof. With $A = c(w)a(v) - c(v)a(w)$ and $B = c(w)b(v) - c(v)b(w)$, the pair (A, B) is nonzero: otherwise the nonzero last coordinates would make v, w proportional. Its value is nonzero by irrationality, its height is at most $2HK$, and

$$|A + B\gamma| \le KH^{-\kappa} + HK^{-\kappa} \le 2KH^{-\kappa}.$$

Combine this with (7.3) and rearrange. □

Consequently, an infinite family of distinct primitive rows with nonzero last coordinates, exponent $\kappa > 3$, and increasing heights satisfying $\log H_{n+1}/\log H_n \to 1$ would exclude algebraic irrationality. Such rows are pairwise independent after identifying opposite signs, and (7.2) excludes rationality when their values are nonzero. In the rational case the nonzero last coordinate already guarantees this. This is another sufficient criterion, not a family produced here.

Proposition 7.6 (An unconditional restriction on paired forms)**.** *Suppose independent rows of heights at most $H_n \to \infty$ have $|L(v_n)|, |L(w_n)| \le H_n^{-\kappa}$. Put $C_n = a_1b_2 - a_2b_1$, retain A_n, B_n above, and let $G_n = \gcd(|C_n|, |A_n|, |B_n|)$. If $G_n \ge H_n^{\tau}$ for a fixed $0 \le \tau < 2$, then $\kappa + 3\tau \le 5$.*

Proof. The nonzero integer vector $(C_n, -A_n, -B_n)/G_n$ has height at most $2H_n^2/G_n$. Direct expansion gives

$$C_n - A_n e - B_n T = L(v_n)(b_2 - c_2 e) - L(w_n)(b_1 - c_1 e).$$

Its value is nonzero by the independence of $1, e, T$. Equation (7.1) therefore implies

$$C_0 H_n^{1-\kappa}/G_n \ge C_\epsilon (H_n^2/G_n)^{-2-\epsilon}, \qquad H_n^{5+2\epsilon-\kappa} \ge C'_\epsilon G_n^{3+\epsilon}.$$

Insert $G_n \ge H_n^{\tau}$, let $n \to \infty$, and then $\epsilon \downarrow 0$. □

Here G_n is the content of all three minors, while g_n in Theorem 7.4 uses only two. They must not be conflated.

Remark 7.7 (What these criteria do not establish)*.* No required infinite family has been constructed. In particular, Theorems 7.2 and 7.4 are *sufficient*, not equivalent, transcendence criteria. Their eliminated primitive pairs give rational approximations to γ with a fixed margin beyond exponent 2, and therefore force $\mu(\gamma) > 2$. Indeed their nonzero values tend to zero, so the primitive pair heights are unbounded; small values make the heights comparable with the reduced denominators. The stated strict exponent inequalities then give the asserted margin. A transcendental γ with $\mu(\gamma) = 2$ would satisfy neither existence target. Thus these criteria rigorously bypass individual nonvanishing tests, but are not evidence that a suitable family exists, and do not supersede the logically different mixed E–D separation target.

8 Exact information carried by the finite collision tower

Let $\mathbf{E}$ denote the ring of E-values. To make the finite-ring statement self-contained, we recall the connection identity and its proof from [1], with the corrected indexing. For $r \ge 1$ set

$$\mathrm{Ein}_r(z) = \sum_{n\ge1} \frac{(-1)^{n-1} z^n}{n^r n!}, \qquad P_r(\lambda) = [t^r]\Gamma(1-t)\lambda^t,$$

and, for positive algebraic λ, define

$$\omega_{\lambda,r} = \int_0^\infty e^{-\xi} \frac{(-1)^{r-1}\log^{r-1}(1+\xi/\lambda)}{\lambda(r-1)!(1+\xi/\lambda)}\, d\xi.$$

Lemma 8.1 (Connection identity)**.** *For all the indicated λ, r,*

$$\omega_{\lambda,r} = e^{\lambda}(\mathrm{Ein}_r(\lambda) - P_r(\lambda)).$$

Proof. Substitute $u = 1 + \xi/\lambda$. For $|t|$ small, absolute convergence permits summation under the integral and gives

$$\sum_{r\ge1} \omega_{\lambda,r} t^r = e^{\lambda} t \int_1^\infty e^{-\lambda u} u^{-t-1}\, du = e^{\lambda} t \lambda^t \Gamma(-t, \lambda),$$

where $\Gamma(\cdot,\lambda)$ is the upper incomplete Gamma function. Its lower-series expansion gives

$$\Gamma(-t,\lambda)=\Gamma(-t)-\sum_{n\geq 0}\frac{(-1)^n\lambda^{n-t}}{n!(n-t)}.$$

Now $t\Gamma(-t)=-\Gamma(1-t)$; the $n=0$ term contributes λ^{-t}; and $-t/(n-t)=-\sum_{k\geq 1}t^k/n^k$ for $n\geq 1$. Therefore

$$t\Gamma(-t,\lambda)=-\Gamma(1-t)+\lambda^{-t}\left(1+\sum_{k\geq 1}\mathrm{Ein}_k(\lambda)t^k\right).$$

After multiplication by $e^{\lambda}\lambda^t$, comparison of the coefficients of t^r proves the identity. □

Proposition 8.2. *If $R\geq 1$ and $\Lambda\subset\overline{\mathbb{Q}}_{>0}$ is finite with $1\in\Lambda$, then*

$$\begin{aligned}&\mathbf{E}[\omega_{\lambda,r}:\lambda\in\Lambda,\ 1\leq r\leq R]\\&\quad=\mathbf{E}[\gamma,\ \log\lambda\ (\lambda\in\Lambda),\ \zeta(2),\dots,\zeta(R)].\end{aligned}$$

For $R=1$ the zeta list is empty. Therefore all the indicated collisions lie in $\mathbf{E}$ if and only if every displayed generator on the right lies in $\mathbf{E}$.

Proof. Put $g_0=1$ and $g_r=\mathrm{Ein}_r(1)-e^{-1}\omega_{1,r}$. The logarithmic generating function gives

$$\gamma=g_1,\qquad \zeta(k)=kg_k-\gamma g_{k-1}-\sum_{j=2}^{k-1}\zeta(j)g_{k-j}.$$

This recursively recovers all the zeta values up to R using ring operations only. The first-level relation also gives

$$\log\lambda=\mathrm{Ein}(\lambda)-e^{-\lambda}\omega_{\lambda,1}-\gamma.$$

Conversely each $P_r(\lambda)$ is a rational polynomial in these generators, and $e^{\pm\lambda}$ and $\mathrm{Ein}_r(\lambda)$ are E-values. Both ring inclusions follow. □

This explains precisely what raising the tower order adds. Algebraicity of γ does not supply membership of the other generators in $\mathbf{E}$. The proposition is a triangular reformulation, not a proof of the missing collisions.

9 Binomial closure and exact arithmetic alternatives

The polynomial ring generated by a number records only polynomial relations with integer coefficients. Adjoining generalized binomial coefficients produces a larger ring whose finite-character theory is more rigid. This section applies the classical theory of integer-valued polynomials; no literature-wide novelty claim is made. The relevant framework is discussed in [9, 10]. All statements hold for arbitrary real numbers before specialization to Euler's constant.

9.1 The enlarged ring and its reductions

Set

$$\mathcal{I}=\{P\in\mathbb{Q}[X]:P(\mathbb{Z})\subseteq\mathbb{Z}\},\qquad b_n(X)=\binom{X}{n},\qquad \mathcal{B}_x=\mathbb{Z}[b_n(x):n\geq 0].$$

The ring $\mathcal{B}_x$ contains $\mathbb{Z}[x]$. Newton expansion gives $\mathcal{I} = \bigoplus_{n\geq 0} \mathbb{Z} b_n$, so evaluation induces a surjection $\mathcal{I} \to \mathcal{B}_x$. If x is transcendental, this map is injective. If x is algebraic with primitive minimal polynomial $f \in \mathbb{Z}[X]$, its kernel is

$$J_f = f\mathbb{Q}[X] \cap \mathcal{I}.$$

For a prime p let $Z_p(f) = \{\beta \in \mathbb{Z}_p : f(\beta) = 0\}$. Write $C(\mathbb{Z}_p, \mathbb{Z}/p^k\mathbb{Z})$ for the ring of continuous functions to the discrete finite ring, with pointwise operations. Such functions are locally constant.

Theorem 9.1 (Exact reduction algebras)**.** *For every prime p and integer $k \geq 1$,*

$$\mathcal{B}_x/p^k\mathcal{B}_x \simeq \begin{cases} C(\mathbb{Z}_p, \mathbb{Z}/p^k\mathbb{Z}), & x \notin \overline{\mathbb{Q}}, \\ \displaystyle\prod_{\beta \in Z_p(f)} \mathbb{Z}/p^k\mathbb{Z}, & x \in \overline{\mathbb{Q}}. \end{cases}$$

The product over the empty set denotes the zero ring. The unital $\mathbb{Z}$-algebra homomorphisms from $\mathcal{B}_x$ to $\mathbb{Z}/p^k\mathbb{Z}$ are consequently indexed by $\mathbb{Z}_p$ in the transcendental case and by $Z_p(f)$ in the algebraic case. Reduction between successive values of k is a bijection on these homomorphism sets.

Proof. Every $P \in \mathcal{I}$ maps $\mathbb{Z}_p$ continuously into $\mathbb{Z}_p$: apply continuity of its rational polynomial expression and density of the integers. Reduction of evaluation modulo p^k has kernel $p^k\mathcal{I}$. Indeed, divisibility by p^k at all integers is precisely the assertion that P/p^k is integer-valued.

We first prove that every locally constant function is represented. It depends on a residue class modulo some p^m. For $a = \sum_{j<m} a_j p^j$ define

$$D_a(X) = \prod_{j=0}^{m-1} \left[1 - \left(\binom{X}{p^j} - a_j\right)^{p-1}\right] \in \mathcal{I}.$$

The digit form of Lucas's identity, $\binom{t}{p^j} \equiv t_j \pmod p$ for $t = \sum_{j\geq 0} t_j p^j \in \mathbb{Z}_p$, shows that D_a is 1 modulo p on $a + p^m\mathbb{Z}_p$ and 0 elsewhere. The identity first holds for nonnegative integers, by comparing coefficients in $(1+T)^t$ modulo p, and extends to $\mathbb{Z}_p$ by continuity. The polynomial

$$E_a(X) = D_a(X)^{p^{k-1}}$$

is the indicator of that ball modulo p^k. To check this, use $(1+pu)^{p^{k-1}} \equiv 1 \pmod{p^k}$ and $p^{k-1} \geq k$; the case $k = 1$ is immediate. Integer linear combinations of the E_a represent all locally constant functions. We have proved

$$\mathcal{I}/p^k\mathcal{I} \simeq C(\mathbb{Z}_p, \mathbb{Z}/p^k\mathbb{Z}).$$

This proves the transcendental assertion.

In the algebraic case let J be the image of J_f in this function ring. Each function in J vanishes on $Z_p(f)$. Conversely, fix $\beta \in \mathbb{Z}_p$ with $f(\beta) \neq 0$, and set $t = v_p(f(\beta)) \geq 0$. On a sufficiently small residue ball U about β, the valuation of f is constantly t. The preceding indicator construction gives $q \in \mathcal{I}$ congruent to 1_U modulo p^{t+1}. Then

$$g = \frac{fq}{p^t} \in J_f, \qquad g(\beta) \in \mathbb{Z}_p^\times.$$

For the first assertion, at integers in U the factor f is divisible by p^t, and outside U the factor q is divisible by p^{t+1}. Thus g is integer-valued; the introduced denominator has no other prime factor. Its rational polynomial expression is a multiple of f.

Let h be locally constant modulo p^k and vanish on $Z_p(f)$. Its support is compact and disjoint from that finite root set. Cover its support by finitely many clopen sets on each of which one of the preceding functions g is a unit. Refine them to a disjoint clopen partition. On each part multiply its g by the locally constant function h/g, extended by zero outside that part. The resulting function belongs to J, and summing the finitely many parts gives h. Therefore J is exactly the ideal of functions vanishing on $Z_p(f)$. Restriction to this finite root set is onto, since disjoint clopen neighborhoods separate its points. This proves the displayed product formula.

A unital homomorphism from a locally constant function ring to the local ring $\mathbb{Z}/p^k\mathbb{Z}$ sends the indicators of a finite clopen partition to idempotents, selecting precisely one part. These selections are compatible under refinement and determine a unique point of the compact space $\mathbb{Z}_p$. Constants then force the homomorphism to be evaluation at that point. In a finite product the same argument selects one factor. These descriptions are independent of k, proving unique lifting. □

Taking inverse limits gives a further explicit consequence:

$$\widehat{\mathcal{B}_x}^{(p)} \simeq \begin{cases} C(\mathbb{Z}_p, \mathbb{Z}_p), & x \notin \overline{\mathbb{Q}}, \\ \mathbb{Z}_p^{\#Z_p(f)}, & x \in \overline{\mathbb{Q}}. \end{cases}$$

Thus the transcendental alternative forces continuum-many characters already modulo p, while the algebraic alternative permits at most $\deg f$. This is a structural consequence of the alternative, not a computation of the characters of the actual gamma ring. For example, $\mathcal{B}_{\sqrt{17}}$ has exactly two characters to $\mathbb{Z}/2^k\mathbb{Z}$ for every k, with no transient congruence-root branches. In $\mathcal{B}_{\sqrt{2}}$ the number 2 is a unit: $\binom{\sqrt{2}}{2} = 1 - \sqrt{2}/2$ yields $\sqrt{2}/2$ and then $1/2$. Hence its reduction modulo 2 is the zero ring.

9.2 An explicit identity forced in the algebraic alternative

For $N \geq 1$ define the subgroup of finite sums

$$T_N(x) = \sum_{n \geq N} \mathbb{Z} \binom{x}{n}.$$

Before evaluation this is the ideal of $\mathcal{I}$ vanishing at $0, 1, \ldots, N-1$: Newton expansion identifies vanishing at these points with vanishing of the first N binomial coefficients. It follows that $T_N(x)$ is an ideal of $\mathcal{B}_x$.

Theorem 9.2 (Tail quotient and constructive unit)**.** *If x is transcendental, then*

$$\mathcal{B}_x / T_N(x) \simeq \mathbb{Z}^N.$$

If x is algebraic and $x \notin \{0, \ldots, N-1\}$, then $T_N(x) = \mathcal{B}_x$. More explicitly, if $f \in \mathbb{Z}[X]$ is its minimal polynomial of degree d and $M = \max_{0 \leq m < N} |f(m)|$, there are computable integers $c_N, \ldots, c_K$ such that

$$1 = \sum_{n=N}^{K} c_n \binom{x}{n}, \qquad K \leq N - 1 + dM.$$

If $x = m \in \{0, \ldots, N-1\}$, the quotient instead equals $\mathbb{Z}$.

Proof. In the transcendental case evaluation is injective. Restriction of $\mathcal{I}$ to the N integer points is onto $\mathbb{Z}^N$, because the binomial interpolation matrix is unit triangular. Its kernel is the tail ideal, proving the first assertion.

In the algebraic case under consideration each $f(m)$ is nonzero. Set

$$L_m(X) = \binom{X}{m}\binom{N-1-X}{N-1-m}, \qquad P_m(X) = \binom{\operatorname{sgn}(f(m))f(X)}{|f(m)|}.$$

Both polynomials are integer-valued. On $0 \le j < N$ one has $L_m(j) = \delta_{mj}$, and $P_m(m) = 1$. Furthermore f divides P_m in $\mathbb{Q}[X]$, since its binomial expression has a factor $f(X)$. Consequently

$$F_N(X) = \sum_{m=0}^{N-1} L_m(X)P_m(X), \qquad Q_N(X) = 1 - F_N(X)$$

satisfy $Q_N(j) = 0$ for $0 \le j < N$, $Q_N(x) = 1$, and $\deg Q_N \le N - 1 + dM$. Expand Q_N in the binomial basis:

$$Q_N(X) = \sum_{n=0}^{\deg Q_N} \Delta^n Q_N(0)\binom{X}{n}.$$

The coefficients are integers, and the first N are zero. This yields the required identity and proves that the tail ideal contains 1. Finally, for $x = m \in \{0, \dots, N-1\}$ the ring is $\mathbb{Z}$ and every tail value is zero. □

Corollary 9.3 (A proper-ideal target at Euler's constant)**.** *For Euler's constant, which satisfies* $0 < \gamma < 1$*, the following are equivalent:*

1. *γ is transcendental;*
2. *$T_N(\gamma)$ is a proper ideal for some $N \ge 1$;*
3. *$\mathcal{B}_\gamma/T_N(\gamma) \simeq \mathbb{Z}^N$ for every $N \ge 1$.*

For any fixed prime p, these conditions are also equivalent to the existence of a unital homomorphism

$$\phi : \mathcal{B}_\gamma \longrightarrow \mathbb{Z}/p\mathbb{Z}, \qquad \phi\left(\binom{\gamma}{n}\right) = 0 \quad (n \ge 1).$$

Proof. The first three equivalences follow from Theorem 9.2. Under transcendence, evaluation at $X = 0$ on $\mathcal{I}$, followed by reduction modulo p, gives the indicated homomorphism. Under algebraicity the tail-unit identity with $N = 1$ would map 1 to 0, so no such homomorphism exists. □

The corollary states an exact remaining target. It does not prove that target for γ. Prescribing images of the generators does not construct a homomorphism: preservation of all their integer relations is essential, and the algebraic tail identity exhibits the precise obstruction.

9.3 Coefficient costs and the topology obstruction

For every real $0 < x < 1$, independently of its arithmetic nature,

$$\left|\binom{x}{n}\right| = \frac{x\Gamma(n-x)}{\Gamma(1-x)\Gamma(n+1)} \quad (n \ge 1).$$

These values decrease strictly, since consecutive magnitudes have ratio $(n-x)/(n+1)$. The exact tail sum is

$$S_N(x) = \sum_{n\ge N} \left|\binom{x}{n}\right| = \frac{\Gamma(N-x)}{\Gamma(1-x)\Gamma(N)} = \prod_{j=1}^{N-1}\left(1-\frac{x}{j}\right).$$

Indeed the product tends to zero, starts at $S_1 = 1$, and its successive difference is $xS_N/N = |\binom{x}{N}|$.

Proposition 9.4 (Necessary and constructive coefficient budgets)**.** *Every finite identity* $1 = \sum_{n=N}^{K} c_n\binom{x}{n}$ *with* $0 < x < 1$ *satisfies*

$$\sum_{n=N}^{K} |c_n| \ge \left|\binom{x}{N}\right|^{-1}, \qquad \max_{N\le n\le K} |c_n| > \frac{\Gamma(1-x)\Gamma(N)}{\Gamma(N-x)}.$$

The respective lower bounds are asymptotic to $\Gamma(1-x)N^{1+x}/x$ *and* $\Gamma(1-x)N^x$.

For the construction of Theorem 9.2, put $K_0 = N-1+dM$ *and* $T = \max_{0\le j\le K_0}|f(j)|$. *There are at most* dM *coefficient positions, and*

$$\max_n |c_n| \le 2^{K_0}\big[1 + N(K_0+N)^{N-1}(T+M)^M\big], \qquad T \le (d+1)H(f)\max(1,K_0)^d.$$

In particular the construction permits $\log\max(1,\max_n|c_n|) = O_f(N^d\log(N+1))$.

Proof. The triangle inequality and the largest tail magnitude prove the first lower bound. With $H = \max|c_n|$, positivity of the infinitely many omitted magnitudes gives

$$1 \le H\sum_{n=N}^{K}\left|\binom{x}{n}\right| < HS_N(x),$$

which proves the strict second bound. The stated asymptotics follow from the usual ratio asymptotic for the Gamma function. For the upper estimate, at $0 \le j \le K_0$ bound $|L_m(j)|$ by $(K_0+N)^{N-1}$ and $|P_m(j)|$ by $(T+M)^M$. Thus $|Q_N(j)| \le 1 + N(K_0+N)^{N-1}(T+M)^M$. Use $|\Delta^n Q_N(0)| \le 2^n \max_{j\le n}|Q_N(j)|$. The bound for T follows by bounding each monomial of f; for fixed f, $M = O_f(N^d)$ and $K_0 = O_f(N^d)$. □

The budgets do not conflict. In particular, tail-unit identities with a coefficient bound independent of N are impossible as $N\to\infty$ for every $x\in(0,1)$, including rational numbers. This does not exclude the algebraic alternative, whose construction permits increasing coefficients. The elementary identity $-8\binom{1/2}{2} = 1$ already disproves the claim that small tail values cannot have integer span containing 1; Theorem 9.2 gives such identities starting arbitrarily far out.

Remark 9.5 (Finite ring operations versus real infinite sums). The positive weights $w_n(x) = (-1)^{n-1}\binom{x}{n}$ satisfy

$$\sum_{n\ge1} w_n(x) = 1, \qquad \sum_{n\ge1} w_n(x)z^n = 1-(1-z)^x \quad (0<x<1).$$

These identities hold for rational and transcendental parameters alike. A character of $\mathcal{B}_x$ acts on finite ring expressions; it need not preserve a real-convergent infinite sum. In the transcendental case the character at the p-adic point 0 sends every $b_n(x)$, $n\ge1$, to zero. Applying that character termwise to the displayed real sum would give a false contradiction. Continuity for the real topology has not been proved and is not part of a ring homomorphism. Similarly, small singularity sets of these generating functions do not supply the integer-coefficient or denominator hypotheses of arithmetic rationality criteria.

The recovered companion archive contains `check_binomial_tail.py`. Its eleven checks for $f = 2X - 1$, $2X^2 - 1$, and $X^2 + X - 1$ were rerun successfully: each checks integer Newton coefficients, vanishing below N, and exact divisibility of $Q_N - 1$ by f. These finite examples supplement the quantified polynomial proof above.

10 Rational Borel kernels and arithmetic division

The earlier Euler-series calculation extends to every rational Borel kernel. The resulting classification is an application of exponential partial-fraction reduction and the regular-singularity theorem for G-operators; it is not a claim of literature-wide priority. Crucially, it classifies formal arithmetic division, not the arithmetic nature of an integral merely because the corresponding analytic quotient is regular.

For a formal series $u(x) = \sum_{n\ge 0} u_n x^n$, put

$$\mathcal{B}u(z) = \sum_{n\ge 0} \frac{u_n}{n!} z^n.$$

We use the convention that u is arithmetic Gevrey of order 1 if $\mathcal{B}u$ is a G-function; see [5]. Let $\alpha \in \overline{\mathbb{Q}} \setminus \{0\}$, let $R(z) = \sum_{n\ge 0} r_n z^n \in \overline{\mathbb{Q}}(z)$ be regular at 0, and define

$$f_R(x) = \sum_{n\ge 0} n! r_n x^n, \qquad g_a(x) = \frac{f_R(x) - a}{x - \alpha}, \qquad G_a = \mathcal{B}g_a, \quad a \in \overline{\mathbb{Q}}. \tag{10.1}$$

The quotient in (10.1) is taken in $\overline{\mathbb{Q}}[[x]]$.

Theorem 10.1 (Complete rational-kernel classification)**.** *The series g_a is arithmetic Gevrey of order 1 if and only if there exists $T \in \overline{\mathbb{Q}}(z)$, regular at 0, such that*

$$R = T - \alpha T', \qquad a = T(0). \tag{10.2}$$

In this case T is unique and $G_a = T'$ is rational.

Proof. Write $g_a(x) = \sum c_n x^n$ and $b_n = c_n/n!$. Coefficient comparison in $(x - \alpha)g_a = f_R - a$ gives

$$b_0 = \frac{a - r_0}{\alpha}, \qquad \alpha n b_n = b_{n-1} - n r_n \quad (n \ge 1).$$

Consequently the convergent germ G_a satisfies

$$\alpha G_a' - G_a = -R', \qquad G_a(0) = \frac{a - R(0)}{\alpha}. \tag{10.3}$$

At each finite pole β of R, write the principal part as $\sum_{j=1}^{m_\beta} r_{\beta,j}(z - \beta)^{-j}$ and put

$$c_\beta = \sum_{j=1}^{m_\beta} \frac{r_{\beta,j}(-1/\alpha)^{j-1}}{(j-1)!}. \tag{10.4}$$

Successively cancelling the highest pole orders, and then the polynomial part, gives a rational function T, regular at 0, with

$$R = T - \alpha T' + \sum_\beta \frac{c_\beta}{z - \beta}. \tag{10.5}$$

For completeness, a pole of order $j \geq 2$ is removed using a multiple of $(z-\beta)^{-j+1}$ in T. The map $T \mapsto T - \alpha T'$ is triangular with diagonal 1 on polynomials. The remaining simple-pole coefficient is exactly (10.4), since

$$\operatorname{Res}_{z=\beta}\bigl(e^{-z/\alpha}R(z)\,dz\bigr) = e^{-\beta/\alpha}c_\beta, \qquad e^{-z/\alpha}(T-\alpha T') = -\alpha\frac{d}{dz}(e^{-z/\alpha}T).$$

These residues also prove uniqueness of the coefficients c_β. The remaining ambiguity in T would solve $T - \alpha T' = 0$; a nonzero solution is a constant multiple of $e^{z/\alpha}$ and is not rational.

An integrating factor in (10.3) yields

$$G_a(z) = e^{z/\alpha}\left(\frac{a-R(0)}{\alpha} - \frac{1}{\alpha}\int_0^z e^{-t/\alpha}R'(t)\,dt\right).$$

Since the residue of a derivative is zero,

$$\operatorname{Res}_\beta(e^{-t/\alpha}R'(t)\,dt) = \alpha^{-1}e^{-\beta/\alpha}c_\beta.$$

Analytic continuation counterclockwise around β therefore gives

$$\Delta_\beta G_a(z) = -\frac{2\pi i}{\alpha^2}e^{(z-\beta)/\alpha}c_\beta. \tag{10.6}$$

If G_a were a G-function and some c_β were nonzero, two continued solutions of its minimal homogeneous differential equation would have a nonzero exponential difference. The minimal operator of a G-function is regular singular, including at infinity; this standard G-operator input is recalled in [7, §1, p. 3]. Every solution of such an operator has power–logarithmic local behaviour after finite ramification. The exponential $e^{z/\alpha}$ has an irregular singularity at infinity and cannot be such a solution. Thus every c_β is zero.

We now have $R = T - \alpha T'$. Substituting in (10.3), including its initial condition, gives

$$G_a(z) = T'(z) + \frac{a-T(0)}{\alpha}e^{z/\alpha}. \tag{10.7}$$

Rational functions regular at 0 are G-functions. If $a \neq T(0)$, closure under subtraction would make $e^{z/\alpha}$ a G-function, contradicting the same regular-singularity property. Hence $a = T(0)$. Conversely, (10.2) makes (10.7) equal to T', which proves arithmetic Gevrey membership and completes the proof. □

Corollary 10.2 (Denominator obstruction)**.** *Fix a number field K containing a, α and the coefficients of R. Let D_N be the least positive rational integer such that $D_N b_0, \ldots, D_N b_N$ belong to the ring of integers of K. If (10.2) does not hold, then*

$$\limsup_{N\to\infty}\frac{\log D_N}{N} = +\infty.$$

Proof. Equation (10.3) gives holonomicity. For every embedding of K into $\mathbb{C}$, its rational forcing term is analytic in a neighbourhood of 0 and its leading coefficient is a nonzero constant. The corresponding solution is analytic there. Since there are finitely many embeddings, Cauchy's estimates give exponential bounds for all conjugates of the b_n. Thus an exponential bound on D_N would supply the remaining G-function condition, contrary to Theorem 10.1. Failure of every such bound is the stated infinite limsup. □

We do not assert a limit, an effective growth rate, or a universal factorial asymptotic for these denominators.

10.1 What analytic evaluation does not supply

Suppose now that $\alpha > 0$ and R has no poles on $[0, \infty)$. The positive-direction Borel sum at α is

$$\mathcal{L}_\alpha(R) = \int_0^\infty e^{-t} R(\alpha t)\, dt.$$

If (10.2) holds, integration by parts gives

$$\mathcal{L}_\alpha(R) = \int_0^\infty e^{-t} \left(T(\alpha t) - \frac{d}{dt} T(\alpha t) \right) dt = T(0) = a. \tag{10.8}$$

The rational function T has no additional finite poles, so the integration is legitimate and its boundary term vanishes at infinity. The converse assertion

$$\mathcal{L}_\alpha(R) = a \in \overline{\mathbb{Q}} \quad \Longrightarrow \quad R = T - \alpha T', \quad a = T(0) \tag{10.9}$$

is *not* proved here. It is a zero-removal assertion for this family, within the framework of [5, Conjecture 2]. For $R(z) = 1/(1+z)$ and $\alpha = 1$, the obstruction is $c_{-1} = 1$, whereas $\mathcal{L}_1(R) = \delta$. Thus (10.9) for this one kernel would already prove the transcendence of δ.

Arithmetic division is not always obstructed: for

$$R(z) = \frac{1}{1+z} + \frac{1}{(1+z)^2}, \quad T(z) = \frac{1}{1+z}, \quad \alpha = a = 1,$$

we have $G_a = T' = -1/(1+z)^2$ and $\mathcal{L}_1(R) = 1$. These examples distinguish the proved formal criterion from its unproved analytic converse.

10.2 The whole one-pole hierarchy has two generators

Proposition 10.3 (Exact one-pole reduction)**.** *Suppose the only possible finite pole of $R \in \overline{\mathbb{Q}}(z)$ is -1. If its principal part is $\sum_{j=1}^m r_j (1+z)^{-j}$, then*

$$R = T - T' + \frac{c}{1+z}, \qquad c = \sum_{j=1}^m \frac{(-1)^{j-1} r_j}{(j-1)!}, \qquad \mathcal{L}_1(R) = T(0) + c\delta.$$

Consequently the value space of the whole family is exactly $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\delta$. Under A*, this space has dimension* 2 *over $\overline{\mathbb{Q}}$ and is contained in the ring* **E** *of E-values.*

Proof. The reduction and integral formula follow from (10.5) and integration by parts. The kernels 1 and $1/(1+z)$ supply the two generators, proving equality of the value spaces. Under A, the function

$$F_\gamma(z) = e^z(\mathrm{Ein}(z) - \gamma)$$

is an E-function with $F_\gamma(1) = \delta$. Lemma 4.1 gives algebraic linear independence of $1, e, T$. If δ were algebraic, the identity $T - \gamma e - \delta = 0$ would contradict that independence. Thus δ is transcendental under A, and the dimension is exactly 2. An explicit E-function witness for $\mathcal{L}_1(R)$ is $T(0) + cF_\gamma(z)$. □

Unconditionally the dimension is *at most* 2, since this argument does not determine the arithmetic nature of δ. Under A we obtain the consistent classification

$$\mathcal{L}_1(R) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad c = 0 \quad \Longleftrightarrow \quad R = T - T'.$$

Increasing the pole order or polynomial numerator in this family creates no further linearly independent collision. In particular it does not contradict Theorem 10.1.

The supplied companion also invokes a mixed function from earlier manuscripts to motivate analytic division at 1. That construction is not reproduced or needed in the rational-kernel proof above. The analytic-to-arithmetic implication in [5, Conjecture 3] remains an additional hypothesis; the classification does not establish it.

11 A stronger deformation and the normalization it still loses

Write $\gamma_{\mathrm{low}}(a, z)$ for the lower incomplete gamma function, to avoid confusion with Euler's constant.

Proposition 11.1 (A periodic deformation)**.** *For real t put*

$$h_t(a) = e^{t\sin(2\pi a)}, \quad G_t(a) = h_t(a)\Gamma(a), \quad U_t(a,z) = G_t(a) - \gamma_{\mathrm{low}}(a,z), \qquad z > 0.$$

Then $G_t(a+1) = aG_t(a)$, $G_t(n+1) = n!$, and

$$G_t(a)G_t(1-a) = \frac{\pi}{\sin \pi a}.$$

It has the same poles and residues as Γ, is positive on $a > 0$, and $U_t(\cdot, z)$ is entire. Moreover

$$U_t(a+1,z) = aU_t(a,z) + z^a e^{-z}, \qquad \partial_z U_t(a,z) = -z^{a-1}e^{-z}, \qquad -G_t'(1) = \gamma - 2\pi t.$$

For $t \neq 0$, it does not preserve the canonical upper-tail limit at infinity or logarithmic convexity.

Proof. The function h_t is entire, zero-free, 1-periodic, equals 1 at integers, and satisfies $h_t(a)h_t(1-a) = 1$. These facts give recurrence, integer values, reflection and unchanged residues. Subtracting the meromorphic lower incomplete function cancels every pole in a. Its recurrence and z-derivative give the two formulas for U_t. Since $h_t'(1) = 2\pi t$, differentiation gives the shifted slope.

For $a > 0$,

$$\lim_{z\to\infty} U_t(a,z) = (h_t(a) - 1)\Gamma(a).$$

If this is zero for every positive a, then $h_t \equiv 1$, forcing $t = 0$. Also $(\log G_t)''(a) = \psi'(a) - 4\pi^2 t \sin(2\pi a)$; since $\psi'(a) \to 0$ on the positive real axis, it becomes negative along an arithmetic progression when $t \neq 0$. □

This strengthens the manuscript's deformation example while retaining more classical identities. It does not claim preservation of all Stokes data.

Lemma 11.2 (All upper-tail positivity fixes the slope)**.** *Let G be differentiable on a real neighborhood of 1, with $G(1) = 1$. If $G(a) - \gamma_{\mathrm{low}}(a,z) \geq 0$ for every $z > 0$ and every a in that neighborhood, then $G'(1) = -\gamma$.*

Proof. Let $z \to \infty$. Then $G(a) \geq \Gamma(a)$ and equality holds at $a = 1$. The differentiable difference has a local minimum there, so its derivative is zero. □

Thus stronger analytic normalization does select the actual constant. The unresolved step is an arithmetic conclusion about that selected derivative. Merely identifying the correct member of a deformed family does not establish transcendence.

12 Gamma multiplication, dynamics, and a finite contradiction

This section imports Fourier analysis and invariant measures into the normalization problem. All its conclusions are unconditional statements in world C. They do not assume either arithmetic alternative for γ.

For a real 1-periodic function p and an integer $m \geq 2$, write

$$(M_m) \qquad \sum_{j=0}^{m-1} p(x + j/m) = p(mx). \tag{12.1}$$

Then $F(x) = \Gamma(x)e^{p(x)}$ retains the Gamma recurrence and its exact Gauss m-multiplication formula. The additional conditions $p(0) = 0$ and $p(-x) = -p(x)$ give normalization at 1 and the reflection formula, respectively. We use the usual multiplication formula from [6].

12.1 A regularity boundary

Theorem 12.1 (Absolute-continuity rigidity)**.** *An absolutely continuous real periodic solution of one equation (M_m), with $m \geq 2$, is identically zero.*

Proof. Integration gives $(m-1)\int_0^1 p = 0$. Let $a_n = \int_0^1 p(x)e^{-2\pi inx}\,dx$. Comparing the coefficient at frequency mn gives $ma_{mn} = a_n$. For $f = p' \in L^1$ we therefore have $\widehat{f}(mn) = \widehat{f}(n)$. The Riemann–Lebesgue lemma along $m^k n \to \infty$ makes every nonzero Fourier coefficient of f vanish; the constant coefficient also vanishes by periodicity. Fourier uniqueness and absolute continuity imply that p is constant, and its mean is zero. □

In particular, a positive C^1 function on $(0, \infty)$ satisfying the Gamma recurrence and one exact Gauss multiplication formula must be Γ: apply the theorem to $\log(F/\Gamma)$ and extend periodically. Analytic Gamma uniqueness has classical antecedents; see [11, Section 2.11]. No originality is claimed for this uniqueness principle.

Theorem 12.2 (Continuous solutions retaining any finite list)**.** *Let S be a finite nonempty set of primes, and let*

$$\mathcal{N}_S = \left\{ \prod_{q \in S} q^{v_q} : v_q \in \mathbb{Z}_{\geq 0} \right\}, \qquad p_S(x) = \sum_{n \in \mathcal{N}_S} \frac{\sin(2\pi nx)}{n}.$$

Then p_S is a nonzero continuous, odd, 1-periodic function with $p_S(0) = 0$. It satisfies (M_m) for every $m \in \mathcal{N}_S$, $m \geq 2$, and is Hölder continuous of every exponent $0 < \alpha < 1$. It is not absolutely continuous.

Proof. The series converges absolutely and uniformly because $\sum_{n \in \mathcal{N}_S} n^{-1} = \prod_{q \in S}(1 - q^{-1})^{-1} < \infty$. Its coefficient at frequency 1 is $1/(2i)$, so it is nonzero. Summing a frequency over $x + j/m$ annihilates it unless $m \mid n$. For the surviving frequencies put $n = m\ell$; division by an S-smooth integer preserves membership in $\mathcal{N}_S$. This proves (M_m) term by term. Moreover,

$$|p_S(x + t) - p_S(x)| \leq C_\alpha |t|^\alpha \prod_{q \in S} (1 - q^{\alpha - 1})^{-1}.$$

This follows from $|\sin(u + v) - \sin u| \leq C_\alpha |v|^\alpha$. Absolute continuity would contradict Theorem 12.1. □

Given any finite list of integer multipliers, choose S to contain their prime factors. Thus $\Gamma(x)\exp(cp_S(x))$, for $c \neq 0$, retains that entire list as well as recurrence, normalization, positivity, and reflection. These are real continuous functions, not meromorphic Gamma functions with all their defining analytic properties. In particular, no derivative at 1 is being assigned to them.

12.2 An exact dictionary with an external open problem

Let $T_m(x) = mx \pmod 1$ and $R(x) = -x \pmod 1$ on the circle, and let λ be normalized Lebesgue measure.

Theorem 12.3 (Invariant-measure dictionary)**.** *The following classes are in bijection:*

1. *atomless Borel probability measures μ invariant under T_2, T_3, and R;*
2. *continuous real 1-periodic functions p with $p(0) = 0$, $p(-x) = -p(x)$, (M_2), (M_3), and $x + p(x)$ nondecreasing on $[0, 1]$.*

The maps are

$$p(x) = \mu([0, x]) - x \quad (0 \le x \le 1), \qquad \mu = dx + dp.$$

Lebesgue measure corresponds exactly to $p = 0$.

Proof. For an atomless probability, its distribution function F is continuous with endpoints $0, 1$. Reflection symmetry gives $F(1 - x) = 1 - F(x)$, so $p = F - x$ extends periodically to an odd function of mean zero. For $n \neq 0$, distributional differentiation gives

$$\widehat{\mu}(n) = 2\pi i n \widehat{p}(n). \tag{12.2}$$

Invariance under T_m is equivalent to $\widehat{\mu}(mn) = \widehat{\mu}(n)$. Consequently $m\widehat{p}(mn) = \widehat{p}(n)$. These are exactly the nonconstant Fourier identities for (M_m); its constant coefficient vanishes because p has mean zero. Fourier uniqueness and continuity prove the identities pointwise.

Conversely, $F = x + p$ is a continuous distribution function. Its Stieltjes measure is an atomless probability. Equation (12.2) and (M_m) prove invariance, and oddness proves reflection symmetry. The constructions are inverse. □

The assertion that class (2) contains only $p = 0$ is equivalent to the atomless form of Furstenberg's $\times 2, \times 3$ measure-rigidity conjecture, stated in [12, Conjecture 1.3]. Indeed, given a non-Lebesgue invariant atomless measure, symmetrize it as $\nu = (\mu + R_*\mu)/2$. If $\nu = \lambda$, then $\mu \le 2\lambda$. The Fourier coefficients of this absolutely continuous measure tend to zero while being constant along powers of 2; hence $\mu = \lambda$, a contradiction. Thus symmetrization preserves nontriviality. This is an equivalent reformulation, not a solution of that conjecture.

12.3 A completed finite contradiction

Theorem 12.4 (Positive-measure obstruction)**.** *For every finite nonempty S and every real $c \neq 0$, the function $x + cp_S(x)$ is not nondecreasing on $[0, 1]$.*

Proof. Suppose it were nondecreasing. Then $\mu = dx + c\,dp_S$ would be a probability measure, with Fourier coefficients

$$\widehat{\mu}(0) = 1, \qquad \widehat{\mu}(n) = \begin{cases} a := \pi c, & |n| \in \mathcal{N}_S, \\ 0, & |n| \notin \mathcal{N}_S, \end{cases} \quad n \neq 0.$$

These coefficients follow by distributional differentiation of the uniformly convergent series; we have not assumed beforehand that dp_S is a measure.

Choose a prime $r \notin S$ and $q \in S$. With $L = r - 1$, set $n_j = q^{Lj}$ for $j \geq 1$. Each n_j is S-smooth, while each nonzero difference $n_j - n_k$ is divisible by r and is not S-smooth. The Gram matrix of $1, e^{2\pi i n_1 x}, \ldots, e^{2\pi i n_K x}$ in $L^2(\mu)$ is therefore

$$M_K = \begin{pmatrix} 1 & a\mathbf{1}^T \\ a\mathbf{1} & I_K \end{pmatrix}. \tag{12.3}$$

Its Schur complement must satisfy $1 - Ka^2 \geq 0$. Choosing $K > 1/a^2$ contradicts this. Explicitly, the nonnegative trigonometric polynomial

$$U_K(x) = \left| 1 - a \sum_{j=1}^{K} e^{2\pi i n_j x} \right|^2$$

would obey $0 \leq \int U_K \, d\mu = 1 - Ka^2 < 0$. □

For $S = \{2, 3\}$ take $r = 7$, $q = 2$, and $L = 3$, so $n_j = 8^j$. At $c = 1/\pi$ already $K = 2$ gives the exact certificate

$$M_2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \det M_2 = -1.$$

This is a genuine contradiction to the added positivity assertion. Neither A nor B has been proved to force that assertion, so it does not reject either arithmetic world. Abstractly, the proof is Bessel's inequality: infinitely many orthogonal unit vectors cannot all have the same nonzero inner product with the vector 1.

Proposition 12.5 (Failure of bounded variation)**.** *The function p_S is not of bounded variation.*

Proof. If it were, $\nu = dp_S/\pi$ would be a finite signed measure with $\widehat{\nu} = 1_{E_S}$, where $E_S = \mathcal{N}_S \cup (-\mathcal{N}_S)$. Fourier uniqueness would give $\nu * \nu = \nu$. The classical idempotent theorem, in the form stated in [13, Theorem 1.1], implies that E_S is in the coset ring of $\mathbb{Z}$. Such a set is periodic outside a finite set: its finite Boolean description uses only residue classes and singleton cosets. Every infinite such set has positive symmetric density. But E_S is infinite and

$$\#(E_S \cap [-X, X]) \leq 2 \prod_{q \in S} \left(1 + \frac{\log X}{\log q} \right) = o(X).$$

This is a contradiction. □

The deeper external theorem is used only in this strengthening. The finite contradiction in Theorem 12.4 is elementary.

12.4 Why extremely small numerical defects do not settle the issue

Proposition 12.6 (Real approximation with any prescribed derivative)**.** *Fix $m \geq 2$ and $a \in \mathbb{R}$. For $N \geq 0$ let*

$$q_N(z) = \sum_{j=0}^{N} m^{-j} \sin(2\pi m^j z), \qquad c_N = \frac{\gamma - a}{2\pi(N+1)}, \qquad G_N(z) = \Gamma(z) e^{c_N q_N(z)}.$$

Then G_N retains recurrence, reflection, normalization, positivity on $(0,\infty)$, and the Gamma divisor and residues. Also $-G_N'(1) = a$. Its real logarithmic multiplication defect is

$$R_N(x) = -c_N m^{-N} \sin(2\pi m^{N+1} x), \qquad \|R_N\|_{\infty,\mathbb{R}} \le \frac{|\gamma - a| m^{-N}}{2\pi(N+1)}.$$

The quotient G_N/Γ converges uniformly to 1 on the real axis.

Proof. The function q_N is entire, odd, periodic, and vanishes at all integers; $q_N'(1) = 2\pi(N+1)$. This proves the exact properties. Root-of-unity averaging gives the telescoping identity

$$\sum_{r=0}^{m-1} q_N(x + r/m) - q_N(mx) = -m^{-N} \sin(2\pi m^{N+1} x).$$

It is precisely the log defect of the Gauss formula after multiplication by c_N. Finally $\|c_N q_N\|_{\infty,\mathbb{R}} \le |\gamma - a|/[2\pi(N+1)(1-m^{-1})] \to 0$. □

When $a \neq \gamma$ these functions do not converge uniformly on a fixed complex neighborhood of 1; their fixed derivative discrepancy already precludes that by Cauchy's estimates. They are not arithmetic models of the actual Gamma function: the parameter c_N uses the actual γ. They prove precisely that tiny real defects cannot be promoted to exact analytic normalization or to an arithmetic conclusion.

The missing implication for an A-world contradiction is therefore substantive: A would have to force a nonzero absolutely continuous solution of (M_m), or some other object actually excluded by a theorem. No such implication has been obtained.

13 A finite algebraic camouflage certificate

The review archive supplies an exact rational-arithmetic certificate for a single degree-twelve example. Its polynomial is unambiguously specified by

$$\begin{aligned} A_0 &= 577215664901532860606512090082, \\ M_0 &= 6220800 \cdot 10^{45}, \\ f(X) &= f_0(X) + M_0(10^{30}X - A_0). \end{aligned}$$

where the coefficients of f_0 in ascending order are

$$\begin{aligned} &(3732480, 1140480, 1104480, 371880, 2140, 38190, 897935, \\ &\ 846090, 606255, 1211490, 1925, 746430, 1). \end{aligned}$$

The script verifies Eisenstein's criterion at 5, coefficientwise congruence to $X(X-1)\cdots(X-11)$ modulo $2^{10}3^5$, and the minimal possible monic degree 12 for that all-residue congruence. All positive-degree coefficients of f are positive. Opposite signs at

$$\frac{A_0}{10^{30}} - 10^{-75}, \qquad \frac{A_0}{10^{30}}$$

therefore isolate its unique positive real root ξ. Defining $N_{p,k}(\xi) = \#\{u \bmod p^k : f(u) \equiv 0 \pmod{p^k}\}$, exhaustive integer enumeration verifies

$$N_{2,10}(\xi) = 1024, \qquad N_{3,5}(\xi) = 243, \qquad N_{2,11}(\xi) = 512.$$

For the real comparison, the certificate uses the positive-real digamma remainder bound in [6], with harmonic index 100 and 20 Bernoulli terms, together with rational logarithm bounds from 150 terms of the expansion

$$\log x = 2\sum_{j=0}^{L-1} \frac{z^{2j+1}}{2j+1} + R_L, \quad z = \frac{x-1}{x+1}, \quad 0 < R_L < \frac{2z^{2L+1}}{(2L+1)(1-z^2)} \quad (x > 1).$$

The decomposition $\log 100 = 6\log 2 + \log(25/16)$ then yields exact rational endpoints for γ and certifies

$$4.02 \cdot 10^{-31} < \gamma - \xi < 4.03 \cdot 10^{-31}.$$

These assertions were rerun successfully from the supplied `verify_camouflage.py`. They establish this finite example; other LLL computations and asymptotic approximation exponents do not follow from it.

Part III
Prime, moment and cross-domain constructions

14 Prime units, minimum certificate degree, and finite camouflage

Write $\mathcal{I} = \mathrm{Int}(\mathbb{Z})$, $\mathcal{B}_x = \mathbb{Z}[\binom{x}{n} : n \geq 0]$, and $U(x) = \{p \text{ prime} : 1/p \in \mathcal{B}_x\}$. Kalinov–Mandelshtam already characterize the algebraic binomial ring by integral p-adic embeddings; their Corollary 6.3 and following paragraph establish prime inverses and their positive density [14]. We do not claim these conclusions as new. Here we give finite certificates, prove a minimum-degree formula, and quantify the limitations of finite prime observations.

14.1 An explicit finite inverse certificate

Proposition 14.1. *Let $f \in \mathbb{Z}[X]$ have degree $d \geq 1$ and no root modulo $q = p^m$. Then*

$$W(X) = \frac{(-1)^{q-1}}{p}\binom{f(X)-1}{q-1} \in \mathcal{I}, \qquad \deg W = d(q-1).$$

At every root x of f, this gives $W(x) = 1/p$, hence an explicit finite integer-binomial expansion of $1/p$ in $\mathcal{B}_x$.

Proof. Put $J_q(Y) = (-1)^{q-1}\binom{Y-1}{q-1}$, so $J_q(0) = 1$. Lucas's digit identity gives, for all integers y,

$$J_q(y) \equiv \begin{cases} 1, & q \mid y, \\ 0, & q \nmid y \end{cases} \pmod{p}.$$

For negative integers the same conclusion follows by periodicity: the coefficient of T^{q-1} in $(1+T)^{y-1}$ modulo p is unchanged by replacing y by $y+q$, because $(1+T)^q \equiv 1+T^q$. The sign in J_q is 1 modulo p. Thus $J_q(f(a))$ is divisible by p for every integer a, proving integer-valuedness of W. Its value at a root is $1/p$. Newton expansion supplies integer coefficients $c_n = \Delta^n W(0)$ with $W(X) = \sum c_n\binom{X}{n}$. □

For an algebraic x with primitive minimal polynomial f, it follows that

$$p \in U(x) \quad \Longleftrightarrow \quad f \text{ has no root in } \mathbb{Z}_p. \tag{14.1}$$

Indeed, a root $\beta \in \mathbb{Z}_p$ gives a well-defined homomorphism $\mathcal{B}_x \to \mathbb{Z}_p$, $P(x) \mapsto P(\beta)$, excluding $1/p$. If no root exists, compactness of $\mathbb{Z}_p$ implies that some finite depth m has no congruence root, so the proposition applies. There is an effective safe depth: choose $A, B \in \mathbb{Z}[X]$ and $0 \neq R \in \mathbb{Z}$ with $Af + Bf' = R$, and put $s = v_p(R)$. A root modulo p^{2s+1} would have $t = v_p(f'(a)) \leq s$; strong Hensel lifting, since $v_p(f(a)) > 2t$, would give a root in $\mathbb{Z}_p$. Thus $m \leq 2s+1$ suffices in the rootless case. One may use $R = \mathrm{Res}(f, f')$.

The certificate also has a crude explicit coefficient budget. If $K = d(q-1)$ and $T = \max_{0\leq j\leq K} |f(j)|$, then

$$\max_n |c_n| \leq \frac{2^K}{p}(T+q)^{q-1}, \qquad T \leq (d+1)H(f)\max(1,K)^d.$$

These follow by bounding $J_q(f(j))$ and using $|\Delta^n W(0)| \le 2^n \max_{j\le n} |W(j)|$. The degree in Proposition 14.1 is an upper bound, not generally an optimum.

14.2 The exact minimum degree at large rootless primes

For $p \in U(x)$ define

$$\kappa_p(x) = \min\{\deg P : P \in \mathcal{I},\ P(x) = 1/p\}.$$

Equivalently, this is the least maximal binomial index of an inverse certificate.

Theorem 14.2. *Let x have primitive minimal polynomial f of degree d. Suppose $p > d$, p does not divide the leading coefficient of f, and f has no root modulo p. Let $A \in \mathbb{F}_p[X]$ be the unique representative of degree less than p of $\bar{f}^{-1}$ modulo $X^p - X$, and set $t = \deg A$. Then*

$$\kappa_p(x) = d + t, \qquad p \le \kappa_p(x) \le p + d - 1.$$

If $d = 2$, this specializes to $\kappa_p(x) = p + 1$.

Proof. The no-root assumption makes $\bar{f}$ invertible modulo $X^p - X$. Lift A to $\widetilde{A} \in \mathbb{Z}[X]$ of the same degree. Then

$$P(X) = \frac{1 - f(X)\widetilde{A}(X)}{p}$$

is integer-valued and has value $1/p$ at x, proving the upper bound $\kappa_p(x) \le d + t$.

Suppose a certificate of smaller degree $K < d+t < 2p$ existed. Because $v_p(n!) \le 1$ for $n \le K$, its integer binomial expansion implies $pP \in \mathbb{Z}_{(p)}[X]$. The relation at x implies

$$Q(X) = \frac{pP(X) - 1}{f(X)} \in \mathbb{Q}[X].$$

It has coefficients in $\mathbb{Z}_{(p)}$: the polynomial f has p-adic Gauss norm 1 by primitivity, and multiplicativity of that norm applied to $fQ = pP - 1$ bounds the norm of Q by 1. Reduction modulo p and evaluation at every $a \in \mathbb{F}_p$ give $\bar{f}(a)\bar{Q}(a) = -1$, since $P(a) \in \mathbb{Z}$. Hence $\bar{Q} \equiv -A \pmod{X^p - X}$. But $\deg Q \le K - d < t < p$, a contradiction. This proves minimality. Finally, the nonzero polynomial $\bar{f}A - 1$ is divisible by $X^p - X$, so $d + t \ge p$, while $t \le p - 1$.

For $d = 2$, complete the square over $\mathbb{F}_p$: after translation $\bar{f}$ is a nonzero constant times $Y^2 - D$ with D a nonsquare. Its inverse function on $\mathbb{F}_p$ is even, and its unique representative of degree less than p is therefore even. Thus t is even. Since $p - 2 \le t \le p - 1$ and p is odd, $t = p - 1$. □

For example, at a root x of $X^2 - 2$ the identity

$$\frac{1}{3} = 1 - 4\binom{x}{2} - 12\binom{x}{3} - 8\binom{x}{4}$$

has the smallest possible maximal binomial index, namely 4. For any fixed quadratic algebraic parameter, the required index at larger rootless odd primes grows exactly as $p + 1$. If the coefficients of $\widetilde{A}$ are chosen in $\{0, \dots, p - 1\}$ and $K = d + t$, the same finite-difference argument gives

$$\max_n |\Delta^n P(0)| \le \frac{2^K}{p}\left[1 + (d+1)(t+1)(p-1)H(f)\max(1, K)^K\right].$$

For fixed f its logarithm is $O_f(p \log p)$.

14.3 Frobenius density and the global alternatives

Let x be algebraic of degree $d > 1$, and let G be the Galois group of the splitting field of f, acting transitively on its roots. Outside finitely many primes, an integral p-adic root exists exactly when Frobenius fixes a root. Chebotarev and (14.1) give the natural density

$$\operatorname{dens} U(x) = \delta(G) := \frac{\#\{g \in G : g \text{ fixes no root}\}}{|G|}, \qquad \frac{1}{d} \le \delta(G) \le 1 - \frac{1}{d}.$$

The Frobenius correspondence and density theorem are standard [16]; their application to prime units is already in [14]. For completeness, the bounds follow from Burnside's counting lemma. If F is the number of fixed points of a uniform element of G, then $\mathbb{E}F = 1$ and $\mathbb{E}[F(F-1)] = r \ge 1$, the number of orbits on ordered distinct pairs. Moreover

$$\mathbb{E}[(F-1)1_{F\ge 1}] = \delta(G), \qquad F(F-1) \le d(F-1) \quad (F \ge 1).$$

Thus $r \le d\delta(G)$; and $1 = \mathbb{E}F \le d\Pr(F \ge 1)$ gives the other bound.

For a reduced rational $x = a/b$, $b > 0$, one has $\mathcal{B}_x = \mathbb{Z}[1/b]$. Indeed Bezout gives $1/b \in \mathbb{Z}[x]$, while every binomial value at a/b is p-integral when $p \nmid b$, so its denominator uses only primes dividing b. For transcendental x, the ring is isomorphic to $\mathcal{I}$, whose evaluation at 0 excludes every prime inverse. Consequently the complete classification is

$$\begin{array}{c|l} U(x) = \varnothing & x \in \mathbb{Z} \text{ or } x \notin \overline{\mathbb{Q}}, \\ 0 < \#U(x) < \infty & x \in \mathbb{Q} \setminus \mathbb{Z}, \\ \#U(x) = \infty & x \in \overline{\mathbb{Q}} \setminus \mathbb{Q}. \end{array}$$

In the last row the density is positive. Since $0 < \gamma < 1$,

$$\gamma \notin \overline{\mathbb{Q}} \iff U(\gamma) = \varnothing, \qquad \gamma \in \overline{\mathbb{Q}} \iff U(\gamma) \ne \varnothing.$$

An exact finite inverse certificate at γ would therefore prove algebraicity. Excluding all such certificates would prove transcendence. Neither conclusion has been obtained. Density zero alone would not settle the issue, since the rational case also has density zero.

14.4 Every finite prime pattern has a quadratic realization

Proposition 14.3. *Let S be a finite set of primes, $V \subseteq S$, and J a nonempty open real interval. There is a quadratic real number $x \in J$ with $U(x) \cap S = V$.*

Proof. For each odd $p \in S$, prescribe a nonzero square residue for D modulo p when $p \notin V$, and a nonsquare residue when $p \in V$. At 2, if present, prescribe $D \equiv 1 \pmod 8$ or $D \equiv 5 \pmod 8$, respectively. Choose an auxiliary prime $\ell \notin S$ and additionally require $D \equiv \ell \pmod{\ell^2}$. The Chinese remainder theorem gives a positive such D. Its valuation at ℓ is one, so it is not a rational square.

Take a bounded subinterval $(u, v) \subseteq J$ and a sufficiently large positive integer L coprime to every prime in S. The interval $(Lu - \sqrt{D}, Lv - \sqrt{D})$ has length exceeding one, so contains an integer a. Then $x = (a + \sqrt{D})/L \in J$ is quadratic. At every $p \in S$, both L and D are units, so an integral p-adic root of its minimal polynomial exists precisely when D is a square in $\mathbb{Q}_p$. The prescribed congruences give this condition exactly at $p \notin V$. Apply (14.1). □

Thus even an arbitrarily long finite list of nonunit primes is compatible with a quadratic parameter in an arbitrarily narrow interval about γ. Unlike camouflage of full residue counts,

this binary statistic needs no increase in degree. This does not exclude an all-primes argument or an argument with independent height control.

The supplied exact-arithmetic program was rerun for the present integration. All eleven checks passed: three finite-depth certificates, five minimum-degree certificates, and three quadratic realizations of prescribed patterns on $\{2,3,5,7\}$. It checks integer Newton coefficients, exact polynomial divisibility, local congruences, and rational real enclosures. The minimality assertions follow from Theorem 14.2, not from a bounded numerical search. A second supplied program also passed nine certificate checks, ten common-denominator snapshots, two hundred rational-denominator formula checks, and the exact Fourier rejection gap 33/200.

15 Binomial denominators and a Galois-density invariant

For $x \in \overline{\mathbb{Q}}$, put $K = \mathbb{Q}(x)$, let $L/\mathbb{Q}$ be its normal closure, and define

$$D_N(x) = \min\left\{ m \in \mathbb{Z}_{>0} : m\binom{x}{n} \in \mathcal{O}_K \quad (0 \le n \le N) \right\}.$$

This is a common denominator into the ring of integers, independent of the choice of an integral basis. It is different from the denominator into $\mathbb{Z}[x]$ introduced below. The results here are deductions from local valuations and Chebotarev; no literature-wide priority is claimed. Related binomial-value valuation descriptions occur in [14, Sections 5–6]; split-prime integrality and exceptional-prime bounds occur explicitly in [18, Lemma 4.4].

Theorem 15.1 (Exact good-prime denominators)**.** *For every $x \in \overline{\mathbb{Q}}$ there is a finite set B_x of rational primes such that, for $p \notin B_x$ and all $N \ge 0$,*

$$v_p(D_N(x)) = \begin{cases} 0, & p \text{ splits completely in } L, \\ v_p(N!), & \text{otherwise.} \end{cases}$$

Moreover,

$$\sum_{p \in B_x} v_p(D_N(x)) \log p = O_x(N).$$

Proof. Choose $m \ge 1$ with $\theta = mx \in \mathcal{O}_K$. Include in B_x all primes dividing m, the index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$, or the discriminant of L. For $p \notin B_x$, every prime $\mathfrak{p}$ of K above p is unramified, x is integral there, and its reduction generates $\mathcal{O}_K/\mathfrak{p}$ over $\mathbb{F}_p$.

If some residue degree $f_\mathfrak{p} > 1$, the reduction of x is not in $\mathbb{F}_p$. Thus every $x - j$, $j \in \mathbb{Z}$, is a $\mathfrak{p}$-unit, and, normalizing $v_\mathfrak{p}(p) = 1$,

$$v_\mathfrak{p}\left(\binom{x}{n}\right) = -v_p(n!).$$

This forces $v_p(D_N) \ge v_p(N!)$. At every prime above p the numerator of the binomial coefficient is integral, so $v_p(N!)$ also suffices to clear all the local denominators. Equality follows.

If all residue degrees are one, every completion $K_\mathfrak{p}$ is $\mathbb{Q}_p$ and $x \in \mathbb{Z}_p$. For fixed n, the polynomial $\binom{X}{n}$ maps $\mathbb{Z}_p$ into $\mathbb{Z}_p$: approximate its argument by nonnegative ordinary integers and use continuity. Thus $v_p(D_N) = 0$ in this case.

The residue degrees are the cycle lengths of Frobenius acting on the conjugates of x. This action of $\mathrm{Gal}(L/\mathbb{Q})$ is faithful, since those conjugates generate L. All degrees are one exactly when Frobenius is the identity, equivalently when p splits completely in L. This proves the good-prime formula, including mixed splitting types.

For the exceptional primes use the normalization $v_{\mathfrak{p}}(p) = e_{\mathfrak{p}}$. If $v_{\mathfrak{p}}(x) \geq 0$, then $v_{\mathfrak{p}}\binom{x}{n} \geq -e_{\mathfrak{p}} v_p(n!)$. If $v_{\mathfrak{p}}(x) < 0$, ultrametricity gives the exact equality

$$v_{\mathfrak{p}}\left(\binom{x}{n}\right) = n v_{\mathfrak{p}}(x) - e_{\mathfrak{p}} v_p(n!).$$

Consequently, with $C_p = \max_{\mathfrak{p}|p} \max(0, -v_{\mathfrak{p}}(x)/e_{\mathfrak{p}})$,

$$0 \leq v_p(D_N) \leq \lceil C_p N \rceil + v_p(N!) = O_{x,p}(N).$$

Summing over the fixed finite set proves the last assertion. In particular, nonintegral algebraic arguments are covered. □

Theorem 15.2 (Normal-closure asymptotic)**.** *For $g = [L : \mathbb{Q}]$ one has*

$$\log D_N(x) = \left(1 - \frac{1}{g}\right) N \log N + O_x(N), \qquad \lim_{N\to\infty} \frac{\log D_N(x)}{N \log N} = 1 - \frac{1}{g}.$$

Proof. Let S be the unramified completely split primes of L. Their Chebotarev density is $1/g$. The fixed-field effective Chebotarev estimate [17, Theorem 1.1], followed by partial summation, gives

$$\sum_{\substack{p \leq X \\ p \in S}} \frac{\log p}{p} = \frac{1}{g} \log X + O_L(1).$$

Its decaying error is integrable in this partial summation; a possible fixed exceptional-zero contribution has exponent $\beta < 1$ and is integrable as well. For $L = \mathbb{Q}$ use the usual prime estimate. Legendre's formula now yields

$$\begin{aligned} \sum_{\substack{p \leq N \\ p \in S}} v_p(N!) \log p &= N \sum_{\substack{p \leq N \\ p \in S}} \frac{\log p}{p} + O(N) \\ &= \frac{1}{g} N \log N + O_L(N). \end{aligned}$$

Indeed the errors from $\lfloor N/p \rfloor - N/p$ total $O(\sum_{p \leq N} \log p) = O(N)$, and the higher prime powers contribute at most $N \sum_p \log p/[p(p-1)] = O(N)$. By Theorem 15.1,

$$\log D_N(x) = \log(N!) - \sum_{\substack{p \leq N \\ p \in S}} v_p(N!) \log p + O_x(N).$$

Apply Stirling's formula. Ordinary asymptotic Chebotarev, without an effective error term, already proves the stated limit with an $o_x(N \log N)$ remainder. □

Remark 15.3 (Mixed splitting and lack of uniformity). The constant is not the density of derangements. For a non-Galois cubic field with normal closure group S_3, the splitting types $(1,1,1)$, $(1,2)$, and (3) have densities $1/6$, $1/2$, and $1/3$. Both latter types force $v_p(D_N) = v_p(N!)$, so the constant is $5/6$, not $1/3$. A single nonlinear completion suffices. All error constants depend on the fixed argument and field; no uniform effective bound in an unknown degree or height of γ is supplied here.

Proposition 15.4 (Exact rational denominator)**.** *If $x = a/b \in \mathbb{Q}$, $(a,b) = 1$ and $b \geq 1$, then*

$$D_N(a/b) = b^N \prod_{p|b} p^{v_p(N!)}, \qquad \lim_{N\to\infty} \frac{\log D_N(a/b)}{N} = \log b + \sum_{p|b} \frac{\log p}{p-1}.$$

These assertions hold for positive, zero, and negative a.

Proof. For $p \nmid b$, the argument belongs to $\mathbb{Z}_p$, so all binomial values are p-integral. If $p \mid b$, every numerator factor $a - jb$ is a p-unit, whence

$$v_p\left(\binom{a/b}{n}\right) = -nv_p(b) - v_p(n!).$$

The largest denominator exponent occurs at $n = N$. This proves the formula, including the empty product for $b = 1$. The limit follows from $v_p(N!)/N \to 1/(p-1)$. □

Corollary 15.5 (The classical order-zero dichotomy)**.** *For any $x \in \mathbb{C}$, the germ $(1+z)^x$ normalized to have value 1 at 0 is a G-function if and only if $x \in \mathbb{Q}$.*

Proof. A G-function must have algebraic coefficients, so its coefficient of z forces $x \in \overline{\mathbb{Q}}$. For algebraic x, the first-order equation $(1+z)F' = xF$ gives holonomicity, and Cauchy's estimates on a small disk give exponential bounds for all conjugate Taylor coefficients. The denominator condition holds for rational x by Proposition 15.4, and fails for algebraic irrational x by Theorem 15.2, since $g \geq 2$. □

Here G means arithmetic Gevrey of order zero; ordinary analytic Gevrey regularity is a different condition. The classical dichotomy is not claimed as a new classification theorem.

15.1 A denominator defined in both arithmetic worlds

For arbitrary complex x, define instead

$$J_N(x) = \min\left\{m \in \mathbb{Z}_{>0} : m\binom{x}{n} \in \mathbb{Z}[x] \quad (0 \leq n \leq N)\right\}.$$

This integer exists and divides $N!$. Unlike D_N, it is defined even when x is transcendental.

Theorem 15.6 (Quantized alternatives)**.** *If x is transcendental, $J_N(x) = N!$ exactly. If x is rational, $J_N(x) = 1$. For algebraic x with normal closure L,*

$$\log J_N(x) = \left(1 - \frac{1}{[L:\mathbb{Q}]}\right) N \log N + O_x(N).$$

In particular the following limit exists for every complex x:

$$\lambda(x) := \lim_{N\to\infty} \frac{\log J_N(x)}{N \log N} = \begin{cases} 0, & x \in \mathbb{Q}, \\ 1 - 1/[L:\mathbb{Q}], & x \in \overline{\mathbb{Q}} \setminus \mathbb{Q}, \\ 1, & x \notin \overline{\mathbb{Q}}. \end{cases}$$

Proof. If x is transcendental and $m\binom{x}{N} = P(x)$ with $P \in \mathbb{Z}[X]$, injectivity of evaluation gives $m\binom{X}{N} = P(X)$. Its leading coefficient $m/N!$ must be an integer. Thus $N! \mid m$, while $N!$ always clears every coefficient under consideration. This proves exact equality.

Now let x be algebraic and use B_x from Theorem 15.1. For every $p \notin B_x$,

$$\mathbb{Z}_{(p)}[x] = \mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}.$$

Indeed m and the index of $\mathbb{Z}[mx]$ in $\mathcal{O}_K$ are p-local units. To justify passage from this equality to denominators, consider the homomorphism of additive $\mathbb{Z}$-modules

$$\phi_N : \mathbb{Z} \longrightarrow \bigoplus_{n=0}^{N} K/\mathbb{Z}[x], \qquad u \longmapsto \left(u\binom{x}{n} + \mathbb{Z}[x]\right)_{n=0}^{N}.$$

Its kernel is $J_N(x)\mathbb{Z}$. Replacing $\mathbb{Z}[x]$ by $\mathcal{O}_K$ defines a homomorphism with kernel $D_N(x)\mathbb{Z}$. Localization is exact and commutes with this finite direct sum, so the displayed equality of localized rings makes their localized kernels equal. Hence

$$v_p(J_N(x)) = v_p(D_N(x)) \qquad (p \notin B_x).$$

No finite-generation assumption on $\mathbb{Z}[x]$ as a $\mathbb{Z}$-module is needed; in particular this covers nonintegral x. At exceptional primes $0 \le v_p(J_N) \le v_p(N!)$, because $J_N \mid N!$. Their contribution is $O_x(N)$, and the same global calculation as before proves the asserted asymptotic.

Finally, for $x = a/b$ in lowest terms, Bezout gives $\mathbb{Z}[x] = \mathbb{Z}[1/b]$. Proposition 15.4 shows that every binomial value lies in this ring, so $J_N = 1$. The limit in the transcendental case follows from Stirling's formula. □

Remark 15.7 (Exact reformulation, not an arithmetic oracle). Theorem 15.6 gives an equivalence: γ is transcendental if and only if $\lambda(\gamma) = 1$. However, computing $J_N(\gamma)$ exactly entails membership questions in $\mathbb{Z}[\gamma]$, hence unknown polynomial relations. Treating γ as a formal indeterminate and assigning $J_N = N!$ would assume the injectivity that must be proved. No independent estimate of $J_N(\gamma)$ is obtained here. Likewise, proving $(1+z)^\gamma$ to be a G-function would prove rationality, whereas proving that it is not a G-function would exclude rationality only. The established denominator alternatives therefore produce no contradiction in either world and settle no designated constant problem.

16 Exact denominator defects and optimal joint camouflage

For $x \in \mathbb{C}$ define

$$J_N(x) = \min\{m \ge 1 : m\tbinom{x}{n} \in \mathbb{Z}[x] \ (0 \le n \le N)\}.$$

It exists, divides $N!$, and $J_0 = J_1 = 1$. Let x now be an algebraic integer, with monic minimal polynomial f of degree d. Write $F_n = X(X-1)\cdots(X-n+1)$ and $R_n = F_n \bmod f$, with $\deg R_n < d$. The integral basis of the order $\mathbb{Z}[x]$, without any maximal-order assumption, gives

$$J_N(x) = \operatorname*{lcm}_{0 \le n \le N} \frac{n!}{\gcd(n!, \operatorname{coeffs}(R_n))}. \tag{16.1}$$

The convention for a zero coefficient vector is $\gcd(n!, 0) = n!$. All quantities are computed recursively by $R_{n+1} = (X-n)R_n \bmod f$. The basic binomial-ring description below is prior work of Kalinov–Mandelshtam (2021); we claim no literature-wide priority for the deductions proved here.

Theorem 16.1 (Exact support of the factorial defect). *For each prime p, put $e_p = \infty$ if $\bar{f} \in \mathbb{F}_p[X]$ has a nonlinear irreducible factor. Otherwise let e_p be its largest linear-root multiplicity. For every $N \ge 0$,*

$$p \mid N!/J_N(x) \quad \Longleftrightarrow \quad pe_p \le N.$$

Consequently

$$\operatorname{rad}(N!/J_N(x)) = \prod_{pe_p \le N} p, \qquad \min\{N \ge 2 : J_N(x) < N!\} = \min_p pe_p.$$

In particular $J_N(x) = N!$ whenever $N < d$.

Proof. Let $k = \lfloor N/p \rfloor$ and $n_0 = kp$. For $n_0 \le n \le N$, $v_p(n!) = v_p(N!)$; earlier indices have smaller valuation. Reduction modulo p gives

$$\overline{F_{kp}} = (X^p - X)^k.$$

If $\bar{f}$ does not divide this polynomial, the remainder R_{kp} has a coefficient which is a p-unit. Equation (16.1) forces $v_p(J_N) = v_p(N!)$. If it does divide, every R_n with $n_0 \le n \le N$ is coefficientwise divisible by p, since F_{n_0} divides F_n. Earlier factorials already have smaller valuation. Thus all the denominators in (16.1) have exponent at most $v_p(N!) - 1$. Finally, $X^p - X$ is the product of all distinct monic linear polynomials over $\mathbb{F}_p$, so divisibility holds exactly when $e_p \le k$. This includes every ramified and nonmaximal-order prime. The minimum is finite because there are completely split good primes, by Chebotarev. For $N < d$, no reduction of F_N by f occurs, and its leading coefficient is 1. □

f	First defect index	Responsible (p, e_p)
$X^2 - 2$	4	$(2, 2)$
$X^2 - X - 1$	10	$(5, 2)$
$X^3 - X - 1$	46	$(23, 2)$
$X^3 - 2$	6	$(2, 3)$

These examples were checked by exact remainder arithmetic through $N = 60$. For $X^3 - X - 1$, the first completely split unramified prime is 59; keeping the ramified prime 23 is essential for the exact answer.

Remark 16.2 (An explicit differential interpretation). Let C be the integral companion matrix of f. For the connection $\nabla_{\partial_z} = \partial_z - C/(1 + z)$, reduction modulo p gives

$$\Psi_p = \frac{C - C^p}{(1 + z)^p}.$$

Indeed, its pth iterate on constant vectors is $(-1)^p C(C + I) \cdots (C + (p - 1)I)/(1 + z)^p$, while $\partial_z^p = 0$ on $\mathbb{F}_p(z)$. Cyclicity of C shows that Ψ_p is nilpotent exactly when $e_p < \infty$, with nilpotency index e_p. Thus the first p-defect is p times that index. This is a concrete logarithmic-connection calculation, not a solution of the general p-curvature conjecture. A companion matrix for γ cannot be supplied without a polynomial relation for it.

Proposition 16.3 (Quadratic denominator camouflage)**.** *Given $N \ge 2$ and a nonempty open real interval I, there is a quadratic algebraic integer $x \in I$ satisfying $J_n(x) = n!$ for all $n \le N$. Its monic minimal polynomial can have coefficient height at most $C_I \prod_{p \le N} p$.*

Proof. Choose a bounded $(u, v) \subset I$ and set $M = \prod_{p \le N} p$. Seek $f = X^2 - AX + B$. At each odd $p \le N$ prescribe $A = 0$ and $-B$ a nonzero nonsquare; at 2 prescribe $A = B = 1$. Every reduction is irreducible. The Chinese remainder theorem gives residue classes for A, B modulo M. Choose A large positive in its class so that

$$A(v - u) - (v^2 - u^2) > M, \qquad A > 2\max(|u|, |v|) + 2.$$

There is an allowed B strictly between $Au - u^2$ and $Av - v^2$. Then $f(u) > 0 > f(v)$, giving a real root in I. Irreducibility modulo 2 implies irreducibility over $\mathbb{Q}$, and Theorem 16.1 proves the entire denominator prefix. The choices require only $A, B = O_I(M)$. □

Define $\mathcal{B}_x = \mathbb{Z}[\binom{x}{n} : n \geq 0]$ and $U(x) = \{p : 1/p \in \mathcal{B}_x\}$. The binomial-ring criterion gives, for algebraic x with primitive minimal polynomial f,

$$p \notin U(x) \quad \Longleftrightarrow \quad f \text{ has a root in } \mathbb{Z}_p. \tag{16.2}$$

For a transcendental parameter t, $U(t) = \varnothing$ and $J_n(t) = n!$. The quadratic construction above does not imitate the first statistic: it has prime inverses at all tested primes. The following result handles both statistics and is sharp in its degree.

Theorem 16.4 (Optimal joint camouflage)**.** *For every $N \geq 2$ and nonempty open real interval I, there is a cubic algebraic integer $x \in I$ with*

$$U(x) \cap \{p : p \leq N\} = \varnothing, \qquad J_n(x) = n! \quad (0 \leq n \leq N).$$

Its monic minimal polynomial can have height at most $C_I M \ell^2$, where $M = \prod_{p \leq N} p$ and $N < \ell \leq 2N$ is prime. For $N \geq 4$, degree three is minimal among algebraic integers.

Proof. Seek $f = X^3 - AX + B$. At 2 prescribe $A = 0, B = 1$, giving $(X+1)(X^2+X+1)$. At every odd $p \leq N$ prescribe A to be a nonzero nonsquare and $B = 0$, giving $X(X^2 - A)$. In every case there is a simple linear factor and an irreducible quadratic factor. The linear root lifts to $\mathbb{Z}_p$ by Hensel, proving the nonunit assertion by (16.2); the nonlinear factor proves the denominator assertion by Theorem 16.1.

At an auxiliary prime $\ell > N$ require $A = 0 \pmod{\ell}$ and $B = \ell \pmod{\ell^2}$. Eisenstein ensures irreducibility. These compatible prescriptions give a class for A modulo $M\ell$ and one for B modulo $M\ell^2$. In a bounded $(u, v) \subset I$, choose A in its class such that

$$A(v-u) - (v^3 - u^3) > M\ell^2, \qquad A > 3\max(u^2, v^2) + 2.$$

Choose B in its class strictly between $Au - u^3$ and $Av - v^3$. Then $f(u) > 0 > f(v)$ and f is strictly decreasing there, so it has a unique root in this interval. Also $A, B = O_I(M\ell^2)$.

For minimality, a degree-one algebraic integer has $J_2 = 1$. For a quadratic algebraic integer, $2 \notin U(x)$ gives a $\mathbb{Z}_2$ root of its monic polynomial. The other root is also in $\mathbb{Z}_2$, since their sum is an integer. Therefore $\bar{f}$ splits fully modulo 2 with $e_2 \leq 2$, and Theorem 16.1 gives $2 \mid 4!/J_4(x)$. This contradicts the required prefix whenever $N \geq 4$. □

Example 16.5 (An exact twenty-step joint witness)**.** The polynomial

$$f(X) = X^3 - 513113715440042X + 296176197922605$$

is Eisenstein at 23. It has one root in $(57721/100000, 28861/50000)$, because its endpoint values are

$$\frac{183023345854948985461636 1}{10^{15}} > 0, \qquad -\frac{412612961981365018758619}{125000000000000} < 0,$$

and its derivative is negative on the interval. At every prime $p \leq 20$ its reduction has splitting type $(1, 2)$. Thus this specified cubic algebraic integer has no tested prime inverses and $J_n = n!$ for every $n \leq 20$, exactly as a transcendental parameter. The accompanying program checks the modular factors, endpoint signs, Eisenstein conditions, and all denominators using only integer and rational arithmetic.

Remark 16.6 (Consequence for the research strategy). The joint construction has $\log H(f) \leq \sum_{p \leq N} \log p + 2\log(2N) + O_I(1)$. It rules out a finite-data inference from a small real interval, absence of all tested prime inverses, and maximal tested denominators to transcendence: a cubic algebraic integer reproduces every such finite observation set. It does not exclude an all-primes argument, independent height control, or additional information specific to γ. Computing (16.1) requires a supplied polynomial f; it supplies no arithmetic oracle for γ.

17 Finite Fourier ranges and robust positivity obstructions

Let S be a finite nonempty prime set, let $\mathcal{N}_S$ be the positive S-smooth integers, and let $\mathcal{R}_S$ be the positive integers coprime to every prime in S. Each positive integer is uniquely ra, with $r \in \mathcal{R}_S$ and $a \in \mathcal{N}_S$. For a continuous real 1-periodic function p, consider

$$(M_q) \qquad \sum_{j=0}^{q-1} p(x + j/q) = p(qx), \quad q \in S.$$

We use $\widehat{\mu}(n) = \int e^{-2\pi i n x}\, d\mu(x)$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$.

Proposition 17.1 (Orbit description)**.** *The equations (M_q) hold precisely when $\widehat{p}(0) = 0$ and there are complex numbers β_r, $r \in \mathcal{R}_S$, such that*

$$2\pi i(ra)\widehat{p}(ra) = \beta_r, \qquad 2\pi i(-ra)\widehat{p}(-ra) = \overline{\beta_r} \quad (a \in \mathcal{N}_S).$$

Proof. Fourier comparison gives $q\widehat{p}(qn) = \widehat{p}(n)$ and zero mean. Equivalently the derivative coefficients are constant on the multiplicative orbits. Unique factorization gives the assertion. Conversely these equations annihilate every Fourier coefficient of each continuous multiplication defect, which is therefore zero. □

This is a criterion for an existing continuous function, not a claim that arbitrary orbit data define one. A useful sufficient realization is

$$U_S(x) = \sum_{a \in \mathcal{N}_S} \frac{\sin(2\pi a x)}{a}, \qquad V_S(x) = \sum_{a \in \mathcal{N}_S} \frac{\cos(2\pi a x)}{a},$$

$$p_\beta(x) = \sum_{r \in \mathcal{R}_S} \frac{\Re\beta_r\, U_S(rx) + \Im\beta_r\, V_S(rx)}{\pi r}.$$

If $\sum_r |\beta_r|/r < \infty$, these series converge absolutely and uniformly and give the specified coefficients. Indeed $\sum_{a \in \mathcal{N}_S} a^{-1} = \prod_{q \in S}(1-q^{-1})^{-1}$, and direct expansion verifies the coefficients. Real β_r give odd functions vanishing at integers. Prime equations suffice for all S-smooth multipliers; a lone composite equation need not imply each prime equation separately.

Theorem 17.2 (Finite-range rigidity)**.** *Let σ be a finite complex atomless measure on $\mathbb{T}$. Suppose its Fourier coefficients take only finitely many values and $(T_m)_*\sigma = \sigma$ for one integer $m \geq 2$, where $T_m(x) = mx \pmod 1$. Then $\sigma = \sigma(\mathbb{T})\lambda$, with λ normalized Lebesgue measure.*

Proof. Let V be the finite Fourier range. For each $v \in V$ the polynomial

$$P_v(t) = \prod_{w \in V \setminus \{v\}} \frac{t - w}{v - w}$$

defines a convolution polynomial $P_v(\sigma)$; a constant term means the corresponding multiple of δ_0. Its Fourier transform is the indicator of $E_v = \{n : \widehat{\sigma}(n) = v\}$. Fourier uniqueness makes this an idempotent measure. Cohen's idempotent theorem, as stated in [13, Theorem 1.1], puts E_v in the coset ring of $\mathbb{Z}$.

Every coset-ring set of $\mathbb{Z}$ is periodic outside a finite set: its finite description uses residue classes and singleton cosets. Taking a common period for the finitely many E_v shows that $\widehat{\sigma}(n)$

equals a periodic sequence b_n except at finitely many integers. If its period is Q, the sequence b_n is the Fourier transform of the finite atomic measure

$$\alpha = \sum_{j=0}^{Q-1} w_j \delta_{j/Q}, \qquad w_j = \frac{1}{Q} \sum_{k=0}^{Q-1} b_k e^{2\pi i k j/Q}.$$

The remaining finitely supported Fourier sequence is the transform of a trigonometric-polynomial density $f\lambda$. Thus $\sigma = \alpha + f\lambda$. Atomlessness forces $\alpha = 0$. Finally invariance gives $\widehat{\sigma}(m^j n) = \widehat{\sigma}(n)$; for $n \neq 0$ the left-hand indices leave the finite Fourier support, so all nonzero coefficients vanish. □

This is an application of a classical structural theorem with an additional finite-range hypothesis. It does not settle the unrestricted $\times 2, \times 3$ problem. The range hypothesis can also be replaced by $\operatorname{dist}(\widehat{\sigma}(n), V) \to 0$ for a fixed finite V: recurrence along $m^j n$ then puts each nonzero-frequency coefficient exactly in V (with the zero-frequency coefficient added as one more value if necessary). A probability measure whose coefficients belong to a fixed uniformly discrete subset of $\mathbb{C}$ also has finite Fourier range: all its coefficients lie in the closed unit disk, and a uniformly discrete set has only finitely many points in a bounded disk.

Corollary 17.3 (All nonzero finite-orbit constructions fail BV)**.** *If only finitely many β_r of a continuous solution are nonzero, then bounded variation implies $p = 0$.*

Proof. The measure dp would be atomless, have finite Fourier range, and be invariant under any T_q, $q \in S$. Its total mass is zero. Theorem 17.2 gives $dp = 0$; the zero-mean condition then gives $p = 0$. □

The same argument excludes every nonzero continuous BV solution whose derivative coefficients have finite range, even if infinitely many orbit labels are present.

In particular, no finite selection of extra amplitudes or phases in the previous smooth-frequency construction repairs its lack of bounded variation. Positivity can be rejected without Cohen's theorem.

Proposition 17.4 (An elementary finite certificate)**.** *Let $p \neq 0$ be a real finite-orbit solution. The distribution $\lambda + dp$ is not a positive probability measure. If $p(0) = 0$, equivalently $x + p(x)$ is not nondecreasing on $[0, 1]$.*

Proof. Choose an orbit r_0 with $\beta_{r_0} = a \neq 0$, and choose a prime ℓ dividing neither the primes in S nor any supported orbit label. Fix $q \in S$ and set $n_j = r_0 q^{j(\ell-1)}$. A proposed measure would satisfy $\widehat{\mu}(n_j) = a$. Every difference $n_i - n_j$, $i \neq j$, is divisible by ℓ, so it belongs to no supported signed orbit and has Fourier coefficient zero. The characters $e_j(x) = e^{-2\pi i n_j x}$ would be orthonormal. For every K this would imply

$$0 \leq \int \left| 1 - \overline{a} \sum_{j=1}^{K} e_j(x) \right|^2 d\mu(x) = 1 - K|a|^2.$$

Choose $K > |a|^{-2}$ for a contradiction. □

For example, with $S = \{2, 3\}$ the two-orbit function $p(x) = U_S(x) - U_S(5x)/5$ has derivative amplitudes π and $-\pi$. The prime 7 and frequencies $n_j = 8^j$ make every nonzero difference divisible by 7, outside both supported signed orbits. Thus the negative second amplitude cannot restore positivity.

Theorem 17.5 (A robust positivity test)**.** *Let μ be a probability measure and let $n_1, \dots, n_K$ be distinct integers. If*

$$|\widehat{\mu}(n_j) - a| \le \varepsilon, \qquad |\widehat{\mu}(n_i - n_j)| \le \eta \quad (i \ne j),$$

where $\varepsilon, \eta \ge 0$, then

$$K(|a| - \varepsilon)_+^2 \le 1 + (K-1)\eta. \tag{17.1}$$

Proof. For $U = \sum_j e^{-2\pi i n_j x}$, the hypotheses give $|\int U\, d\mu| \ge K(|a| - \varepsilon)_+$ and $\int |U|^2 d\mu \le K + K(K-1)\eta$. Cauchy–Schwarz on the probability space proves the result. □

When $0 \le \eta < 1$ and $(|a| - \varepsilon)_+^2 > \eta$, rearrangement gives

$$K \le \frac{1 - \eta}{(|a| - \varepsilon)_+^2 - \eta}.$$

A convenient sufficient condition for rejection is $\varepsilon \le |a|/2$, $\eta \le |a|^2/8$, and $K > 8/|a|^2$: then the difference between the two sides of the necessary inequality is at least $K|a|^2/8 - 1 > 0$.

For example, $a = 1/2$, $\varepsilon = 1/20$, $\eta = 1/100$, and $K = 6$ give a left side $243/200$ and a right side $21/20$, with positive gap $33/200$. Such certified Fourier data cannot come from any probability measure. This test applies directly to truncated moment and spectral reconstruction problems. It is an elementary Gram inequality, not a newly claimed general positivity theorem.

Corollary 17.6 (An arithmetic sufficient condition)**.** *An atomless probability measure invariant under one T_m whose Fourier coefficients have uniformly bounded algebraic degree and absolute multiplicative Weil height is Lebesgue measure.*

Proof. Degree at most D and height at most B bound the primitive minimal-polynomial coefficients by $2^D B^D$, using elementary symmetric functions and the product definition of height. Only finitely many integer polynomials and roots remain. Apply Theorem 17.2. □

Algebraicity alone, or membership in one number field, does not give this height bound. Neither arithmetic alternative for γ has been shown to force a nonzero measure in the excluded class.

18 A complete transfer to finite Fourier moment reconstruction

The following is a concrete use of classical Gram and circulant matrix methods. It closes a finite moment problem and strengthens our earlier necessary test; it is not presented as a new general moment theorem. Write $c_n = \int_{\mathbb{T}} e^{-2\pi i n x}\, d\mu(x)$ for a positive probability measure.

Theorem 18.1 (Exact parity-moment feasibility)**.** *Fix $K \ge 2$ and $a, b \in \mathbb{R}$. The data*

$$c_{2j+1} = a \ (0 \le j < K), \qquad c_{2j} = b \ (1 \le j < K)$$

have a representing probability measure if and only if

$$b \le 1, \qquad K|a| \le 1 + (K-1)b. \tag{18.1}$$

One representing measure on $x = j/(2K)$ has weights

$$w_0 = \frac{1 + (K-1)b + Ka}{2K}, \qquad w_K = \frac{1 + (K-1)b - Ka}{2K}, \qquad w_j = \frac{1-b}{2K} \quad (j \notin \{0, K\}).$$

Proof. The moment matrix $T = (c_{r-s})_{0\le r,s<2K}$ is positive semidefinite. Reordering the monomials by parity gives

$$T \sim \begin{pmatrix} (1-b)I_K + bJ_K & aJ_K \\ aJ_K & (1-b)I_K + bJ_K \end{pmatrix}.$$

Its eigenvalues are $1-b$ with multiplicity $2K-2$ and $1+(K-1)b \pm Ka$. This proves necessity. The displayed weights are nonnegative under (18.1) and sum to one. Finite Fourier summation, comparing with equal weights $(1-b)/(2K)$ at all grid points, gives

$$c_n = (b+a)/2 + (-1)^n(b-a)/2 \qquad (0 < |n| < 2K).$$

This proves sufficiency without a general extension theorem. □

The feasible set is the triangle with vertices $(1,1)$, $(-1,1)$, and $(0,-1/(K-1))$. For $b=0$ the exact bound is $|a| \le 1/K$. The earlier star Gram test, using only 1 and the odd characters, gave $|a| \le 1/\sqrt{K}$. Thus using the full matrix extracts a strictly stronger conclusion from the same moment data.

Proposition 18.2 (Boundary uniqueness and support size)**.** *Every boundary point of this triangle has a unique representing measure, given by the weights above. Every interior point has multiple representing measures. The smallest atomic support is* $\operatorname{rank} T$*: it is* $2K$ *in the interior,* $2K-1$ *on either open sloping edge,* $2K-2$ *at the bottom vertex,* 2 *on the open top edge, and* 1 *at either top vertex.*

Proof. Put $z = e^{-2\pi ix}$, $E(z) = \sum_{j=0}^{K-1} z^{2j}$, and $O(z) = zE(z)$. On the sloping edge with $a \ge 0$,

$$\int |E - O|^2 d\mu = 2K\{1 + (K-1)b - Ka\} = 0.$$

The measure is supported on the zeros of $(1-z)E(z)$: the $2K$-th roots except -1. Finite Fourier inversion fixes the weights. The other edge uses $E+O$ and excludes $+1$. At the bottom vertex both exclusions hold. On the top edge, $c_2 = 1$ implies support on $\{1,-1\}$, and c_1 fixes the weights.

In the interior choose $0 < \theta < \min\{1, 1-b, 1+(K-1)b - K|a|\}$. The parameters $(a/(1-\theta), b/(1-\theta))$ are feasible; let ν represent them. Both $\theta\lambda + (1-\theta)\nu$ and $\theta\rho_M + (1-\theta)\nu$ represent the original data, where λ is normalized Haar measure on the circle and ρ_M is uniform on M-th roots and $M > 2K-1$. They are distinct. Finally an r-atomic representation makes T a sum of r rank-one matrices, so $r \ge \operatorname{rank} T$. The explicit measure attains the bound in every listed case. □

Theorem 18.3 (Exact feasibility with errors)**.** *Fix* $K \ge 2$*. Let* $a, b \in \mathbb{R}$ *and* $\varepsilon, \eta \ge 0$*. The constraints*

$$|c_{2j+1} - a| \le \varepsilon \ (0 \le j < K), \qquad |c_{2j} - b| \le \eta \ (1 \le j < K)$$

admit a probability measure if and only if

$$b - \eta \le 1, \qquad K(|a| - \varepsilon)_+ \le 1 + (K-1)\min\{1, b+\eta\}. \tag{18.2}$$

The unknown moments may be complex.

Proof. Put $A = (|a| - \varepsilon)_+$ and $d = \min\{1, b+\eta\}$. Necessarily $b - \eta \le 1$. With E, O as above,

$$\int |E|^2 d\mu = K + 2\sum_{j=1}^{K-1} (K-j)\Re c_{2j} \le K + K(K-1)d.$$

When $A > 0$, the K^2 terms of $\int E\overline{O}\,d\mu$ all have real parts of the sign of a and magnitude at least A. Therefore its modulus is at least K^2A. Cauchy–Schwarz and $|E| = |O|$ bound this modulus by $\int |E|^2 d\mu$. If $A = 0$, positivity of the latter integral suffices instead. This proves the second necessary condition.

Conversely choose $a' = \operatorname{sgn}(a)A$, with $\operatorname{sgn}(0) = 0$, and $b' = d$. Then $|a'-a| \le \varepsilon$ and $|b'-b| \le \eta$; the latter uses $b - \eta \le 1$. Theorem 18.1 applies and explicitly constructs a measure satisfying all the error bounds. □

For $a = 1/2$, $b = 0$, $\varepsilon = 1/20$, and $\eta = 1/100$, $K = 3$ already fails (18.2) by $33/100$. The previous star test does not reject until $K = 6$ for this pattern. At $K = 2$ feasibility holds, with weights $(191, 99, 11, 99)/400$ on the fourth roots of unity. This is an exact change in the number of moments needed to reject the data, not an asymptotic or numerical optimization claim. No consequence of either arithmetic alternative for γ has been shown to force these forbidden data.

19 Several poles: cancellation and logarithmic rank

Let Λ be a finite set of distinct positive algebraic numbers and put

$$I_\lambda = \int_0^\infty \frac{e^{-t}}{t+\lambda}\,dt = e^\lambda E_1(\lambda), \qquad I_1 = \delta.$$

For $R \in \overline{\mathbb{Q}}(z)$ with finite poles only at $-\lambda$, $\lambda \in \Lambda$, write its principal part there as $\sum_{j=1}^{m_\lambda} r_{\lambda,j}(z+\lambda)^{-j}$ and set

$$c_\lambda = \sum_{j=1}^{m_\lambda} \frac{(-1)^{j-1} r_{\lambda,j}}{(j-1)!}.$$

The rational-Borel reduction proved in [19] gives a unique rational function T, with no other finite poles, such that

$$R = T - T' + \sum_{\lambda\in\Lambda} \frac{c_\lambda}{z+\lambda}, \qquad \mathcal{L}(R) := \int_0^\infty e^{-t}R(t)\,dt = T(0) + \sum_{\lambda\in\Lambda} c_\lambda I_\lambda. \tag{19.1}$$

All residues c_λ are effectively computable algebraic numbers. The integral identity follows by integration by parts; the exponential kills the rational boundary term at infinity.

19.1 The canonical gamma coefficient

The connection formula gives

$$\mathcal{L}(R) = E_R - A_R\gamma - B_R, \tag{19.2}$$

where

$$E_R = T(0) + \sum_\lambda c_\lambda e^\lambda \operatorname{Ein}(\lambda), \qquad A_R = \sum_\lambda c_\lambda e^\lambda, \qquad B_R = \sum_\lambda c_\lambda e^\lambda \log\lambda.$$

Here E_R belongs to the ring $\mathbf{E}$ of E-values.

Theorem 19.1 (Exact canonical cancellation criterion)**.** *The coefficient A_R vanishes if and only if every c_λ vanishes, equivalently if and only if $R = T - T'$ for a rational T. In this case $\mathcal{L}(R) = T(0) \in \overline{\mathbb{Q}}$. Equivalently, if $R(z) = \sum_{n\ge0} r_n z^n$ at zero and $\mathfrak{f}_R(x) = \sum_{n\ge0} n! r_n x^n$, then there exists $a \in \overline{\mathbb{Q}}$ for which $(\mathfrak{f}_R(x) - a)/(x-1)$ is arithmetic Gevrey of order one. The only possible a is $T(0)$.*

Proof. The numbers e^{λ} for distinct algebraic λ are linearly independent over $\overline{\mathbb{Q}}$ by Lindemann–Weierstrass. Thus $\sum c_{\lambda}e^{\lambda} = 0$ forces every algebraic coefficient to be zero. Use the unique reduction (19.1) for rational exactness and the value formula. The arithmetic-division equivalence, including uniqueness of $a = T(0)$, is the $\alpha = 1$ case of the full rational-Borel classification in Theorem 10.1. □

This is a criterion for the coefficient in the particular identity (19.2), not uniqueness of every possible representation of its value. It does not prove that every nonexact kernel has a transcendental integral: that statement includes the unknown arithmetic nature of δ. The analytic-to-arithmetic implication remains the zero-removal problem described in [5].

Proposition 19.2 (Collision locations under A)**.** *Assume $\gamma \in \overline{\mathbb{Q}}$. For every positive algebraic λ,*

$$I_{\lambda} \in \mathbf{E} \quad \Longleftrightarrow \quad \log\lambda \in \mathbf{E}.$$

Consequently the set of such locations is a multiplicative subgroup of $\overline{\mathbb{Q}}_{>0}$ closed under positive rational powers. For fixed algebraic $b > 0$, $b \neq 1$, either all I_{b^j}, $j \in \mathbb{Z}\setminus\{0\}$, belong to $\mathbf{E}$, or none do, according as $\log b$ does or does not belong to $\mathbf{E}$.

Proof. Use $e^{-\lambda}I_{\lambda} = \mathrm{Ein}(\lambda) - \gamma - \log\lambda$ and the facts that $e^{\pm\lambda}$ and algebraic numbers belong to $\mathbf{E}$. The subgroup assertion follows from the real logarithm laws and closure of $\mathbf{E}$ under rational linear combinations. Finally $\log(b^j) = j\log b$, with $j \neq 0$. □

When $1 \in \Lambda$, there is also the unconditional ring identity

$$\mathbf{E}[I_{\lambda} : \lambda \in \Lambda] = \mathbf{E}[\gamma, \log\lambda : \lambda \in \Lambda].$$

Indeed $\gamma = \mathrm{Ein}(1) - e^{-1}I_1$, and then $\log\lambda = \mathrm{Ein}(\lambda) - \gamma - e^{-\lambda}I_{\lambda}$; the reverse inclusion follows from the connection formulas. This restates the first-level collision-ring mechanism in the multipole notation. Thus A does not force every new pole to supply an E/D collision. Nor may Baker's logarithm estimates be applied to B_R as if its coefficients $c_{\lambda}e^{\lambda}$ were algebraic. For geometric locations $\lambda = b^j$, one can at least write

$$B_R = (\log b)\sum_j jc_j e^{b^j};$$

Lindemann–Weierstrass shows that this vanishes exactly when $c_j = 0$ for all $j \neq 0$. This does not settle membership of B_R in $\mathbf{E}$.

19.2 Independent residuals after removing common logarithms

Define the field

$$K_{\exp} = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}}).$$

This field is not being identified with the ring $\mathbf{E}$. We use the previously proved *multipoint* Ein *input*: for distinct nonzero algebraic λ_i, the values $\mathrm{Ein}(\lambda_i)$ are algebraically independent over $K_{\exp}$; see the foundations reproduced with [19]. This is an explicit proof dependency. Its route is to place finitely many exponentials in a Laurent differential field, exclude rational primitives of the relevant distinct characters divided by z, apply primitive differential-field independence, and then apply Siegel–Shidlovskii/Beukers specialization at the nonsingular point 1 [4]. Enlarging the finite exponential field covers every hypothetical relation over $K_{\exp}$. We do not reproclaim that foundation as a new result here.

Theorem 19.3 (Logarithmic rank bound)**.** *Let $\lambda_1, \dots, \lambda_m$ be distinct positive algebraic numbers. Choose a basis $\rho_1, \dots, \rho_r$ for the multiplicative group they generate and write*

$$\lambda_i = \prod_{h=1}^{r} \rho_h^{n_{ih}}, \qquad N = (n_{ih}), \qquad s = \operatorname{rank}_{\mathbb{Q}}[\mathbf{1} \mid N].$$

Then $\operatorname{rank}_{\mathbb{Q}} N = r$, $s \le r+1$, and

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(I_{\lambda_1}, \dots, I_{\lambda_m}) \ge m - s \ge m - r - 1. \tag{19.3}$$

Under A *the lower bound improves to $m - r$.*

Proof. The finitely generated subgroup of $\overline{\mathbb{Q}}_{>0}$ is torsion-free, so it has the indicated basis. Put $y_i = e^{-\lambda_i} I_{\lambda_i}$, $U_i = \operatorname{Ein}(\lambda_i)$ and $\ell_h = \log \rho_h$. The vector connection identity is

$$y = U - \gamma \mathbf{1} - N\ell.$$

Choose a rational matrix B of row rank $m - s$ whose rows annihilate $[\mathbf{1} \mid N]$. Then $By = BU$. A full-row-rank linear image of an algebraically independent tuple is algebraically independent: extend its rows to an invertible matrix and apply the corresponding linear change of indeterminates. Thus the field generated by y contains $m - s$ algebraically independent elements over $K_{\exp}$. Individual multiplication by $e^{\lambda_i} \in K_{\exp}^{\times}$ does not change the field generated. This proves (19.3).

Under A, instead take B to have row rank $m-r$ and annihilate only N. Now $By = BU - \gamma B\mathbf{1}$; the translation belongs to $\overline{\mathbb{Q}}^{m-r}$ and preserves algebraic independence. □

Corollary 19.4 (Geometric dilations)**.** *Fix algebraic $b > 0$, $b \ne 1$. Among the sequence $(I_{b^j})_{j \ge 0}$, removal of at most two terms leaves an algebraically independent family over $K_{\exp}$. Under* A*, removal of at most one term suffices. Moreover the explicit values*

$$\begin{aligned} D_j &= E_1(b^j) - 2E_1(b^{j+1}) + E_1(b^{j+2}) \\ &= \operatorname{Ein}(b^j) - 2\operatorname{Ein}(b^{j+1}) + \operatorname{Ein}(b^{j+2}), \qquad j \ge 0, \end{aligned}$$

are all strictly positive and algebraically independent over $K_{\exp}$.

Proof. For every prefix of length $m \ge 2$, the logarithmic exponent column is $(0, 1, \dots, m-1)^t$, so $r = 1$ and $s = 2$. The rank bounds are $m - 2$ and, under A, $m - 1$. Process the sequence in increasing order, retaining a term exactly when it is transcendental over the field generated by retained terms. Every skipped term is algebraic over that field, so the number retained in each prefix is its transcendence degree. The bounds permit at most two skips in total, or at most one under A. The retained family is algebraically independent.

The common γ term and the affine term $j \log b$ cancel in each D_j. For every finite prefix the second-difference matrix has full row rank, so the multipoint input proves independence of the D_j. Finally $h(t) = E_1(e^t)$ satisfies $h''(t) = e^t e^{-e^t} > 0$. Strict convexity with the nonzero step $\log b$ proves $D_j > 0$. □

The exceptional indices are not identified; the corollary therefore does not decide the arithmetic nature of $I_1 = \delta$ or any other designated member. The D_j are E-values. Expressing them in terms of I_{b^j} uses coefficients e^{-b^j}, which are not algebraic, so this does not put them in $\mathbf{D}$ or contradict Theorem 19.1.

Positivity by itself also cannot provide the missing separation. For positive algebraic weights a_λ, the strictly positive kernel

$$R(z) = \sum_\lambda a_\lambda \left(\frac{1}{z+\lambda} + \frac{1}{(z+\lambda)^2} \right)$$

has algebraic integral $\sum_\lambda a_\lambda/\lambda$, since $R = T - T'$ with $T = \sum_\lambda a_\lambda/(z+\lambda)$. The closed statements in this section concern canonical coefficients and family ranks; no implication here excludes either gamma world.

20 A prime decomposition and density-one independence

Write

$$K_{\exp} = \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}}), \qquad U_n = \mathrm{Ein}(n), \quad y_n = E_1(n), \quad I_n = e^n y_n.$$

Our explicit input is the order-one multipoint theorem proved in the foundational part of this volume: the U_n are algebraically independent over $K_{\exp}$ in the finite-subset sense. Its differential-field proof and Siegel–Shidlovskii/Beukers specialization step are required dependencies of the following arithmetic deductions. No analogous specialization for E_1 or for γ is assumed. The connection identity is

$$y_n = U_n - \gamma - \log n. \tag{20.1}$$

The deductions below make no claim of priority.

20.1 Explicit positive residuals for every composite integer

For composite n, set $\Omega(n) = \sum_p v_p(n)$ and

$$R_n = y_n - \sum_{p|n} v_p(n) y_p + (\Omega(n) - 1) y_1. \tag{20.2}$$

Theorem 20.1. *Every R_n is strictly positive. The family of all R_n, indexed by composite positive integers, is algebraically independent over $K_{\exp}$. Each R_n is an E-value.*

Proof. The identity $\log n = \sum_p v_p(n) \log p$ and (20.1) give

$$R_n = U_n - \sum_{p|n} v_p(n) U_p + (\Omega(n) - 1) U_1.$$

In any finite collection, the composite coordinate U_n has coefficient one in its own row and zero in every other row. All remaining columns are prime coordinates and U_1. Thus the rows have full rank, and a full-row-rank linear image of an algebraically independent tuple is algebraically independent. The expression is also a rational linear combination of E-values.

For positivity put $h(s) = E_1(e^s)$. Direct differentiation gives

$$h'(s) = -e^{-e^s}, \qquad h''(s) = e^s e^{-e^s} > 0.$$

Consequently, for $a, b > 0$, $h(a+b) - h(a) > h(b) - h(0)$. Iteration gives

$$h(a_1 + \cdots + a_k) - \sum_{j=1}^{k} h(a_j) + (k-1) h(0) > 0 \quad (k \geq 2, a_j > 0).$$

Take a_j to be the logarithms of the prime factors of n, counted with multiplicity. The left side is R_n. □

For example,

$$R_6 = E_1(6) - E_1(2) - E_1(3) + E_1(1), \qquad R_{12} = E_1(12) - 2E_1(2) - E_1(3) + 2E_1(1).$$

Here $\pi(N)$ denotes the number of primes at most N. This construction combines different geometric bases with a single triangular independence proof. Indeed the rows

$$c_n = (1, (v_p(n))_{p \le N}), \qquad 1 \le n \le N,$$

have rank $\pi(N) + 1$, already witnessed by $n = 1$ and the primes. The residual rows for the composite $n \le N$ form a basis of the left kernel. Hence they describe the entire rational linear cancellation space for the common constant and the prime logarithms on this range.

Corollary 20.2. *For $N \ge 1$,*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(I_1, \dots, I_N) \ge N - \pi(N) - 1.$$

There is a subset $S \subseteq \mathbb{N}$ of natural density one such that $\{I_n : n \in S\}$ is algebraically independent over $K_{\exp}$. At most $\pi(N) + 1$ of $I_1, \dots, I_N$ can be algebraic over $K_{\exp}$.

Proof. The field generated by $y_1, \dots, y_N$ contains the $N - \pi(N) - 1$ independent composite residuals. Multiplication by $e^n \in K_{\exp}^{\times}$ does not change this field. Process the I_n in increasing order, retaining I_n exactly when it is transcendental over the field generated by the retained predecessors. The retained terms are algebraically independent and their number in each prefix is that prefix's transcendence degree. Thus at most $\pi(N) + 1 = o(N)$ terms are omitted. Every term algebraic over $K_{\exp}$ is omitted. □

The set S is not identified. Density one does not decide the arithmetic nature of the designated term $I_1 = \delta$.

20.2 A sharper consequence in the algebraic-gamma world

Proposition 20.3. *If γ is algebraic over $K_{\exp}$, then*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(I_1, \dots, I_N) \ge N - \pi(N),$$

and δ is transcendental over $K_{\exp}$. In particular these conclusions hold under A*, the assumption $\gamma \in \overline{\mathbb{Q}}$.*

Proof. Over the algebraic extension $K_{\exp}(\gamma)$, annihilate only the $\pi(N)$ logarithm columns in (20.1). This leaves $N - \pi(N)$ independent combinations, since algebraic translation preserves algebraic independence. Algebraic base extension preserves transcendence degree. Finally $U_1 = \gamma + \delta/e$ is transcendental over $K_{\exp}$; both summands cannot be algebraic over that field. □

This is a consequence of A, not a contradiction to it.

20.3 Why combining bases does not remove the exceptional term

Let $\rho_h > 0$ be algebraic, let $n_{ih} \in \mathbb{Z}$, and assume the locations $\lambda_i = \prod_h \rho_h^{n_{ih}}$ are distinct. The connection identity reads

$$y_i = U_i - \gamma - \sum_h n_{ih} \log \rho_h.$$

If selected y_i are all algebraic over $K_{\exp}$, their rows $(1, n_{i1}, \dots, n_{ir})$ must be rationally independent. Otherwise a nonzero rational relation would express a nonzero linear combination of the independent U_i as an algebraic number over $K_{\exp}$. Under the hypothesis of Proposition 20.3, the exponent rows $(n_{ih})_h$ alone must be independent.

These constraints and the preceding bounds are sharp for the stated algebraic inputs, as the following two formal models demonstrate. They are not replacements for the actual analytic Euler integrals. Let K be any characteristic-zero field, take independent indeterminates $(U_n)_{n\ge1}$, and prescribe constants $a_p \in K$ for all primes.

In the first model set $g = a \in K$, and define

$$\ell_p = U_p - g - a_p, \quad \ell_n = \sum_p v_p(n)\ell_p, \quad y_n = U_n - g - \ell_n.$$

Then every $y_p = a_p$, whereas y_1 and all composite y_n are algebraically independent: each has a unique corresponding variable U_n. Every prefix has transcendence degree $N - \pi(N)$.

In the second model choose also $a_1 \in K$, set $g = U_1 - a_1$, and use the same definitions. Then g is transcendental, $y_1 = a_1$, and $y_p = a_p$ for every prime. The composite y_n are algebraically independent, and every prefix has transcendence degree $N - \pi(N) - 1$. Both models obey all multiplicative logarithm-addition relations and composite cancellation identities simultaneously. The symbols ℓ_p are formal logarithmic coordinates, not actual real logarithms; no analytic exponential equation has been imposed.

The countermodel below gives a stronger analytic warning using actual E-values: algebraically independent positive approximants can have an algebraic limit with the same leading dyadic error rate. Hence the missing input must bind the exact Euler normalization to arithmetic, not only to independence and convergence. The formal models above assert no logical independence from a foundational axiom system.

Finally, the expression of R_n through I_n uses the coefficients e^{-n}. It does not show membership in the D-value ring, whose closure under arbitrary E-value multiplication is unavailable. Consequently the positive independent residuals do not by themselves contradict an E/D intersection conjecture.

21 A Hankel bootstrap: gamma as an isolated moment atom

This construction imports the classical Christoffel variational principle into the several-pole Laplace family. It produces a computable infinite sequence with a positive, geometrically contracting error. Its inputs are E-values and logarithms, rather than rational numbers; no arithmetic conclusion for gamma is asserted. Christoffel minimization also appeared in the earlier integrated manuscript for a different, rational Gregory moment family. The present result concerns the tensor determinants of the exponential-integral periods.

Fix positive algebraic numbers a, h. For $u > 0$ put

$$I_u = \int_0^\infty \frac{e^{-t}}{t+u}\,dt = e^u E_1(u), \qquad F(u) = \mathrm{Ein}(u) - \log u = \gamma + E_1(u).$$

For $n \ge 1$, with row and column indices $0, \dots, n-1$, define

$$H_n = (E_1(a + (i+j)h))_{i,j}, \quad F_n = (F(a + (i+j)h))_{i,j}, \quad v_n = (1, \dots, 1)^{\mathsf{T}}.$$

Thus $F_n = H_n + \gamma v_n v_n^{\mathsf{T}}$.

Theorem 21.1 (Positive affine collapse of the tensor tower)**.** *The matrices H_n are positive definite. Set*

$$c_n = v_n^{\mathsf{T}} \operatorname{adj}(H_n) v_n.$$

Then $c_n > 0$, and

$$\det(F_n - s v_n v_n^{\mathsf{T}}) = \det H_n + (\gamma - s) c_n, \tag{21.1}$$

$$c_n = v_n^{\mathsf{T}} \operatorname{adj}(F_n) v_n. \tag{21.2}$$

Consequently

$$\det(I_{a+(i+j)h})_{i,j=0}^{n-1} = e^{na+hn(n-1)} \det H_n > 0. \tag{21.3}$$

Under the canonical substitution $I_u = e^u(\mathrm{Ein}(u) - \log u - \gamma)$, this determinant has degree exactly one in the displayed gamma parameter, for every n.

Proof. Put $q = e^{-h} \in (0,1)$ and introduce the finite positive measure

$$d\mu_{a,h}(x) = \frac{x^{a/h-1}}{-\log x}\, dx, \qquad 0 < x < q.$$

The substitutions $E_1(u) = \int_1^\infty e^{-ut}\, dt/t$ and $x = e^{-ht}$ give

$$E_1(a + kh) = \int_0^q x^k \, d\mu_{a,h}(x).$$

The measure is positive on an interval, so the squared norm of any nonzero polynomial is positive. This proves positive definiteness. Now $\operatorname{adj}(H_n) = \det(H_n) H_n^{-1}$, whence $c_n > 0$. The rank-one determinant identity gives (21.1); applying it once with base matrix F_n and comparing the coefficient of s proves (21.2). Finally factor e^{a+ih} from row i and e^{jh} from column j to obtain (21.3). □

Theorem 21.2 (Gamma-free formula and certified error)**.** *The quantities*

$$A_n = \frac{\det F_n}{v_n^{\mathsf{T}} \operatorname{adj}(F_n) v_n}$$

are defined by the entire series for Ein *and ordinary real logarithms, without taking gamma as input. They satisfy*

$$A_n - \gamma = \varepsilon_n = \frac{1}{v_n^{\mathsf{T}} H_n^{-1} v_n} = \min_{\substack{\deg p < n \\ p(1)=1}} \int_0^q |p(x)|^2 \, d\mu_{a,h}(x) > 0, \tag{21.4}$$

$$0 < \varepsilon_{n+1} < q^2 \varepsilon_n, \tag{21.5}$$

$$\varepsilon_n \le \frac{E_1(a)}{T_{n-1}(2e^h - 1)^2}. \tag{21.6}$$

Here T_k is the Chebyshev polynomial. In particular A_n decreases strictly to gamma. The fully explicit weaker upper bound

$$\varepsilon_n \le \frac{e^{-a}/a}{T_{n-1}(2e^h - 1)^2}$$

provides certified intervals for gamma using no E_1 input.

Proof. Equation (21.1) at $s = 0$, together with (21.2), gives $A_n = \gamma + \det(H_n)/c_n$. For a coefficient vector b of p, the objective in (21.4) is $b^* H_n b$ and the constraint is $v_n^{\mathsf{T}} b = 1$. The unique minimizer is $b = H_n^{-1} v_n / (v_n^{\mathsf{T}} H_n^{-1} v_n)$; this proves the variational identity. If p_n is the minimizing polynomial, xp_n is feasible at level $n+1$. Since $0 < x < q$ on the support,

$$\varepsilon_{n+1} \le \int x^2 |p_n(x)|^2 d\mu_{a,h}(x) < q^2 \varepsilon_n.$$

For the sharper bound, use the feasible polynomial

$$p(x) = \frac{T_{n-1}(2x/q - 1)}{T_{n-1}(2/q - 1)}.$$

Its numerator has modulus at most one on $[0, q]$; the measure's total mass is $E_1(a)$. Finally $E_1(a) < e^{-a}/a$ follows immediately from its integral over $[a, \infty)$. □

The rate in (21.6) is exponential: if $\xi = 2e^h - 1$ and $\rho = \xi + \sqrt{\xi^2 - 1} > 1$, then $T_k(\xi) = (\rho^k + \rho^{-k})/2$. The moments $F(a + kh)$ are precisely those of

$$\nu_{a,h} = \mu_{a,h} + \gamma \delta_1.$$

Thus the construction recovers the mass of an isolated atom at 1. This is an application of the classical Christoffel principle, not a claim of a new principle in moment theory. A primary reference is B. Simon, *The Christoffel–Darboux Kernel* [21].

21.1 The smallest case and the precise arithmetic gap

At $n = 2$, put $u_j = F(a + jh)$ for $j = 0, 1, 2$. Then

$$c_2 = u_0 - 2u_1 + u_2 = E_1(a) - 2E_1(a + h) + E_1(a + 2h) > 0,$$

and

$$A_2 = \frac{u_0 u_2 - u_1^2}{u_0 - 2u_1 + u_2}, \qquad A_2 - \gamma = \frac{E_1(a)E_1(a + 2h) - E_1(a + h)^2}{c_2} > 0.$$

The positivity of the numerator is the strict Cauchy–Schwarz inequality for the two monomials $1, x$ in $L^2(\mu_{a,h})$. For $a = h = 1$, this tests the algebraic-coefficient tensor $I_1 I_3 - I_2^2$. Its quadratic gamma coefficient cancels because $1 + 3 = 2 + 2$, but its linear coefficient is $-e^4 c_2 \neq 0$. Thus the assertion that the highest gamma coefficient can never cancel in tensor products is false; the full tower nevertheless retains gamma.

For $a = h = 1$ and $n = 8$, the numerical diagnostic recorded in the Korean companion gives $\varepsilon_8 \simeq 1.94354 \times 10^{-14}$ and the fully explicit upper bound 9.40656×10^{-14}. Independent 160-digit arithmetic reproduces these figures. These decimal values are diagnostics; the rigorous upper bound is the formula in Theorem 21.2.

The input count is $2n - 1$ values of Ein and their associated logarithms. This formula does not produce rational approximants or algebraic coefficients with a height bound. The accompanying independence argument in fact proves transcendence for a density-one subfamily when $a = h = 1$. Therefore the positive error and its exponential decrease are not small integer polynomial values at gamma. An arithmetic compression theorem producing such polynomials with an adequate height budget would be an additional theorem, not a consequence of (21.4). Under an algebraicity assumption on gamma the measure $\nu_{a,h}$ remains perfectly legitimate, so its atom does not itself yield a contradiction.

21.2 What algebraic-coefficient formal tensor cancellation allows

Let $\lambda_1, \dots, \lambda_m$ be distinct algebraic numbers and $v_i = e^{\lambda_i}$. For an algebraic-coefficient polynomial $P \in \overline{\mathbb{Q}}[X_1, \dots, X_m]$,

$$P(X + sv) = P(X) \text{ in } \mathbb{C}[X, s] \quad \Longleftrightarrow \quad P \text{ is constant.} \tag{21.7}$$

Indeed differentiation gives $\sum_i e^{\lambda_i} \partial_i P = 0$. The coefficient of each monomial is an algebraic linear combination of the distinct exponentials. Lindemann–Weierstrass forces every one of its summands to vanish; hence every $\partial_i P$ is zero. The same argument works for rational functions: for coprime $A, B \in \overline{\mathbb{Q}}[X]$ with $B \neq 0$, differentiate A/B and multiply by B^2. Lindemann–Weierstrass then gives $(\partial_i A)B - A\partial_i B = 0$ for every i, so A/B is constant.

The coefficient-field hypothesis matters. Over $K_{\exp}$, the nonconstant polynomial $e^{-\lambda_1} X_1 - e^{-\lambda_2} X_2$ is invariant. Also (21.7) is a formal identity in independent X_i. It is not a theorem that specialization at the actual period values creates no additional identities. Such a claim would require arithmetic information not supplied by this calculation.

21.3 Synthesis: the Hankel approximants themselves have density-one independence

The multipoint Ein input and the Christoffel construction combine to give arithmetic information about the approximants themselves. Specialize $a = h = 1$, put

$$m_k = \mathrm{Ein}(k+1) - \log(k+1), \qquad F_n = (m_{i+j})_{0 \le i,j < n}, \qquad A_n = \frac{\det F_n}{v_n^{\mathsf{T}} \operatorname{adj}(F_n) v_n},$$

and retain the field $K_{\exp}$ from Section 20.

Theorem 21.3. *For every $N \ge 1$,*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(A_1, \dots, A_N) \ge N - \pi(2N-1). \tag{21.8}$$

There is a subset $S \subseteq \mathbb{N}$ of natural density one for which $\{A_n : n \in S\}$ is algebraically independent over $K_{\exp}$. At most $\pi(2N-1)$ of the first N approximants are algebraic over $K_{\exp}$.

Proof. First we show that the A_n recover every even-index moment once the odd-index moments are supplied. Fix n, put $d = n-1$, and regard the largest moment $t = m_{2d}$ as variable, keeping the preceding moments fixed. It occurs only in the bottom-right matrix entry. Both the determinant and the sum of cofactors are affine in t, so

$$A_n = \frac{\alpha t + \beta}{\kappa t + \lambda}, \qquad \alpha, \beta, \kappa, \lambda \in K_{\exp}(m_0, \dots, m_{2d-1}). \tag{21.9}$$

For $n = 1$ this reads $A_1 = m_0$ over the empty preceding moment field.

At the actual moments F_n is positive definite, since $\gamma > 0$. Set

$$b = \frac{F_n^{-1} v_n}{v_n^{\mathsf{T}} F_n^{-1} v_n}, \qquad p(x) = \sum_{j=0}^{d} b_j x^j.$$

This polynomial minimizes the squared norm for $\nu = \mu_{1,1} + \gamma\delta_1$ subject to $p(1) = 1$. The atom adds the constant γ to the objective, so p also minimizes the norm for $\mu_{1,1}$. It must have degree

exactly d. Otherwise, for $d \geq 1$, $(x-1)p$ would be an admissible variation of degree at most d vanishing at 1. Stationarity would give

$$\int_0^{e^{-1}} (x-1)p(x)^2 \, d\mu_{1,1}(x) = 0,$$

which is impossible: the integrand is strictly negative away from finitely many zeros of the nonzero real polynomial p. For $d = 0$, $p = 1$ directly. Thus $b_d \neq 0$.

Since $A_n = (v_n^{\mathsf{T}} F_n^{-1} v_n)^{-1}$, differentiation with respect to the bottom-right entry gives

$$\frac{\partial A_n}{\partial t} = b_d^2 > 0.$$

Together with (21.9), this proves $\alpha\lambda - \beta\kappa \neq 0$ at the preceding moment values. The exact inverse

$$t = \frac{\beta - A_n\lambda}{A_n\kappa - \alpha}$$

has nonzero denominator there, since $A_n\kappa - \alpha = (\kappa\beta - \alpha\lambda)/(\kappa t + \lambda)$. Induction on n therefore proves the field identity

$$K_{\exp}(m_0, \ldots, m_{2N-2}) = K_{\exp}(A_1, \ldots, A_N, m_1, m_3, \ldots, m_{2N-3}). \tag{21.10}$$

On adjoining the $\pi(2N-1)$ numbers $\log p$ for primes $p \leq 2N-1$, the moment field contains all $\mathrm{Ein}(1), \ldots, \mathrm{Ein}(2N-1)$, because $\log j = \sum_p v_p(j) \log p$. Their independence implies

$$\mathrm{trdeg}_{K_{\exp}} K_{\exp}(m_0, \ldots, m_{2N-2}) \geq 2N - 1 - \pi(2N-1).$$

In (21.10) there are only $N-1$ extra odd-index moments. Subtracting that possible transcendence-degree contribution proves (21.8). Greedy retention of a transcendence basis, as in Corollary 20.2, omits at most $\pi(2N-1) = o(N)$ terms and proves the remaining assertions. □

This is a synthesis of the two constructions: the new Hankel approximants cannot all be rational, or even algebraic over $K_{\exp}$. They contain a density-one independent subfamily while decreasing to γ. This does not force their limit to be transcendental. It rules out treating the raw A_n as a sequence of rational or algebraic approximants with an ordinary Diophantine height. Any subsequent rationalization would still require its own error and height analysis.

22 An explicit countermodel to independence-based bootstrapping

The following construction uses the actual E-values already available in the program. It identifies a collection of hypotheses that does not determine the arithmetic nature of a limit. Write $U_N = \mathrm{Ein}(2^N)$, $u_N = E_1(2^N)$, and

$$D_N = U_N - 2U_{N+1} + U_{N+2} = u_N - 2u_{N+1} + u_{N+2}.$$

Here D_N denotes a difference, not a binomial denominator.

Proposition 22.1 (Algebraic limits with the same leading error)**.** *For any real algebraic number a, the sequence*

$$Z_N = a + (N+1)D_N, \qquad N \geq 1,$$

consists of E-values algebraically independent over $K_{\exp}$. It decreases strictly to a from above, and

$$Z_N - a \sim \frac{N+1}{2^N} e^{-2^N}.$$

Its arguments have logarithmic height $O(N)$ and its linear coefficients have logarithmic height $O(\log(N+1))$.

Proof. Every finite second-difference matrix has full row rank, so the multipoint input proves independence of its D_N values. Algebraic translations and nonzero rational rescalings preserve independence; closure of E-values under algebraic linear combinations gives the first assertion.

Put $u(t) = E_1(t)$ and $d(t) = u(t) - 2u(2t) + u(4t)$. Substitution gives $u(2t) \le e^{-t}u(t)$. For $t \ge 2$ this implies

$$(1 - 2e^{-t})u(t) \le d(t) < u(t), \qquad \frac{d(2t)}{d(t)} \le \frac{e^{-t}}{1 - 2e^{-t}}.$$

Consequently

$$\frac{Z_{N+1} - a}{Z_N - a} \le \frac{3}{2} \frac{e^{-2}}{1 - 2e^{-2}} < 1 \qquad (N \ge 1).$$

All errors are positive. The standard integration-by-parts estimate $E_1(t) \sim e^{-t}/t$ and the same ratio bound give the asymptotic. The displayed arguments and coefficients prove the height assertions. □

This is the same leading error as

$$Y_N = (N+1)\operatorname{Ein}(2^N) - N\operatorname{Ein}(2^{N+1}) = \gamma + (N+1)u_N - Nu_{N+1}.$$

Even the polynomial slope detector survives: for every nonzero $P \in \mathbb{C}[X]$,

$$\lim_{N\to\infty} \frac{-\log|P(Z_N)|}{2^N} = \operatorname{ord}_{X=a} P.$$

Indeed, factor $P(X) = (X-a)^m Q(X)$ with $Q(a) \ne 0$. Thus independence, actual E-value membership, positivity, monotonicity, small coefficient heights, and the known leading decay do not together force a transcendental limit. The countermodel changes the exact balanced formula: its Ein coefficient sum is zero, whereas the sum for Y_N is one. It does not rule out a future argument using this normalization and additional arithmetic information.

Lemma 22.2 (Algebraic tail intersection)**.** *Let $x_1, x_2, \dots$ be algebraically independent over a field $K \subset \mathbb{C}$, and put $F_M = K(x_n : n \ge M)$. Then*

$$\bigcap_M F_M = K, \qquad \bigcap_M \{z \in \mathbb{C} : z \text{ is algebraic over } F_M\} = \{z \in \mathbb{C} : z \text{ is algebraic over } K\}.$$

Proof. Every expression involves finitely many variables. Disjoint independent finite tuples A, B have $K(A) \cap K(B) = K$, by clearing denominators and comparing the resulting polynomial identity. This proves the first claim. If z is algebraic over both $K(A)$ and $K(B)$ but transcendental over K, transcendence-degree submodularity gives

$$|A| + |B| \ge \operatorname{trdeg}_K K(A, B, z) + \operatorname{trdeg}_K K(z) \ge |A| + |B| + 1,$$

a contradiction. Taking one finite early tuple and one sufficiently late tuple proves the second claim; the converse is immediate. □

For the independent Y_N over $\overline{\mathbb{Q}}$, a proof that γ is algebraic over every tail field would prove algebraicity, not transcendence. Ordinary convergence establishes no such algebraic membership. These are elementary deductions used as stopping rules; no novelty is asserted for the field-intersection lemma, and no gamma alternative is rejected here.

23 Importing zero theorems: a fresh-prime obstruction

Put $D = d/dz$ and, for $r \in \overline{\mathbb{Q}}$, define

$$F_r(z) = e^z(\mathrm{Ein}(z) - r) = \sum_{n\geq 0}(H_n - r)\frac{z^n}{n!}.$$

The identity follows from $F_r' = F_r + (e^z - 1)/z$ and $F_r(0) = -r$. In the algebraic gamma world one may take $r = \gamma$. Throughout this section a rational differential operator is held fixed when its input parameter r is varied.

An E-function $G(z) = \sum g_n z^n/n!$ is *globally bounded* in the convention of [20, Introduction] if some positive integer M makes every $M^{n+1}g_n$ an algebraic integer. Thus global boundedness is imposed on the *normalized* coefficients g_n, equivalently on the coefficients of the associated ordinary G-series $\sum g_n z^n$. The following result uses the weaker condition of integrality outside a fixed finite set of primes.

Theorem 23.1 (Finite prime support erases the input parameter)**.** *Let $r \in \overline{\mathbb{Q}}$ and $L \in \overline{\mathbb{Q}}(z)[D]$. Suppose $G = L(F_r)$ is regular at 0, and write $G = \sum_{n\geq 0} g_n z^n/n!$. Suppose there are a number field K containing every g_n and a finite set S of rational primes such that every g_n is integral at every prime of K above every $p \notin S$. Then*

$$L(e^z) = 0.$$

In particular, for this fixed L the expression $L(F_s)$ is independent of $s \in \mathbb{C}$. The conclusion applies whenever $L(F_r)$ is a globally bounded E-function.

Proof. Successive differentiation, using the displayed equation for F_r, gives

$$L(F_r) = a(z)F_r + b(z)e^z + c(z), \qquad a, b, c \in \overline{\mathbb{Q}}(z), \qquad a(z)e^z = L(e^z).$$

Indeed differentiation sends a to $a' + a$, which is exactly the rule for the coefficient of e^z when differentiating ae^z. Enlarge K to contain r and all these rational functions' coefficients. Integrality away from S is preserved by extension of number fields. Choose a nonzero $q \in K[z]$ that clears their denominators, giving

$$qG = A(z)F_r + B(z)e^z + C(z), \qquad A, B, C \in K[z].$$

Multiplication by the fixed polynomial q only enlarges S by finitely many primes: normalized coefficients transform into finite sums of algebraic coefficients times falling factorials and previous normalized coefficients.

Suppose $A = \sum_{k=0}^{d} a_k z^k \neq 0$. There is an $m \in \{0, \dots, d\}$ such that

$$A_m = \sum_{k=0}^{\min(d,m)} a_k m^{\underline{k}} \neq 0,$$

because the matrix $(m^{\underline{k}})_{0\leq m,k\leq d}$ is triangular with nonzero diagonal. Write $qG = \sum h_n z^n/n!$. Take any sufficiently large rational prime p. Exclude the finitely many primes needed for integrality of the fixed coefficients and r, and the numerator and denominator primes of the norm of A_m. For any prime $\mathfrak{p}$ of K above p, A_m is then a unit. For $p > m + d$, harmonic numbers satisfy

$$pH_{p+m-k} \equiv \begin{cases} 1, & k \leq m, \\ 0, & k > m \end{cases} \pmod{\mathfrak{p}}.$$

Only the denominator p contributes to this congruence, since all indices in question are positive and less than $2p$. The normalized coefficient of AF_r at $n = p + m$ is

$$\sum_{k=0}^{d} a_k (p+m)^{\underline{k}} (H_{p+m-k} - r).$$

The coefficients supplied by $Be^z + C$ are integral at $\mathfrak{p}$. Reducing after multiplication by p therefore gives

$$p h_{p+m} \equiv A_m \not\equiv 0 \pmod{\mathfrak{p}}.$$

This contradicts integrality of h_{p+m}. Thus $A = 0$, so $a = 0$ and $L(e^z) = 0$. Finally $F_s - F_r = (r - s)e^z$ proves independence of s. □

If $L(e^z) \neq 0$, the proof gives more than failure of global boundedness. After the fixed polynomial multiplication, every sufficiently large prime p occurs in the denominator of the single coefficient h_{p+m}, for one fixed offset $m \leq \deg A$. At unramified primes above such a p its valuation is exactly -1. No claim of literature-wide priority is made for this deduction.

The condition is necessary, not sufficient. The operator $L = D - 1$ erases the parameter but produces $(e^z - 1)/z$, whose normalized coefficients $1/(n + 1)$ have infinitely many denominator primes. The operator $L = z(D - 1)$ does produce the globally bounded function $e^z - 1$. For the intended gamma application, L must be constructed with coefficients independent of gamma: allowing $L = \gamma z(D - 1)$ under the algebraic hypothesis simply reinserts gamma as a coefficient.

23.1 The recent zero-factorization theorem: exact applicability

Fischler–Rivoal's [20, Theorem 1.5] assumes a nonzero globally bounded E-function over $\mathbb{Q}$, with every zero of one common multiplicity $m \geq 2$ and a rational differential operator of order at most $m + 1$ annihilating it. Its conclusion is an mth-power factorization by a globally bounded E-function of order at most two. An output merely defined over $\overline{\mathbb{Q}}$ would not by itself meet the rational-coefficient hypothesis. Their [20, Theorem 1.2 and Corollary 1.3] also establish a special case of entire quotient rigidity when the numerator is obtained by a rational differential operator applied to the denominator. Theorem 23.1 closes one proposed preprocessing step: no fixed gamma-independent rational differential operator makes F_γ globally bounded while retaining its input gamma parameter. This conclusion does not exclude nonlinear constructions, several independent framings, or a different input family. Nor does it claim that global boundedness is necessary for every zero theorem.

23.2 The exact mixed specialization and its logical cost

Define

$$\widehat{d}(z) = \sum_{n \geq 0} (-1)^n n! z^{-n-1}, \qquad \Psi_r(z) = F_r(z) - \widehat{d}(z).$$

Positive-direction summation yields

$$d_0(z) = e^z E_1(z), \qquad \Psi_{r,0}(z) = e^z(\gamma + \log z - r), \qquad \Psi_{r,0}(1) = e(\gamma - r).$$

Proposition 23.2 (Formal nondivisibility)**.** *For every algebraic r, there is no mixed arithmetic Gevrey function Q such that $(z - 1)Q = \Psi_r$.*

Proof. Write $Q = G(z) + h(1/z)$ with G an E-function and $h(1/z)$ having only negative powers. If $c \in \overline{\mathbb{Q}}$ is the coefficient of z^{-1} in h, comparison of nonnegative powers gives

$$F_r(z) = (z-1)G(z) + c.$$

Consequently $F_r(1) = c$. But e and $\mathrm{Ein}(1)$ are algebraically independent by the established E-function input; thus $e(\mathrm{Ein}(1) - r)$ is transcendental, a contradiction. □

It follows that the statement

$$\forall r \in \overline{\mathbb{Q}}, \quad \Psi_{r,0}(1) = 0 \implies \Psi_r \text{ is divisible by } z-1 \text{ in the mixed class}$$

is *equivalent* to the transcendence of gamma. One direction uses the proposition; the reverse direction is a vacuous implication. This is the exact family specialization of [5, Conjecture 3], not a proved weakening of that conjecture. Pure E-function division is a theorem [4, Proposition 4.1]; replacing its input class by mixed functions is the missing arithmetic step.

23.3 Worked p-curvature calculation

Put $Y = (\Psi_{r,0}, e^z)^t$. Then

$$Y' = A(z)Y, \qquad A(z) = I + N/z, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

Its fundamental matrix is

$$e^z \begin{pmatrix} 1 & \log z \\ 0 & 1 \end{pmatrix}.$$

Neither the coefficient matrix nor this chosen fundamental matrix depends on r or gamma. The constant vector selecting Y is $(\gamma - r, 1)^t$.

Define $A_0 = I$ and $A_{n+1} = A_n' + A_n A$, so that $Y^{(n)} = A_n Y$. Direct differentiation gives

$$A_n = I + N \sum_{k=1}^{n} \binom{n}{k} (-1)^{k-1} (k-1)! z^{-k}.$$

Modulo every prime p, the interior binomial coefficients disappear and Wilson's congruence gives

$$A_p \equiv I - N z^{-p}.$$

This matrix is nonzero and invertible. With the opposite connection sign convention the corresponding curvature changes sign; its nonvanishing and gamma-independence are unchanged. Tensoring out the exponential rank-one factor leaves the logarithmic curvature $-Nz^{-p}$, which is nonzero nilpotent.

The effective order-one theorem of Fürnsinn–Pannier [18, Theorem 1.3] concerns algebraicity of functions solving a specified rational differential equation. It supplies no arithmetic evaluation rule for the constant vector selecting a particular solution of the present rank-two system. Thus its use here does not distinguish the two gamma worlds. A result controlling the arithmetic of the connection framing is still required.

24 An imported categorical alternative and a failed positivity shortcut

Let $\mathcal{D}_t = \mathrm{Rep}(S_t)$ denote the complex-linear additive Karoubi envelope of the partition category, with its standard symmetric tensor structure. This is not the representation category of a literal group with nonintegral size. Put

$$R_+(t) = \left\{ \sum_{n=0}^{N} a_n \binom{t}{n} : a_n \in \mathbb{Z}_{\geq 0},\ N < \infty \right\}.$$

We import two established results. For parameters outside $\mathbb{Z}_{\geq 0}$, Harman–Kalinov characterize the existence of complex-linear strong symmetric monoidal functors $\mathcal{D}_s \to \mathcal{D}_t$ by the condition $s \in R_+(t)$ [15, Corollary 3.2.4]. Kalinov–Mandelshtam prove that $R_+(t)$ is a ring precisely when t is algebraic and not a nonnegative integer [14, Theorem 1.3]. Here and below the parameters in the functor statement are outside $\mathbb{Z}_{\geq 0}$.

Corollary 24.1 (A categorical reformulation)**.** *For the actual $0 < \gamma < 1$, the following are equivalent:*

$$\gamma \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad -1 \in R_+(\gamma) \quad \Longleftrightarrow \quad \exists\, \mathcal{D}_{-1} \longrightarrow \mathcal{D}_\gamma$$

where the arrow is a complex-linear strong symmetric monoidal functor.

Proof. The semiring $R_+(t)$ contains 1, and is a ring exactly when it contains -1, since multiplication by -1 then supplies every additive inverse. Apply the two imported results. The finite relation $1 + \sum a_n \binom{\gamma}{n} = 0$ is a nonzero rational polynomial relation, since its polynomial has value $1 + a_0 > 0$ at 0. □

This is a direct corollary of known work, not an independently solved problem. Under such a functor the source fundamental object maps to a commutative Frobenius algebra of categorical dimension -1 in $\mathcal{D}_\gamma$; this is not ordinary vector-space dimension. A tempting positivity contradiction fails for an exact reason. If X is the fundamental object of $\mathcal{D}_t$, symmetric categorical trace gives

$$\dim(\Lambda^2 X) = \tfrac{1}{2}\left(\mathrm{Tr}(\mathrm{id}_{X\otimes X}) - \mathrm{Tr}(c_{X,X})\right) = \frac{t(t-1)}{2} < 0 \quad (0 < t < 1).$$

Thus no realization preserving this trace can land in a category where all such dimensions are nonnegative. This obstruction applies to *both* arithmetic alternatives, and is not an axiom of $\mathcal{D}_t$ itself. At $t = 1/2$ the exact identity $8\binom{1/2}{2} = -1$ gives the stated functor, even though $\dim(\Lambda^2 X) = -1/8$.

The word "unitary" alone cannot supply the missing trace-positivity assumption: intrinsic C^* dimensions and dimensions from a chosen symmetric trace need not agree, as super symmetry illustrates. The remaining arithmetic task is to prove existence or nonexistence of the specified functor for γ, not to impose a positivity axiom that already excludes valid algebraic examples.

Part IV
Functional transcendence and arithmetic costs

25 From arithmetic rank to differential transcendence

Set $K = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}})$. The multipoint and Hankel recovery arguments reproduced in Section 2 give, for $N \geq 1$,

$$\operatorname{trdeg}_K K(I_1, \dots, I_N) \geq N - \pi(N) - 1, \qquad I_n = e^n E_1(n), \tag{25.1}$$

$$\operatorname{trdeg}_K K(A_1, \dots, A_N) \geq N - \pi(2N-1). \tag{25.2}$$

Here A_n denotes the $a = h = 1$ Hankel approximant: with $m_k = \operatorname{Ein}(k+1) - \log(k+1)$, $F_n = (m_{i+j})_{0 \leq i,j < n}$ and $v_n = (1, \dots, 1)^{\mathsf{T}}$,

$$A_n = \frac{\det F_n}{v_n^{\mathsf{T}} \operatorname{adj}(F_n) v_n}.$$

We transfer these rank bounds to differential equations and nonlinear recurrences. No novelty claim is made for the general principles or for their applications below.

25.1 A coefficient-field lemma, including singular equations

Lemma 25.1. *Let $L \subseteq \mathbb{C}$ be a characteristic-zero field. If a formal power series $f(z) = \sum_{n \geq 0} a_n z^n$ satisfies a nonzero polynomial differential equation over $L(z)$, then for some M every a_n belongs to $L(a_0, \dots, a_M)$.*

Proof. Clear denominators and choose a nonzero annihilator $F \in L[z, Y_0, \dots, Y_r]$ of least total degree in the Y variables. This degree is positive. At least one series

$$S_j(z) = \partial_{Y_j} F(z, f, \dots, f^{(r)})$$

is nonzero: a nonzero partial polynomial has smaller total degree and cannot itself annihilate f. Put

$$v = \min_{S_j \neq 0} (\operatorname{ord}_z S_j - j), \qquad b_j = [z^{v+j}] S_j, \qquad B(T) = \sum_{j=0}^{r} b_j (T)_j.$$

Negative coefficient indices mean zero. At least one b_j is nonzero, and the falling factorials $(T)_j$ have distinct degrees, so $B \neq 0$. Its coefficients depend on only finitely many initial a_n.

Write $f_{<n} = \sum_{k<n} a_k z^k$ and $h = f - f_{<n}$. For sufficiently large n, polynomial Taylor expansion gives

$$0 = F(z, f_{<n}, \dots, f_{<n}^{(r)}) + \sum_{j=0}^{r} S_j(z) h^{(j)}(z) + O(z^{2(n-r)}).$$

Evaluating S_j at f instead of $f_{<n}$ changes the linear term only by $O(z^{2(n-r)})$; every higher Taylor term has at least that order. If $n \geq r$ and $n > v + 2r$, the coefficient of z^{n+v} gives

$$B(n) a_n = -[z^{n+v}] F(z, f_{<n}, \dots, f_{<n}^{(r)}). \tag{25.3}$$

Indeed terms of h with index greater than n have order greater than $n+v$ after linearization. The nonzero polynomial B has only finitely many nonnegative integer roots. Beyond these roots and the finite initial indices defining B, equation (25.3) determines each coefficient rationally from earlier ones. Induction proves the assertion. □

This is the classical nonlinear coefficient-recursion mechanism. For a primary modern account of Mahler's recurrence, see Krattenthaler–Rivoal, *Arithmetic properties of the Taylor coefficients of differentially algebraic power series*, `https://arxiv.org/html/2502.09259v1`, Section 1, equation (1.4). The proof above supplies the precise general coefficient-field version needed here; no assumption that the a_n are algebraic is used.

Corollary 25.2. *If the coefficient field of $f \in \mathbb{C}[[z]]$ has infinite transcendence degree over a fixed characteristic-zero subfield $K \subseteq \mathbb{C}$, then f is differentially transcendental over $\mathbb{C}(z)$.*

Proof. A putative differential equation contains only finitely many complex constants $c_1, \dots, c_s$ in its coefficients. Apply Lemma 25.1 over $L = K(c_1, \dots, c_s)$. All the Taylor coefficients would then lie in $L(a_0, \dots, a_M)$, a field of finite transcendence degree over K, a contradiction. □

Multiplication of the individual Taylor coefficients by prescribed nonzero elements of K leaves their generated field unchanged. In particular ordinary and exponential generating series have the same obstruction.

25.2 Explicit differentially transcendental integral transforms

Theorem 25.3. *The analytic function and the entire function*

$$\mathcal{I}(z) = \int_0^1 \frac{dx}{(1-zx)(1-\log x)} \qquad (|z|<1), \tag{25.4}$$

$$\widehat{\mathcal{I}}(z) = \int_0^1 \frac{e^{zx}}{1-\log x}\,dx \qquad (z \in \mathbb{C}) \tag{25.5}$$

are differentially transcendental over $\mathbb{C}(z)$. The ordinary and exponential generating functions of $E_1(n)$, $\mathrm{Ein}(n)$, and A_n have the same property.

Proof. Tonelli's theorem and $1/(n+t) = \int_0^1 x^{n+t-1}dx$ give

$$I_n = \int_0^\infty \frac{e^{-t}}{n+t}\,dt = \int_0^1 \frac{x^{n-1}}{1-\log x}\,dx.$$

Geometric and exponential summation, uniform on compact subsets of the respective domains, identify (25.4) with $\sum_{n\ge1} I_n z^{n-1}$ and (25.5) with $\sum_{n\ge1} I_n z^{n-1}/(n-1)!$. Equation (25.1) and Corollary 25.2 apply. Multiplication by $e^{-n} \in K^\times$ gives the $E_1(n)$ assertion. The $\mathrm{Ein}(n)$ are already algebraically independent over K. Use (25.2) for A_n. All these ordinary generating series have positive radius of convergence, and their exponential versions are entire: $I_n = O(1/n)$, $E_1(n) = O(e^{-n}/n)$, $\mathrm{Ein}(n) = O(\log(n+1))$, and A_n is bounded. □

These are generating transforms of sampled values. The functions $E_1(z)$ and $\mathrm{Ein}(z)$ themselves satisfy linear differential equations; no contrary claim is being made. Nor does functional differential transcendence decide the arithmetic nature of an individual value such as $I_1 = \delta$. In particular, if $H(z) = \widehat{\mathcal{I}}(z)$, then

$$H(0) = \int_0^1 \frac{dx}{1-\log x} = I_1 = eE_1(1) = \delta.$$

The theorem classifies H as a function, including its analytic germ at zero; it does not classify the number $H(0)$. Differential transcendence alone imposes no arithmetic classification on a prescribed value.

25.3 Removing the gamma pole does not remove the obstruction

Put

$$\mathcal{A}(z)=\sum_{n\ge 1}A_n z^{n-1},\qquad \widehat{\mathcal{A}}(z)=\sum_{n\ge 1}\frac{A_n z^{n-1}}{(n-1)!},\quad \xi=2e-1,\quad \rho=\xi+\sqrt{\xi^2-1}.$$

The proved Christoffel bound is

$$0<A_n-\gamma\le\frac{E_1(1)}{T_{n-1}(\xi)^2}\le 4E_1(1)\rho^{-2(n-1)}.$$

Consequently

$$\mathcal{E}_A(z)=\mathcal{A}(z)-\frac{\gamma}{1-z}=\sum_{n\ge 1}(A_n-\gamma)z^{n-1}$$

is analytic for $|z|<\rho^2$, and the entire function

$$\widehat{\mathcal{E}}_A(z)=\widehat{\mathcal{A}}(z)-\gamma e^z$$

satisfies $|\widehat{\mathcal{E}}_A(z)|\le 4E_1(1)\exp(|z|/\rho^2)$. Both functions remain differentially transcendental over $\mathbb{C}(z)$. Indeed finite transcendence degree for their coefficient field would, on adjoining the single constant γ, contradict (25.2). Apply Corollary 25.2. Thus the simple pole at 1 does not account for the differential transcendence. Its residue's arithmetic nature is still not determined.

26 Quantitative exclusion of nonlinear recurrence templates

The argument below does not assume that a recurrence can be solved for its last variable. It therefore includes singular or degenerate specializations.

Theorem 26.1. *Suppose $(a_n)_{n\ge 1}\subseteq\mathbb{C}$ satisfies*

$$\operatorname{trdeg}_K K(a_1,\dots,a_N)\ge N-D(N).$$

Let $L=K(c_1,\dots,c_s)$, $d=\operatorname{trdeg}_K L$, and let $P(T,X_0,\dots,X_r)\in L[T,X_0,\dots,X_r]$ be nonzero, with positive degree in at least one X_j. Let e be the number of positive integers n for which $P(n,X)$ is the zero polynomial; then $e\le\deg_T P$. The number $Z_P(N)$ of indices $1\le n\le N-r$ satisfying $P(n,a_n,\dots,a_{n+r})=0$ obeys

$$Z_P(N)\le(r+1)(D(N)+d)+e. \tag{26.1}$$

Proof. Omit the identically specialized equations. Greedily select disjoint satisfying intervals $[n,n+r]$ in increasing order of their left endpoint. Each selected interval excludes at most $r+1$ possible starts. If M intervals are selected, then $Z_P(N)-e\le(r+1)M$. Each selected block has a nonzero algebraic relation over L, so its $r+1$ entries contribute transcendence degree at most r. Taking the union of disjoint blocks and adjoining all uncovered entries yields

$$\operatorname{trdeg}_L L(a_1,\dots,a_N)\le N-M.$$

By the tower formula and monotonicity,

$$N-D(N)\le\operatorname{trdeg}_K K(a_1,\dots,a_N)\le d+N-M.$$

Thus $M\le D(N)+d$, proving (26.1). □

Rational-function coefficients in T are covered after clearing denominators and excluding their finitely many poles. Arbitrary complex constants are covered because a single polynomial uses only finitely many of them.

For I_n and $E_1(n)$ take $D(N) = \pi(N)+1$; for A_n take $D(N) = \pi(2N-1)$. In either case every fixed finite-order algebraic recurrence over $\mathbb{C}(n)$ can hold only on a density-zero set of windows, and cannot hold eventually. This excludes linear P-recursiveness and nonlinear polynomial/rational-in-n templates. For the completely independent sequence $\mathrm{Ein}(n)$, or for the positive composite residuals enumerated increasingly, take $D(N) = 0$: any fixed such relation can hold on only finitely many windows. This is an exact algebraic exclusion, not a test of approximate numerical fits.

If a polynomial relation fits every window through N, the budget gives

$$N - r \le (r+1)(D(N)+d)+e, \qquad r+1 \ge \frac{N+1-e}{D(N)+d+1}.$$

A genuine forward algebraic generation rule obeys a stronger constraint. If its first N outputs are algebraic over a field generated over K by d independent parameters and r initial seeds, then necessarily

$$r + d \ge N - D(N).$$

A step at which its specialized defining polynomial vanishes identically introduces an additional free seed, which must be counted. These results do not exclude growing-order recurrences, new special values introduced at each step, nonalgebraic coefficient sequences, or the integral representations already available. They exclude an entire class of fixed algebraic compression methods, without deciding the arithmetic nature of γ.

27 Nonlinear fresh-prime descent

Set $E = e^z$ and, for $r \in \overline{\mathbb{Q}}$, set

$$F_r = e^z(\mathrm{Ein}(z) - r) = \sum_{n\ge 0}(H_n - r)\frac{z^n}{n!}.$$

Then $E' = E$ and $F_r' = F_r + (E-1)/z$. Consequently

$$\mathcal{A}_r = \overline{\mathbb{Q}}(z)[E, E^{-1}, F_r]$$

is a differential algebra. It includes fixed polynomial nonlinear combinations of all derivatives of its generators. For a germ $G = \sum g_n z^n/n!$, write $[G]_n = g_n$. Integrality of normalized coefficients will always mean integrality at every prime above a specified rational prime, in one number field.

27.1 The fresh-prime derivation

Lemma 27.1. *Let $P \in K[z, T, T^{-1}, Y]$, where K contains r, and put $G = P(z, E, F_r)$. For every fixed $m \ge 0$ and every sufficiently large rational prime p, at every prime $\mathfrak{p}$ of K above p,*

$$p[G]_{p+m} \equiv [EP_Y(z, E, F_r)]_m \pmod{\mathfrak{p}}.$$

The same formula holds for $P \in K[z, T, T^{-1}, Y, V]$ evaluated at $V = J(z)$ if the normalized coefficients $[J]_0, \dots, [J]_{p+m}$ are integral at $\mathfrak{p}$. There is then no derivative term in V.

Proof. It suffices to check $z^a E^b F_r^c J^d$. All fixed algebraic coefficients are integral after excluding finitely many primes. Products of exponential generating series use integral multinomial coefficients. Since $p + m < 2p$, at most one F_r factor can have index at least p. Terms without such a factor are integral and disappear after multiplication by p. For its index $p + k$,

$$p(H_{p+k} - r) \equiv 1 \pmod{\mathfrak{p}}.$$

Lucas's congruence reduces the remaining multinomial coefficient to the one of total index m, with the large index replaced by k. There are c choices of the large factor, giving $E \cdot cF_r^{c-1}$. The remaining E^b and J^d factors have only low indices in these terms. Multiplication by z^a introduces a falling factorial. For $a \le m$ it reduces to $m^{\underline{a}}$; for $a > m$ both sides vanish. This proves the assertion. In the case with J, no factor of J can supply a denominator, by the stated integrality hypothesis. □

The polynomial and its degree are fixed independently of p. Choosing a different polynomial such as F_r^p for each tested prime is not a uniform application of this lemma: its derivative coefficient vanishes at that particular prime.

Lemma 27.2. *The functions E and F_r are algebraically independent over $\overline{\mathbb{Q}}(z)$.*

Proof. The exponential E is transcendental over $\overline{\mathbb{Q}}(z)$. Put $K_0 = \overline{\mathbb{Q}}(z, E)$, with $DE = E$. If $\mathrm{Ein}(z)$ were algebraic over K_0, averaging its trace over that finite extension would yield $v \in K_0$ with

$$Dv = (1 - E^{-1})/z.$$

In the Laurent ring $\overline{\mathbb{Q}}(z)[E, E^{-1}]$, a nonunit irreducible factor B cannot divide DB: comparing two of its nonzero Laurent coefficients would give a rational function with nonzero constant logarithmic derivative. Thus any nonunit denominator factor of v would acquire a higher pole under differentiation, which the Laurent-polynomial right side does not have. Hence v is a Laurent polynomial in E. Its coefficient of E^0 would be a rational primitive of $1/z$, impossible by the residue at zero. Since $F_r = E(\mathrm{Ein} - r)$, the assertion follows. □

Theorem 27.3 (Algebraic descent from finite prime support)**.** *Let $K_0 = \overline{\mathbb{Q}}(z, e^z)$ and let $J(z) = \sum_{n \ge 0} j_n z^n/n!$ be a germ regular at zero, algebraic over $K_0(F_r)$. Assume all j_n belong to one number field and are integral outside a fixed finite set of rational primes. Then J is algebraic over K_0. In particular, if $J \in K_0(F_r)$, then $J \in K_0$.*

More generally, retain regularity, algebraicity over $K_0(F_r)$, and the one-number-field coefficient hypothesis, but drop the finite-prime support assumption. If J is not algebraic over K_0, there is a fixed integer $m \ge 0$ such that for every sufficiently large rational prime p, at least one of $j_0, \ldots, j_{p+m}$ is nonintegral at a prime above p.

Proof. By Lemma 27.2, evaluation

$$K_0[Y, V] \longrightarrow K_0(F_r, J), \qquad (Y, V) \longmapsto (F_r, J)$$

has a height-one prime kernel generated by an irreducible polynomial $P(Y, V)$. Clear its K_0 coefficient denominators; the multiplier is independent of Y, V and is a unit in K_0. Irreducibility and the kernel-generator assertion are used over $K_0[Y, V]$, not over the polynomial coefficient ring after clearing. All coefficients then lie in $K[z, T, T^{-1}, Y, V]$ for a number field K, where $T = E$. The algebraic extension of the field containing the j_n preserves integrality outside the original finite set.

Apply Lemma 27.1 to the identically zero series $P(F_r, J)$. For each fixed m the algebraic number $[EP_Y(F_r, J)]_m$ reduces to zero at every sufficiently large prime. It is therefore zero, since a nonzero algebraic number has only finitely many prime divisors. Thus $P_Y(F_r, J) = 0$. The kernel description implies that P divides P_Y. If $P_Y \neq 0$, its degree in Y is smaller than that of P, which is impossible. Hence $P_Y = 0$ and $P \in K_0[V]$, proving the first assertion. The rational case uses the fact that K_0 is relatively algebraically closed in the pure transcendental field $K_0(F_r)$.

For the last assertion, if J is not algebraic over K_0, then $EP_Y(F_r, J)$ is nonzero. Choose a nonzero coefficient at a fixed index m. At every sufficiently large prime it is a unit. If all $j_0, \dots, j_{p+m}$ were integral there, the finite version of Lemma 27.1 would reduce that unit to zero. This contradiction proves the asserted denominator appearance. □

This is a statement about an algebraic direction in a differential field, not the arithmetic nature of gamma. Under the hypothesis that gamma is algebraic one may set $r = \gamma$, consistently with the theorem. If a construction inserts gamma into its algebraic coefficients or branch-selection rule, the theorem does not erase that external input.

27.2 Complete classification in the polynomial differential algebra

Theorem 27.4. *Let $\mathcal{E}_{\mathrm{gb}}$ be the ring of globally bounded E-functions, where global boundedness concerns normalized Taylor coefficients. Then*

$$\mathcal{A}_r \cap \mathcal{E}_{\mathrm{gb}} = \overline{\mathbb{Q}}[z, e^z, e^{-z}].$$

More generally, for $G \in \mathcal{A}_r$ regular at zero, integrality of normalized coefficients outside finitely many primes already forces $G \in \overline{\mathbb{Q}}(z)[e^z, e^{-z}]$.

Proof. Clear rational z denominators and write $qG = P(z, E, F_r)$. Multiplication by the fixed polynomial q adds only finitely many denominator primes to normalized coefficients. Lemma 27.1 implies $EP_Y(z, E, F_r) = 0$, and Lemma 27.2 gives $P_Y = 0$ formally. Hence the displayed general assertion holds, with Laurent-polynomial dependence on E.

Now suppose G is a globally bounded E-function, hence entire. Write $G = \sum_j a_j(z)e^{jz}$ with rational functions a_j. At a nonzero finite algebraic pole α, cancellation of the highest principal parts would give $\sum_j c_j e^{j\alpha} = 0$ with algebraic c_j. Lindemann–Weierstrass forces all $c_j = 0$. Repeating at each possible pole shows that each a_j is a Laurent polynomial in z.

Suppose a negative power remains, and choose the maximal $M \geq 1$:

$$G = \sum_{a=-M}^{D} z^a Q_a(e^z), \qquad Q_{-M}(T) = \sum_j c_j T^j \neq 0.$$

For fixed $k \geq 1$ and large prime p, put $n = kp - M$. Only the highest negative power has a factor p in its denominator $(n+1)\cdots(n+M)$; all smaller negative powers have integral coefficients there. Fermat's congruence, including negative integer frequencies, gives

$$p[G]_{kp-M} \equiv \frac{\sum_j c_j j^k}{k(-1)^{M-1}(M-1)!} \pmod{\mathfrak{p}}.$$

Thus $\sum_j c_j j^k = 0$ for every $k \geq 1$. Vandermonde invertibility at the distinct nonzero integer frequencies forces $c_j = 0$ for $j \neq 0$. The remaining term $c_0 z^{-M}$ cannot be canceled at zero by any lower negative power, contradicting entire regularity. Hence no negative power remains. The reverse inclusion follows directly: polynomial multiples of e^{jz}, $j \in \mathbb{Z}$, have globally bounded normalized coefficients. □

For example, $(e^z-1)^2/z$ has normalized coefficients $(2^{n+1}-2)/(n+1)$. The apparent cancellation at $n=p-1$ is misleading: at $n=2p-1$ every odd prime gives $p[G]_{2p-1}\equiv 1 \pmod p$. The second prime block detects the surviving denominator.

Finite prime support alone does not imply that a rational expression in E is an E-function. For example $1/(2-e^z)$ has integral normalized coefficients $\sum_{k=0}^{n} k!S(n,k)$, obtained by expanding the geometric series in e^z-1, but has poles at $\log 2+2\pi i\mathbb{Z}$. This distinguishes the weak arithmetic hypothesis in Theorem 27.3 from the entire-function hypothesis used in Theorem 27.4.

27.3 Import boundary and remaining scope

The convention for global boundedness above is exactly that used in the introduction of Fischler–Rivoal [20]. Their Theorem 1.5 uses this condition in a zero-factorization result; Theorem 1.2 and Corollary 1.3 concern a different entire-quotient theorem. The present classification concerns our preprocessing algebra, and does not claim to generalize those theorems or establish historical priority.

The descent theorem goes beyond fixed linear operators to rational and algebraic operations. It still does not cover arbitrary new transcendental functions, varying-degree constructions with uniform height control, several independently normalized connection constants, or a direct arithmetic theorem for mixed Gevrey values. None of the results here decides the algebraic or transcendental world for gamma.

Exact finite checks are supplied in `check_nonlinear_imports.py`: 288 nonlinear fresh-prime congruences and seven second-block Laurent checks. These are checks of the formulas, not substitutes for the proofs.

28 Sharp decay and a two-point inverse moment theorem

The previous Hankel construction gave positive upper approximants to gamma. Here its exponential rate is determined exactly, and a second Christoffel trace is shown to recover all the moment information lost by a single trace. The arguments use classical polynomial inequalities, orthogonal polynomials, and Jacobi matrices; no priority claim is made for these methods or their consequences.

28.1 The exact exponential rate

Fix $a,h>0$, put $q=e^{-h}$, and let

$$d\mu(x)=\frac{x^{a/h-1}}{-\log x}\,dx \quad (0<x<q), \qquad \lambda_n^{\mu}(z)=\min_{\substack{\deg p<n\\ p(z)=1}}\int |p(x)|^2\,d\mu(x).$$

For $z>q$ set

$$\xi_q(z)=\frac{2z}{q}-1, \qquad \rho_q(z)=\xi_q(z)+\sqrt{\xi_q(z)^2-1}.$$

Theorem 28.1 (Sharp root asymptotic)**.** *For every fixed $z>q$,*

$$\lim_{n\to\infty}\lambda_n^{\mu}(z)^{1/n}=\rho_q(z)^{-2}. \tag{28.1}$$

In particular the earlier approximants $A_n=\gamma+\varepsilon_n$ satisfy

$$\lim_{n\to\infty}-\frac{1}{n}\log\varepsilon_n=2\log\rho_q(1)=4\,\operatorname{arcosh}(e^{h/2}).$$

The exponent is independent of a. This is a fixed-parameter limit; it is not asserted uniformly when a or h varies with n.

Proof. The normalized Chebyshev polynomial on $[0, q]$ gives

$$\lambda_n^{\mu}(z) \le \frac{E_1(a)}{T_{n-1}(\xi_q(z))^2},$$

so the upper root bound in (28.1) follows from $T_m(\xi) = (\rho^m + \rho^{-m})/2$.

For the lower bound fix $0 < \delta < q$. The density has a positive minimum w_δ on $[\delta, q]$, whose length is $L = q - \delta$. Let p be real, of degree at most $m = n - 1$, with $p(z) = 1$, and put $M = \max_{[\delta,q]} |p|$. Chebyshev's exterior extremal inequality gives

$$M \ge \frac{1}{T_m(\xi_\delta(z))}, \qquad \xi_\delta(z) = \frac{2z - q - \delta}{q - \delta}.$$

For $m \ge 1$, Markov's inequality gives $\max |p'| \le 2m^2 M/L$. From a point where $|p| = M$ there is a one-sided interval of length $L/(4m^2)$ contained in $[\delta, q]$ on which $|p| \ge M/2$. Therefore

$$\int |p|^2 \, d\mu \ge \frac{w_\delta L}{16 m^2 T_m(\xi_\delta(z))^2}.$$

Complex coefficients cannot improve the minimum: taking the real part preserves the constraint and can only decrease the objective. Taking nth roots yields a lower limit at least $(\xi_\delta(z) + \sqrt{\xi_\delta(z)^2 - 1})^{-2}$. Let $\delta \downarrow 0$ to finish the proof. □

Corollary 28.2 (The second trace has its own sharp rate)**.** *Let $\nu = \mu + \gamma\delta_1$, and let $d > 1$. Then*

$$\lim_{n\to\infty} \lambda_n^{\nu}(d)^{1/n} = \rho_q(d)^{-2}.$$

Moreover $0 < \lambda_{n+1}^{\nu}(d) < d^{-2}\lambda_n^{\nu}(d)$. For $n \ge 2$ one has the explicit upper bound

$$\lambda_n^{\nu}(d) \le \frac{E_1(a)}{(d-1)^2 T_{n-2}(\xi_q(d))^2}.$$

Proof. Adding the positive atom gives $\lambda_n^{\nu}(d) \ge \lambda_n^{\mu}(d)$. If p is its minimizing polynomial, xp/d is feasible at the next level. Since $x^2 \le 1$ on the support, with strict inequality on the continuous part, this proves the strict contraction. For the reverse root bound use

$$p(x) = \frac{x-1}{d-1} \frac{T_{n-2}(2x/q - 1)}{T_{n-2}(2d/q - 1)}.$$

It has degree at most $n - 1$, equals one at d, and vanishes at the atom. Since $|x - 1| \le 1$ on $[0, q]$, it gives the stated bound. □

Only the exponential rate has been proved here. A full leading constant or a ratio asymptotic is not inferred from a root limit. The density is also in the classical Szegő class after an affine change to $[-1, 1]$, but no unverified normalization of a stronger Szegő asymptotic is needed for these conclusions.

The root limit also proves the exact convergence radius

$$R\left(\sum_{n\ge1} (A_n - \gamma) z^{n-1}\right) = \rho_{e^{-1}}(1)^2$$

for $a = h = 1$, by Cauchy–Hadamard. The series remains differentially transcendental by the coefficient-field argument above.

28.2 Two exterior traces determine the measure

Let σ be a finite positive measure with infinite support contained in a compact interval $[L, U]$. Let p_n be its orthonormal polynomials with positive leading coefficient, and let

$$K_n(t,t) = \sum_{j=0}^{n-1} p_j(t)^2, \qquad \lambda_n(t) = K_n(t,t)^{-1}, \qquad K_0(t,t) = 0.$$

The convention uses polynomials of degree strictly below n.

Theorem 28.3 (Two-trace reconstruction)**.** *Fix $U \le c < d$. The two sequences $(\lambda_n(c))_{n\ge1}$ and $(\lambda_n(d))_{n\ge1}$ uniquely determine σ. The first N entries of both traces determine its moments through degree $2N-2$ using arithmetic operations and positive square roots. No rational-field equality is asserted.*

Proof. Every zero of every p_n lies strictly between the extreme points of the support. Thus $p_n(c), p_n(d) > 0$, even when $c = U$. The data recover these values without a sign ambiguity:

$$p_n(t) = \sqrt{\lambda_{n+1}(t)^{-1} - \lambda_n(t)^{-1}}, \quad t \in \{c, d\}, \quad n \ge 0, \tag{28.2}$$

where $\lambda_0(t)^{-1} = 0$.

Write the Jacobi recurrence as

$$a_{n+1}p_{n+1}(t) + b_n p_n(t) = t p_n(t) - a_n p_{n-1}(t), \qquad a_{n+1} > 0,$$

with $a_0 p_{-1} = 0$. Suppose a_n is already known. Evaluating at c, d gives a two-by-two linear system for a_{n+1}, b_n. Its determinant is

$$p_n(c)p_n(d)\left(\frac{p_{n+1}(c)}{p_n(c)} - \frac{p_{n+1}(d)}{p_n(d)}\right) < 0.$$

Indeed each ratio $r_n(t) = p_n(t)/p_{n-1}(t)$ is positive and strictly increasing for $t \ge U$. Induction follows from

$$a_n r_n(t) = t - b_{n-1} - \frac{a_{n-1}}{r_{n-1}(t)}, \qquad a_n r_n'(t) = 1 + \frac{a_{n-1} r_{n-1}'(t)}{r_{n-1}(t)^2} > 0,$$

with $r_1'(t) = 1/a_1$. Thus every reconstruction system is nonsingular.

The mass $M = \sigma(\mathbb{R}) = \lambda_1(c) = \lambda_1(d)$ is known. The first N trace entries give $a_1, \dots, a_{N-1}$ and $b_0, \dots, b_{N-2}$. Complete the $N \times N$ truncated Jacobi matrix by assigning any value to the last diagonal entry. For $0 \le k \le 2N-2$ its moments are

$$\int x^k \, d\sigma(x) = M\,(J_N^k)_{00}.$$

A walk starting and ending at index zero cannot use the last diagonal entry in fewer than $2N-1$ steps, or reach the next omitted index in fewer than $2N$ steps. This proves the finite assertion. All moments determine a compactly supported measure uniquely by polynomial density in continuous functions, proving uniqueness of σ. □

The endpoint allowance is essential for the gamma application: $\nu = \mu + \gamma\delta_1$ has support bounded above by 1 and $\lambda_n^\nu(1) = A_n$.

Proposition 28.4 (An admissibility criterion)**.** *Fix $L < U \le c < d$. Two positive proposed traces arise from a positive infinite-support measure on $[L, U]$ if and only if all of the following hold: their first entries agree; their reciprocal increments in (28.2) are positive; every reconstruction system is nonsingular and gives $a_n > 0$, $b_n \in \mathbb{R}$; and the reconstructed Jacobi matrix satisfies $LI \le J \le UI$ on finitely supported vectors. The resulting measure is unique.*

Proof. Necessity follows from the previous proof and the representation of J as multiplication by x on the polynomial subspace of $L^2(\sigma)$. Conversely the quadratic-form bounds produce a bounded selfadjoint Jacobi operator with spectrum in $[L, U]$. Its spectral measure at the first coordinate, multiplied by the common prescribed mass, has infinite support because every a_n is positive. Its orthonormal polynomials satisfy the reconstructed recurrence. Induction makes their values at c, d equal the positive roots prescribed by (28.2). Summing their squares recovers the reciprocal traces by telescoping. □

This is a countable exact characterization, not a finite stopping algorithm for arbitrary real input sequences.

28.3 One trace is provably insufficient

Proposition 28.5 (An explicit continuum of identical traces)**.** *There is a continuum of distinct probability measures, all with infinite support contained in $[5/16, 319/400]$, for which*

$$\lambda_N(1) = \frac{15}{16^N - 1} \qquad (N \ge 1).$$

Proof. For $9/100 \le t \le 11/100$, take the bounded Jacobi matrix with

$$a_n = t \ (n \ge 1), \qquad b_0 = 1 - 4t, \qquad b_n = 1 - \frac{17t}{4} \ (n \ge 1).$$

The diagonal-minus/plus-row-radius bounds put its spectrum in $[5/16, 319/400]$. Its first-coordinate spectral measure is a probability measure with infinite support. The recurrence, starting with $p_0 = 1$, gives $p_n(1) = 4^n$ for every n, because $4t + (1 - 17t/4) + t/4 = 1$. Hence the kernel is $\sum_{j=0}^{N-1} 16^j = (16^N - 1)/15$. The measures are distinct since their means are $b_0 = 1 - 4t$. □

28.4 The second trace preserves the full arithmetic rank

Specialize $a = h = 1$ and set

$$m_k = \mathrm{Ein}(k+1) - \log(k+1), \quad F_n = (m_{i+j})_{0 \le i,j < n}, \quad v_n(z) = (1, z, \dots, z^{n-1})^{\mathsf{T}}.$$

These are moments of $\nu = \mu_{1,1} + \gamma\delta_1$. Define

$$A_n = \frac{1}{v_n(1)^{\mathsf{T}} F_n^{-1} v_n(1)}, \qquad B_n = \frac{1}{v_n(2)^{\mathsf{T}} F_n^{-1} v_n(2)}.$$

The duplicate first value is $A_1 = B_1 = m_0$. Let $K_{\exp} = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}})$ and retain the explicitly stated companion input that distinct nonzero algebraic arguments give algebraically independent Ein values over this field.

Theorem 28.6 (Lossless rank transfer through two traces)**.** *For every $N \geq 1$,*

$$\begin{aligned}&\operatorname{trdeg}_{K_{\exp}} K_{\exp}(A_1, \ldots, A_N, B_2, \ldots, B_N)\\ &\quad = \operatorname{trdeg}_{K_{\exp}} K_{\exp}(m_0, \ldots, m_{2N-2}) \geq 2N - 1 - \pi(2N-1).\end{aligned}$$

In the ordered list $A_1, A_2, B_2, A_3, B_3, \ldots$, there exists a set of indices of natural density zero whose deletion leaves an algebraically independent family over $K_{\exp}$.

Proof. The trace field is a subfield of the moment field because its entries are rational expressions in the moments. By Theorem 28.3 with $c = 1, d = 2$, the moments are algebraic over the trace field: the reconstruction uses positive square roots followed by rational operations. Thus their transcendence degrees agree. Adjoining the $\pi(2N-1)$ prime logarithms $\log p$, $p \leq 2N-1$, to the moment field recovers the $2N-1$ independent values $\mathrm{Ein}(1), \ldots, \mathrm{Ein}(2N-1)$. This proves the bound. Greedy retention of an algebraically independent subfamily skips at most $\pi(2N-1) = o(N)$ entries in each completed block; an incomplete block differs by only one entry. □

This is stronger than the single-trace dimension bound: no $N-1$ odd-moment allowance is lost. The two limits nevertheless satisfy $A_n \downarrow \gamma$ and $B_n \downarrow 0$; independence of these finite data does not establish the arithmetic nature of either limit.

28.5 Parameter tuning and the arithmetic cost that is actually proved

At fixed h, the gamma error has exponent $c(h) = 4\operatorname{arcosh}(e^{h/2})$. It obeys

$$c(h) \sim 4\sqrt{h} \quad (h \downarrow 0), \qquad c(h) = 2h + 4\log 2 + O(e^{-h}) \quad (h \to \infty).$$

Larger spacing therefore increases the limiting exponent per Hankel dimension, but also increases the largest input argument $a+(2n-2)h$. No uniform fixed-parameter asymptotic is being applied to moving h_n.

There is a rigorous algebraic information cost: any collection of constants from which all $2N-1$ nonduplicate trace entries can be obtained by algebraic operations over $K_{\exp}$ must generate a field of transcendence degree at least $2N-1-\pi(2N-1)$. A fixed finite list of auxiliary constants cannot encode this entire tower algebraically. This is an algebraic-rank statement, not a lower bound for the Diophantine heights of subsequent rational approximations.

For comparison, if independently supplied rational moments have common denominator Q and absolute value at most $C \geq 1$, their A_n, when its denominator is nonzero, is a rational number with an unreduced representation whose numerator and denominator have absolute values at most

$$n!(QC)^n, \qquad Q\,n^2(n-1)!(QC)^{n-1},$$

respectively. This follows by applying the Leibniz bound to the integer matrix QF_n and its cofactors. It is an upper bound for one rationalization procedure, with possible further gcd cancellation; it neither proves a height obstruction nor supplies the rational moments or their required error tolerance. Obtaining simultaneous height and error bounds remains the missing gamma bridge.

References and scope. For the classical variational principle, Jacobi recurrence, and kernel machinery see B. Simon, *The Christoffel–Darboux Kernel*, https://arxiv.org/abs/0806.1528. The Szegő-class comparison is described in G. Szegő, *Orthogonal Polynomials*, fourth edition,

Chapter 12, and in the NIST DLMF, https://dlmf.nist.gov/18.2#xi. The exact rate proof above is self-contained apart from the classical Chebyshev and Markov inequalities; the reconstruction and counterexample are proved in full. No named external open problem, or the arithmetic nature of gamma, is claimed to be settled.

29 Global camouflage densities and complete local denominators

Keep $J_N(x) = \min\{m \geq 1 : m\binom{x}{n} \in \mathbb{Z}[x],\ 0 \leq n \leq N\}$ and $U(x) = \{p : 1/p \in \mathbb{Z}[\binom{x}{n} : n \geq 0]\}$. Define the simultaneous camouflage primes by

$$\mathcal{C}(x) = \{p : p \notin U(x), \quad p \nmid p!/J_p(x)\}.$$

For transcendental x this is the set of all primes. For rational x it is empty, because $J_p(x) = 1$. The prime-unit description used below is prior work of Kalinov–Mandelshtam; the Frobenius interpretation is the standard local factorization theorem. No literature-wide priority is claimed for the following deductions.

Theorem 29.1 (Sharp joint density bound)**.** *Let x be algebraic of degree $d \geq 2$, and let G be the group of the normal closure, in its faithful transitive action on the d roots. Put $g = |G|$ and let $\delta(G)$ be the proportion of derangements. Then $\mathcal{C}(x)$ has natural density*

$$c(x) = 1 - \delta(G) - \frac{1}{g} \leq \frac{d-2}{d-1}.$$

Equality holds if and only if the action is sharply 2-transitive.

Proof. Outside a finite prime set, absence of a prime inverse means that Frobenius has a fixed point, while absence of a p-defect at index p means that Frobenius is not the identity. Chebotarev gives the stated density.

For the bound, let $F(s)$ be the number of fixed points of $s \in G$. The case $d = 2$ is immediate. For $d \geq 3$, Burnside's lemma gives

$$\sum_s F(s) = g, \qquad r := g^{-1} \sum_s F(s)(F(s) - 1) \geq 1,$$

where r counts the orbits on ordered distinct pairs. The identity contributes d to the first sum, so $c \leq 1 - d/g$. If $g \leq d(d-1)$ this is at most $(d-2)/(d-1)$, strictly so for $g < d(d-1)$.

A nonidentity permutation has at most $d-2$ fixed points. Hence

$$r \leq (d-2)\delta(G) + \frac{2(d-1)}{g} = (d-2)(1-c) + \frac{d}{g}.$$

For $g \geq d(d-1)$ this yields

$$c \leq \frac{d-3+d/g}{d-2} \leq \frac{d-2}{d-1},$$

strictly if $g > d(d-1)$. Equality therefore requires $g = d(d-1)$ and equality in the first fixed-point estimate. Thus every nonidentity with a fixed point fixes exactly one point. Every two-point stabilizer is trivial, and its orbit has $g = d(d-1)$ elements; the action is sharply 2-transitive. Conversely, a sharply 2-transitive action has $d(d-2)$ nonidentity elements with a fixed point, giving the stated equality. □

This is a direct application of the classical fixed-point moment method; compare the Cameron–Cohen bound discussed in Burness–Tong-Viet. For every prime q, the splitting field of $X^q - 2$ has the affine group $\mathrm{AGL}_1(\mathbb{F}_q)$, so the bound is sharp for actual algebraic numbers in infinitely many degrees. Indeed, for odd q the polynomial remains Eisenstein at a prime above 2 in $\mathbb{Q}(\zeta_q)$, which is unramified at 2. Its splitting field therefore has degree $q(q-1)$ and the full affine action.

G	d	$\delta(G)$	$1/g$	c
C_2	2	1/2	1/2	0
C_3	3	2/3	1/3	0
S_3	3	1/3	1/6	1/2
A_4	4	1/4	1/12	2/3
S_4	4	3/8	1/24	7/12

The preceding cubic camouflage construction has splitting type $(1, 2)$ at a tested prime, so its irreducible cubic has group S_3. It can imitate every prescribed finite initial observation set, but its global camouflage density is exactly $1/2$. There is no contradiction: the first visible discrepancy depends on arithmetic height. The density $c(x)$ exists for every complex x, and

$$x \notin \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad c(x) = 1.$$

This separates rational values too, unlike prime-unit density alone. It does not compute $c(\gamma)$ from real approximations.

29.1 Complete exponents at maximal-order primes

Kalinov–Mandelshtam already use the local valuation method in Section 5.3, Lemmas 5.12–5.14. Here it gives an explicit common denominator formula when the local order is maximal.

Theorem 29.2 (Exact local denominator formula)**.** *Let x be an algebraic integer with monic minimal polynomial f and $K = \mathbb{Q}(x)$. Suppose $p \nmid [\mathcal{O}_K : \mathbb{Z}[x]]$, and factor $\bar{f} = \prod_i \varphi_i^{e_i}$ over $\mathbb{F}_p$, with $f_i = \deg \varphi_i$. For all $N \geq 0$,*

$$v_p(J_N(x)) = \begin{cases} v_p(N!), & \text{some } f_i > 1, \\ 0, & \text{all } f_i = e_i = 1, \\ v_p(N!) - \lfloor N/(pe) \rfloor, & \text{all } f_i = 1,\ e = \max_i e_i \geq 2. \end{cases}$$

This includes wild ramification. If $\mathbb{Z}[x] = \mathcal{O}_K$, it gives the exact denominator at every prime.

Proof. The index assumption identifies $\mathbb{Z}_p[x]$ with the product of the full local integer rings. The numbers f_i, e_i are their residue degrees and ramification indices. If some $f_i > 1$, the reduction of x in that residue field is not in $\mathbb{F}_p$. Every $x - j$ is a unit, so the corresponding valuation of $\binom{x}{n}$, normalized by $v(p) = 1$, is $-v_p(n!)$. This gives the first case. A component with $f_i = e_i = 1$ is $\mathbb{Q}_p$, where $x \in \mathbb{Z}_p$ and all binomial values are integral.

Consider a component with $f_i = 1$ and $e_i = e \geq 2$. Let a be an integer lifting its residue root. Its integer ring is $\mathbb{Z}_p[x_i]$, whose maximal ideal is $(p, x_i - a)$: the quotient by those generators is $\mathbb{F}_p$. In the discrete valuation normalization $v_\mathfrak{p}(p) = e$, this implies $v_\mathfrak{p}(x_i - a) = 1$. Put $\pi = x_i - a$.

The common denominator through index N is unchanged by integer translation of its argument. The binomial Vandermonde formulas for the shifts a and $-a$ give inverse triangular

integer transformations. We may thus use π in this component. Since $v(\pi) = 1/e < 1$,

$$v(\pi - j) = \begin{cases} 1/e, & p \mid j, \\ 0, & p \nmid j, \end{cases} \qquad -v\binom{\pi}{n} = v_p(n!) - \frac{1}{e}\lceil n/p \rceil.$$

Within the block $pt \le n < p(t+1)$ the maximum is at pt. The successive block maxima increase by $v_p(pt) - 1/e > 0$. Writing $k = \lfloor N/p \rfloor$, the maximum through N is $v_p(N!) - k/e$. Its ceiling is $v_p(N!) - \lfloor N/(pe) \rfloor$. Taking the maximum over local components selects the largest ramification index e. □

In particular, for a root of an Eisenstein polynomial at p of degree $d \ge 2$, its local order is maximal and

$$v_p(J_N) = v_p(N!) - \lfloor N/(pd) \rfloor.$$

For $\sqrt{2}$ this gives the full exponent $v_2(N!) - \lfloor N/4 \rfloor$. For a root of $X^3 - X - 1$, whose order is maximal, the ramified contribution at 23 is $v_{23}(N!) - \lfloor N/46 \rfloor$.

Example 29.3 (Why the index hypothesis is necessary)**.** For $x = \sqrt{17}$, the order has index 2 and both local roots are in $\mathbb{Z}_2$. Thus all binomial values are integral in the maximal order, and the index gives $v_2(J_N) \le 1$. But $\binom{x}{2} = (17 - x)/2 \notin \mathbb{Z}_2[x]$, so $v_2(J_N) = 1$ for every $N \ge 2$. Applying the excluded formula only to $\bar{f} = (X + 1)^2$ would give a linearly growing answer. In general the maximal-order denominator D_N satisfies

$$v_p(D_N) \le v_p(J_N) \le v_p(D_N) + v_p([\mathcal{O}_K : \mathbb{Z}[x]]).$$

29.2 Finite detection with supplied degree and height

Write $\nu(x) = \min\{N \ge 2 : J_N(x) < N!\}$. For an algebraic integer of degree $d \ge 2$ and normal-closure absolute discriminant Δ_L,

$$\nu(x) \le d\,\Delta_L^{310}. \tag{29.1}$$

Indeed, Kadiri–Wong, Theorem 1 supplies an unramified completely split prime $p \le \Delta_L^{310}$. All conjugates of x lie in $\mathbb{Z}_p$, so $\bar{f}$ splits linearly, with maximal multiplicity at most d. The exact first-defect theorem gives a defect by $N = pd$, including possible nonmaximal-order primes. The exponent 310 is an explicit literature bound, not a claim of optimality or practical efficiency.

For completeness a bound using just monic degree d and coefficient height $H \ge 1$ is fully explicit. Put $D = d!$ and

$$\mathfrak{D}(d, H) = [dD(D-1)(1+H)]^{D(D-1)}.$$

Then $\Delta_L \le \mathfrak{D}(d, H)$, whence $\nu(x) \le d\,\mathfrak{D}(d, H)^{310}$. To see the discriminant bound, let $g = [L : \mathbb{Q}] \le D$ and $M = g(g-1)/2$. Among the vectors $c \in \{0, \dots, M\}^d$ there is one such that $\theta = \sum c_i \alpha_i$, for the roots α_i of f, has g distinct conjugates. Each equality between two embeddings is a nonzero linear equation excluding at most $(M+1)^{d-1}$ vectors; the M equations cannot cover the grid. This integral primitive element has conjugates bounded by $dM(1+H)$, so its discriminant, divisible by Δ_L, is at most $[2dM(1+H)]^{g(g-1)} \le \mathfrak{D}(d, H)$.

For nonintegral algebraic irrational parameters one can use the prime-unit statistic directly. Jordan's theorem gives a derangement conjugacy class. Kadiri–Wong supplies an unramified prime in that class with $p \le \Delta_L^{310}$. There is no $\mathbb{Q}_p$ root, hence no $\mathbb{Z}_p$ root, and therefore

$$\min U(x) \le \Delta_L^{310}.$$

If the primitive minimal polynomial has degree d and height H, with leading coefficient a, the algebraic integer ax has the same normal closure and a monic polynomial of height at most H^d. Consequently $\min U(x) \le \mathfrak{D}(d, H^d)^{310}$. For a nonintegral rational of height at most H, a prime dividing its denominator is at most H. These are finite certificate ranges when bounds and exact prime-unit observations are supplied.

No independent degree, height, or normal-closure discriminant bound for an allegedly algebraic γ has been supplied. Such bounds would be new arithmetic input; normalization of the existing Ein differential system does not provide them. Thus effective Chebotarev gives a finite certificate range for bounded algebraic candidates, not a finite decision algorithm for γ.

The absence of a height bound from the common differential system can itself be demonstrated. Every algebraic r gives an E-function $F_r = e^z(\mathrm{Ein}(z) - r)$ satisfying

$$D(D-1)\big[z(D-1)F_r\big] = 0.$$

These functions are uniformly bounded on each fixed complex compact set when r ranges in a bounded real interval. Nevertheless, $r_n = (\lfloor nc \rfloor + \sqrt{2})/n$, for a rational interior point c, gives distinct degree-two parameters in that interval for all sufficiently large n, with unbounded minimal-polynomial height. Thus the common equation, the E-function property, and single-embedding analytic bounds alone cannot provide the missing height bound. Extra uniform arithmetic data could change this conclusion and are not excluded here.

The accompanying exact-arithmetic program checks the density bound and equality condition on 29 permutation groups, and 50,128 local denominator equalities through $N = 240$ across four maximal orders, together with the excluded $\sqrt{17}$ example. These checks supplement the quantified proofs above.

30 Residue-profile moments and joint transfer stability

These are structured applications of classical finite Fourier and positive-matrix methods. They generalize the earlier parity example; no priority claim or implication for the arithmetic nature of γ is made.

30.1 Complete reconstruction for arbitrary cyclic profiles

Fix $d, K \ge 2$, put $M = dK$, and let $a_0 = b \in \mathbb{R}$, $a_{d-r} = \overline{a_r}$. For $c_n = \widehat{\mu}(n)$ prescribe

$$c_0 = 1, \qquad c_n = a_{n \bmod d} \quad (1 \le n < M), \qquad \lambda_l = \sum_{r=0}^{d-1} a_r e^{2\pi i l r/d} \in \mathbb{R}.$$

The Hermitian restriction on the target profile is part of the model.

Theorem 30.1 (Residue-profile simplex)**.** *The data admit a positive probability measure if and only if*

$$1 - b \ge 0, \qquad 1 - b + K\lambda_l \ge 0 \quad (0 \le l < d). \tag{30.1}$$

An explicit representing measure at $x = j/M$ has weights

$$w_{Kl} = \frac{1 - b + K\lambda_l}{M}, \qquad w_j = \frac{1-b}{M} \quad (K \nmid j).$$

The feasible profiles form a d-simplex. Every boundary profile has a unique representing measure; every interior profile has more than one. The minimum atomic support is

$$d(K-1)\mathbf{1}_{b<1} + \#\{l : 1 - b + K\lambda_l > 0\}.$$

Proof. Order the moment matrix by exponent residues modulo d. It becomes

$$T = (1-b)I_M + A \otimes J_K, \qquad A_{rs} = a_{r-s \bmod d}.$$

Its eigenvalues are $1-b$, with multiplicity $d(K-1)$, and $1-b+K\lambda_l$. Necessity follows from positivity. The weights above are nonnegative, sum to one because $\sum_l \lambda_l = db$, and yield the prescribed moments by Fourier inversion.

The simplex vertices are the profiles of the d point masses at l/d and the profile $b = -1/(K-1)$, $a_r = 0$ for $r \neq 0$. The latter is represented by the uniform measure on the M-th roots other than the d-th roots. The barycentric weights are respectively w_{Kl} and $(K-1)(1-b)/K$.

Use $z = e^{-2\pi i x}$ on the circle. Put $\zeta_j = e^{-2\pi i j/M}$ and $q_j(z) = \sum_{n=0}^{M-1} \zeta_j^{-n} z^n$. Its prescribed squared integral is $M^2 w_j$. At a boundary point some $w_j = 0$, forcing the support onto the zeros of q_j, namely the other M-th roots. All zero weights can be excluded in this way, and finite Fourier inversion fixes the remaining weights. In the interior choose

$$0 < \theta < \min\{1, 1-b, \min_l(1-b+K\lambda_l)\}.$$

The rescaled profile $a/(1-\theta)$ is feasible. Mix its grid measure with weight θ of either Lebesgue measure or the uniform measure on R-th roots, $R > M-1$. The resulting distinct measures represent the original data. Finally, any s-atomic representation has $\operatorname{rank} T \leq s$; the grid measure attains that rank. □

A failed inequality in (30.1) supplies a j with $w_j < 0$. Then $|q_j|^2$ is an explicit nonnegative polynomial whose proposed integral is negative. Thus the same calculation gives both reconstruction and a finite dual rejection certificate. Feasibility costs one length-d Fourier transform and $d+1$ comparisons, rather than a general $M \times M$ positivity test.

Theorem 30.2 (Exact reduction of noisy moments)**.** *Assume $\varepsilon_r \geq 0$ and $\varepsilon_{d-r} = \varepsilon_r$. There is a probability measure such that*

$$|c_n - a_{n \bmod d}| \leq \varepsilon_{n \bmod d} \quad (1 \leq n < M)$$

if and only if there is a Hermitian profile a' satisfying (30.1) and $|a'_r - a_r| \leq \varepsilon_r$ for every r.

Proof. For necessity average the feasible moment matrix under arbitrary permutations within its d residue classes and under cyclic permutation of the classes. The result is positive semidefinite, has diagonal one, and has the form $(1-b')I_M + A' \otimes J_K$ with a Hermitian residue profile. Each entry is an average of entries in the same target disk; for negative frequency differences the Hermitian target and symmetric radius assumptions give the same conclusion. Convexity of the disks preserves every error bound. Theorem 30.1 now applies. Sufficiency follows from its explicit grid measure. □

This is a convex feasibility problem in d real dimensions: a second-order cone problem for complex disks, or a linear program for real symmetric targets after averaging with complex conjugation. It is exact as a mathematical reduction; numerical boundary decisions still require certification. For completeness, let C be the feasible simplex and use $\langle u, a\rangle = \sum_r \Re(\overline{u_r} a_r)$. Its support function is

$$h_C(u) = \max\left\{ -\frac{u_0}{K-1},\ \max_l \sum_r \Re(\overline{u_r} e^{-2\pi i l r/d}) \right\}.$$

Convex separation gives the equivalent dual criterion

$$\langle u, a\rangle \le h_C(u) + \sum_r \varepsilon_r |u_r| \quad \text{for every Hermitian } u.$$

A violating u certifies infeasibility; a feasible profile constructs a measure. With real algebraic inputs, finite semialgebraic decision methods can resolve even boundary cases.

30.2 The exact joint stability constant for two multipliers

For $N \ge 1$ and distinct primes p, q define

$$(P_p f)(x) = \frac{1}{p}\sum_{j=0}^{p-1} f((x+j)/p), \qquad \widehat{P_p f}(n) = \widehat{f}(pn).$$

Let $C_N(p,q)$ be the best constant, over trigonometric polynomials of degree at most N, in

$$\|f - \widehat{f}(0)\|_2 \le C_N(p,q)\big(\|f - P_p f\|_2^2 + \|f - P_q f\|_2^2\big)^{1/2}. \tag{30.2}$$

Put $D_N = \{(a,b) \ge 0 : p^a q^b \le N\}$ and let H_N be the matrix of the energy

$$\sum_{(a,b)\in D_N} \big(|u_{a,b} - u_{a+1,b}|^2 + |u_{a,b} - u_{a,b+1}|^2\big),$$

where u is extended by zero outside D_N. Its diagonal entry at (a,b) is $2 + \mathbf{1}_{a>0} + \mathbf{1}_{b>0}$, and neighboring entries are -1.

Theorem 30.3 (Joint spectral reduction and optimal order)**.** *One has $C_N(p,q) = \lambda_{\min}(H_N)^{-1/2}$, with*

$$\dim H_N = \frac{(\log N)^2}{2\log p \log q} + O_{p,q}(\log N) \quad (N \longrightarrow \infty).$$

Set $L_p = 1 + \lfloor \log_p N \rfloor$, $L_q = 1 + \lfloor \log_q N \rfloor$, and $L = 1 + \lfloor \log_{pq} N \rfloor$. Then

$$\frac{1}{2\sqrt{2}\sin(\pi/(4L+2))} \le C_N(p,q) \le \frac{1}{2\sqrt{\sin^2(\pi/(4L_p+2)) + \sin^2(\pi/(4L_q+2))}}.$$

In particular $C_N(p,q) = \Theta_{p,q}(\log N)$. The exact optimum and the lower-bound examples can be realized by nonnegative real probability densities.

Proof. Decompose every nonzero frequency uniquely as $rp^a q^b$, with $p \nmid r$ and $q \nmid r$. Parseval splits the joint defect into the energies on $D_{N/|r|}$. These are lower ideals; zero extension from $D_{N/|r|}$ into D_N preserves energy. Hence their least Rayleigh quotients are no smaller than that on D_N. The orbit $r = 1$ attains the minimum. The matrix is positive definite, since zero energy propagates a constant along the positive coordinate edges to the zero exterior. This proves the spectral formula. The dimension estimate is the elementary lattice-point count in $a\log p + b\log q \le \log N$.

On a one-prime chain of length s, the difference matrix $I - S$ has least squared singular value $4\sin^2(\pi/(4s+2))$. Indeed the vector $u_j = \cos((j+1/2)\pi/(2s+1))$ solves the second-difference equation with first endpoint $u_{-1} = u_0$ and last endpoint $u_s = 0$. Applying this to each p-chain and each q-chain and adding the resulting inequalities proves the upper bound. For the

lower bound, the square $0 \le a, b < L$ lies inside D_N. The tensor product of the two length-L minimizing vectors, extended by zero, has energy-to-norm ratio $8\sin^2(\pi/(4L+2))$.

A real minimizing vector supplies a cosine polynomial g on its Fourier orbit. For sufficiently small $t \neq 0$, $1 + tg$ is a nonnegative probability density, with the same homogeneous norm ratio. This also applies to the square test vector. □

Thus two maps improve constants but cannot give a uniform or $o(\log N)$ homogeneous Lipschitz estimate of the form (30.2), even for positive densities. This is a bandlimited norm statement. It does not settle an unrestricted invariant-measure conjecture or provide the missing arithmetic bridge for γ.

31 Moving-degree fresh-prime removal: an explicit arithmetic budget

31.1 Scope

This note addresses polynomial preprocessing of

$$E = e^z, \qquad F_r = e^z(\mathrm{Ein}(z) - r) = \sum_{n\ge 0}(H_n - r)\frac{z^n}{n!},$$

where r is algebraic. Write $[G]_n = n![z^n]G$. The preprocessing polynomial is allowed to change with the prefix length, and both its degree and coefficient height may grow.

The conclusion concerns **removing prime denominators from normalized coefficients**, not merely obtaining an E-function. In fact, F_r is already an E-function for algebraic r; its harmonic denominators have exponential least-common-multiple growth. A lower bound for the cost of making a finite prefix integral is an arithmetic budget, not by itself an obstruction to every E-function argument or a proof about gamma.

The functional algebraic independence of E, F_r over $\mathbb{C}(z)$ also has a short direct proof. As $z \to +\infty$,

$$F_r(z) = e^z(\log z + \gamma - r) + O(1/z).$$

In a nonzero polynomial relation, collect the largest total (E, F_r)-degree s. Its contribution is e^{sz} times a nonzero polynomial in z and $\log z + \gamma - r$; all lower-degree contributions, and errors from the displayed expansion, are exponentially smaller. A nonzero polynomial in $z, \log z + c$ has a leading power of z times a nonzero polynomial in $\log z$, so it cannot be exponentially small. This excludes the relation, after clearing any rational z-denominators. It uses no arithmetic assumption on γ. The ingredients below are an explicit uniform zero estimate, an integer jet certificate independent of r, and a degree-height bound valid for changing polynomials.

31.2 The fresh-prime congruence is uniform in polynomial degree

Let $P \in \mathbb{Z}[z, T, Y]$, put $G = P(z, E, F_r)$, and let $\mathfrak{p}$ lie over a rational prime p at which r is integral. For **every** $0 \le m < p$,

$$\boxed{p[G]_{p+m} \equiv [EP_Y(z, E, F_r)]_m \pmod{\mathfrak{p}}.} \tag{MD.1}$$

No bound on the degree of P is needed for this identity. Its right side may vanish because the degree is large, as in $P = Y^p$, but that is a detected arithmetic possibility rather than a failure of the identity.

Proof. It is enough to expand $z^a E^b F_r^c$. Since $p+m<2p$, at most one factor F_r can receive an index at least p in its exponential-generating-series convolution. All terms with only smaller indices are $\mathfrak{p}$-integral and disappear after multiplication by p. For the unique large index $p+k$,

$$p(H_{p+k}-r)\equiv 1 \pmod{\mathfrak{p}}.$$

Lucas reduction removes the leading base-p digit from the multinomial coefficient. There are c choices of the large factor, yielding $E\,cF_r^{c-1}$. When the factor z^a is restored, its falling factorial reduces to $m^{\underline{a}}$ for $a\le m$. If $a>m$, it either contributes a factor p or the remaining total degree is below p; both sides of (MD.1) are zero modulo $\mathfrak{p}$. This also covers $a\ge p$ and $c\ge p$. □

31.3 An explicit zero estimate without a Padé-perfectness assumption

Let $D\ge 1, A\ge 0$, and set

$$L=\frac{(D+1)(D+2)}{2},\qquad M_*=LA+\frac{L(L-1)}{2}.$$

Theorem 31.1 (Uniform polynomial zero estimate)**.** *For every nonzero polynomial $Q(z,T,Y)$ of z-degree at most A and total (T,Y)-degree at most D,*

$$\boxed{\operatorname{ord}_{z=0} Q(z,E,F_r)\le M_*.} \tag{MD.2}$$

The bound is independent of r, of the coefficient height, and of the chosen polynomial.

Proof. Let V be the column of the L monomials $E^bF_r^c$, $b+c\le D$, and let $\theta=z\,d/dz$. Their equations are

$$\theta(E^bF_r^c)=z(b+c)E^bF_r^c+c(E^{b+1}F_r^{c-1}-E^bF_r^{c-1}).$$

Thus

$$\theta V=(B_0+zB_1)V,$$

where B_0 is nilpotent: each of its nonzero transitions strictly lowers the exponent c.

Write $G=Q(z,E,F_r)=\ell_0(z)V$, with $\deg\ell_0\le A$, and recursively set

$$\ell_{j+1}=\theta\ell_j+\ell_j(B_0+zB_1).$$

Then $\ell_jV=\theta^jG$ and $\deg\ell_j\le A+j$. Let $h\le L$ be the first index for which $\ell_0,\dots,\ell_h$ are dependent over $\mathbb{C}(z)$. The first h rows are independent. Their rational span W is connection-stable: the first dependence puts the connection of ℓ_{h-1} in that span, and the other rows map to their successors; the Leibniz rule handles rational coefficients.

Over the discrete valuation ring $R=\mathbb{C}[[z]]$, take the saturation

$$\mathcal{S}=(W\otimes_{\mathbb{C}(z)}\mathbb{C}((z)))\cap R^L.$$

It is free of rank h, is a direct summand of R^L, and is stable under the row connection. Reduction modulo z embeds it into the ambient residue space; the induced residue is a restriction of the nilpotent matrix B_0, and is therefore nilpotent.

Choose a basis matrix S of $\mathcal{S}$, and put $y=SV$. It is a formal power-series solution of $\theta y=C(z)y$, with $C(z)\in\operatorname{Mat}_h(R)$ and $C(0)$ nilpotent. It is nonzero because $G\neq 0$. Here $G\neq 0$ follows from the previously established algebraic independence of E,F_r. A nonzero solution y must have at least one unit coordinate: if its minimum order were $m>0$, comparison of leading terms would make m an eigenvalue of the nilpotent matrix $C(0)$.

Let s be the index valuation of the row module generated by $\ell_0, \dots, \ell_{h-1}$ inside its saturation. By Smith normal form,

$$s = \min\{\operatorname{ord}_0 \Delta : \Delta \text{ a nonzero maximal minor of } (\ell_j)_{j<h}\}.$$

The degree of every such minor is at most

$$\sum_{j=0}^{h-1}(A+j) = hA + h(h-1)/2,$$

so s is no larger. The adjugate formula also gives $z^s\mathcal{S}$ inside the original row module. If G has order m, every $\theta^j G$ has order at least m. Therefore every coordinate of $z^s y$ has order at least m. Since one coordinate of y is a unit, $m \le s$. Finally $h \le L$ proves (MD.2). □

This argument uses the nilpotent regular-singular residue, not an assertion that the first as-many-as-dimension jets are nonsingular. That stronger perfectness assertion is neither needed nor claimed.

The same proof is reusable beyond these functions: if an analytic vector V satisfies $\theta V = (B_0 + zB_1)V$ with nilpotent B_0, then every **nonzero evaluated** expression $\ell_0(z)V$, $\deg \ell_0 \le A$, has order at most $LA + L(L-1)/2$. Algebraic independence is used only to ensure nonvanishing for all nonzero polynomial inputs in the present application. No priority claim relative to general multiplicity-estimate literature is made. For established effective zero estimates and their Shidlovskii–Bertrand–Chirskii–Yebbou background, see Bostan–Rivoal–Salvy, §1, "Minimal differential equations", equation (4). This note gives a self-contained special-case argument; it does not claim to have invented the zero-estimate method or established priority for the combined arithmetic budget.

31.4 A finite integer certificate independent of the algebraic shift

Use the full basis

$$\mathcal{B}_{A,D} = \{z^a T^b Y^c : 0 \le a \le A,\ b + c \le D\}, \qquad B = (A+1)L.$$

Theorem 31.1 makes evaluation into jets of orders $0, \dots, M_*$ injective. Select any B independent rows with indices

$$0 \le m_1 < \cdots < m_B \le M_*; \qquad M = \max_i m_i.$$

For each basis monomial form its normalized coefficient at order m_i, multiply that row by $m_i!$, and compute the determinant at $r = 0$. Denote this nonzero integer by $\Delta_{A,D}$.

Two facts make this a reusable certificate.

Integrality. The ordinary coefficients of F_0 lie in $(n!)^{-2}\mathbb{Z}$, because H_n has denominator dividing $n!$. The same is true of E, and this coefficient class is closed under products: the factorial ratios produce squares of integer multinomial coefficients. Hence a normalized coefficient of any integral polynomial in z, E, F_0 has denominator dividing $n!$. The row scaling therefore gives an integer matrix.

Shift invariance. The substitution $F_r = F_0 - rE$ acts on the complete monomial basis by a triangular matrix with determinant one, separately at each z-degree. Thus exactly the same row determinant is $\Delta_{A,D}$ for every algebraic r. When r is integral at $\mathfrak{p}$, this change of basis and its inverse are integral there.

An explicit height bound follows from Hadamard. Put $H_M = \sum_{j=1}^{M} 1/j$. For a basis monomial, its normalized coefficient through order M has absolute value at most

$$M^A D^M H_M^D.$$

Indeed the harmonic factors in the exponential convolution are at most H_M, the sum of multinomial coefficients is at most D^M, and multiplication by z^a contributes at most M^A. Therefore

$$\boxed{\log|\Delta_{A,D}| \le B\left\{\tfrac{1}{2}\log B + \log(M!) + A\log M + M\log D + D\log H_M\right\}.} \tag{MD.3}$$

Here the selected M can be used; replacing it by M_* gives a closed-form bound. In particular,

$$\log|\Delta_{A,D}| = O\big((A+1)D^4(A+D^2)\log(A+D+2)\big). \tag{MD.4}$$

The implicit constant is absolute. Keeping the exact determinant can be much stronger than this deliberately coarse estimate.

31.5 A degree-height budget for changing polynomials

For $P \in \mathbb{Z}[z,T,Y]$ retaining Y, define

$$C_Y(P) = \gcd\{c\, p_{abc} : c \ge 1\} > 0,$$

where p_{abc} are its monomial coefficients. Let $H(P)$ be their maximum absolute value. Then $C_Y(P) \le DH(P)$.

Theorem 31.2 (Finite-prefix prime-removal budget)**.** *Assume* $\deg_z P \le A$, $\deg_{T,Y} P \le D$, *and let* $G = P(z,E,F_r)$. *Let* $\mathcal{P}$ *be any set of rational primes satisfying*

$$M < p \le T - M,$$

such that for each $p \in \mathcal{P}$ *there is a prime* $\mathfrak{p} \mid p$ *at which* r *and every* $[G]_n$, $0 \le n \le T$, *are integral. Then*

$$\boxed{\prod_{p\in\mathcal{P}} p \;\mid\; \Delta_{A,D}\, C_Y(P).} \tag{MD.5}$$

Consequently,

$$\boxed{\sum_{p\in\mathcal{P}} \log p \le \log|\Delta_{A,D}| + \log D + \log H(P).} \tag{MD.6}$$

Proof. Put $Q = TP_Y$. It has the same degree caps as P; its coefficient vector in $\mathcal{B}_{A,D}$ is integral and has content $C_Y(P)$. For $m = m_i$, congruence (MD.1) and the integral prefix give $[Q(z,E,F_r)]_{m_i} = 0$ modulo $\mathfrak{p}$, since $p + m_i \le T$ and $m_i < p$. Multiplying by $m_i!$, the integer-scaled jet matrix at shift r kills the vector of Q modulo $\mathfrak{p}$. Its determinant is $\Delta_{A,D}$. The adjugate gives

$$\Delta_{A,D}\, u_j \equiv 0 \pmod{\mathfrak{p}}$$

for every integer coefficient u_j of Q. These are rational integers; hence p divides every $\Delta_{A,D}u_j$, and so divides $\Delta_{A,D}C_Y(P)$. Multiply over the distinct primes. □

Only prime factors of Δ exceeding M can matter in (MD.5); stripping its small-prime factors sharpens the numerical budget. The displayed height estimate simply uses the full determinant to avoid factoring it.

31.5.1 Combined denominator-height budget

Let $K = \mathbb{Q}(r)$, and let d_T be the least positive rational integer such that

$$d_T[G]_n \in \mathcal{O}_K \qquad (0 \le n \le T).$$

It exists because the prefix is finite and consists of algebraic numbers. Let S_r be the finite set of rational primes at which r is not integral at at least one prime of K. Then

$$\boxed{\prod_{\substack{M<p\le T-M\\ p\notin S_r}} p \;\mid\; d_T\,\Delta_{A,D}\,C_Y(P).} \tag{MD.7}$$

Indeed, if $p \nmid d_T$, the full prefix is integral at every prime above p, so Theorem 31.2 applies. If $p \mid d_T$, the prime is already accounted for. Consequently, with $\vartheta(x) = \sum_{p\le x} \log p$,

$$\boxed{\log(d_T H(P)) \ge \vartheta(T-M) - \vartheta(M) - \sum_{p\in S_r} \log p - \log|\Delta_{A,D}| - \log D.} \tag{MD.8}$$

Thus the bound measures the combined cost of residual denominators and preprocessing coefficients, rather than assuming that preprocessing removes every denominator.

31.5.2 Corollary (slowly growing nonlinear preprocessing still costs $e^{T-o(T)}$)

Fix algebraic r. Suppose P_T retains Y and has degree caps A_T, D_T. Define d_T for its normalized prefix as above. If

$$(A_T+1)D_T^4(A_T+D_T^2)\log(A_T+D_T+2) = o(T), \tag{MD.9}$$

then

$$\boxed{\log\big(d_T H(P_T)\big) \ge T - o(T).} \tag{MD.10}$$

In particular, if preprocessing makes the entire prefix integral, then $\log H(P_T) \ge T - o(T)$. For $A_T = 0$, the sufficient growth condition simplifies to

$$D_T^6 \log(D_T+2) = o(T).$$

Proof. Condition (MD.9) implies both $M = o(T)$ and $\log|\Delta| = o(T)$. Apply (MD.8) and use the prime number theorem in the form $\vartheta(T-M) - \vartheta(M) = T - o(T)$. Also $\log D = o(T)$. □

The leading constant one is sharp in either allocation of cost. For algebraic integral r, $P_T(Y) = Y$ has height one and $d_T = \operatorname{lcm}(1,\dots,T)$. The latter equality follows by subtracting successive normalized coefficients, whose difference is $1/n$. Alternatively,

$$P_T(Y) = \operatorname{lcm}(1,\dots,T)Y$$

has integral normalized coefficients through order T, and its logarithmic height is $\psi(T) = T + o(T)$. This is a sharp asymptotic cost statement within the specified regime.

31.5.3 Degree escapes are accounted for, not assumed absent

If $P = Y^L$, then $C_Y(P) = L$. Thus a prime dividing L can make the first fresh-prime derivative vanish even at height one; removing many such primes requires their product to divide L, so it costs degree. More generally, $P_Y = 0$ modulo p means P mod p is a polynomial in Y^p. Formula (MD.5) captures this mechanism through derivative content. The theorem does not pretend to exclude all fast-degree constructions outside (MD.9).

31.6 The sharper pure-polynomial subfamily

For comparison, when $P \in \mathbb{Z}[Y]$ has degree d, a direct jet argument is stronger. Exclude primes at which r is nonintegral or $1-r$ is not a unit; assume $r \neq 1$. If $p > d$ and $p + d - 1 \leq T$, an integral prefix gives

$$[EP'(F_r)]_m \equiv 0 \quad (0 \leq m < d).$$

The germ $F_r + r$ has linear coefficient $1 - r$, a unit. Formal inversion modulo p, together with the unit factor E, forces the polynomial P' of degree at most $d-1$ to be zero modulo p. Since $p > d$, every nonconstant coefficient of P is divisible by p. Hence

$$\log H(P) \geq \vartheta(T - d + 1) - \vartheta(d) - O_r(1).$$

For $d = o(T)$, this again gives $\log H \geq T - o(T)$, with a much wider allowed degree regime. The multivariate theorem above pays for separating the additional z, E directions and does not claim the same optimal degree threshold.

The exceptional shift $r = 1$ can also be covered: $F_1 + 1 = z^2/4 + O(z^3)$. Away from $p = 2$, a nonzero polynomial $P' \bmod p$ of degree at most $d-1$ has composition order at most $2(d-1)$. Test those jets instead, requiring $p > \max\{d, 2(d-1), 2\}$ and $p + 2(d-1) \leq T$. This retains the same $d = o(T)$ asymptotic conclusion.

32 A fixed-shift obstruction for the Laguerre approximants

We use the Hessami Pilehrood $k!$ family, which is distinct from Aptekarev's original $(n+k)!$ family:

$$q_n = \sum_{k=0}^{n} \binom{n}{k}^2 k! = n! L_n(-1), \qquad p_n = \sum_{k=0}^{n} \binom{n}{k}^2 k!(2H_{n-k} - H_k), \qquad E_n = p_n - \gamma q_n.$$

The source is Kh. Hessami Pilehrood and T. Hessami Pilehrood, *On a continued fraction expansion for Euler's constant*, arXiv:1010.1420, equations (21)–(23), Lemmas 2–3, and Theorem 1. Writing $Su_n = u_{n+1}$ and

$$R = S^2 - 2(n+2)S + (n+1)^2,$$

their formulas give $Rq = 0$, $Rp = RE = -n/(n+2)$, and

$$E_n = n! n^{-1/4} e^{-2\sqrt{n}} \left(\sqrt{\pi/e} + O(n^{-1/2}) \right). \tag{32.1}$$

In particular E_n is eventually positive and grows faster than any power of n.

Theorem 32.1 (Fixed rational-function shifts)**.** *Let $T \in \mathbb{Q}(n)[S]$ be fixed. If Tq is not the zero sequence, then for some $c \neq 0$ and $\rho \in \frac{1}{2}\mathbb{Z}$,*

$$(TE)_n = c n^{\rho} E_n (1 + o(1)).$$

Consequently $|(TE)_n| \to \infty$. Adding a fixed rational function of n does not change this conclusion.

Proof. We first prove

$$\frac{E_{n+1}}{E_n} = n - \sqrt{n} + o(\sqrt{n}). \tag{32.2}$$

The positive Whittaker integral in the cited paper has the form

$$I_n = \frac{n!}{e}\int_0^\infty \frac{e^{-t}}{1+t}\left(\frac{t}{1+t}\right)^n dt, \qquad I_n - E_n = O((n+1)^{-2}).$$

The integral identity follows by substituting the standard integral for $U(n+1,1,1)$ into its equation (28); the error bound is its Lemma 2. Set $J_n = I_n/n!$. The positive moment representation implies $J_n^2 \le J_{n-1}J_{n+1}$ by Cauchy–Schwarz. Therefore $f_n = \log J_n$ is convex. Equation (32.1) gives

$$f_n = -2\sqrt{n} - \tfrac{1}{4}\log n + \tfrac{1}{2}\log(\pi/e) + O(n^{-1/2}).$$

For $h = \lfloor n^{3/4} \rfloor$, the backward and forward h-step secant slopes bound $f_{n+1} - f_n$. Substitution yields

$$f_{n+1} - f_n = -n^{-1/2} + O(n^{-3/4}),$$

and the negligible correction $I_n - E_n$ gives (32.2).

Right division by the monic operator R in the shift Ore ring gives

$$T = UR + A(n)S + B(n), \qquad A, B \in \mathbb{Q}(n).$$

The hypothesis and $Rq = 0$ imply $(A, B) \neq (0, 0)$. Since RE is a rational function,

$$(TE)_n = A(n)E_{n+1} + B(n)E_n + C(n), \qquad C \in \mathbb{Q}(n).$$

If $A = 0$, the conclusion follows from the leading term of B. Otherwise let $a = \deg A$ and $B_0 = B + nA$. Equation (32.2) gives

$$A(n)\frac{E_{n+1}}{E_n} + B(n) = B_0(n) - A(n)\sqrt{n}\,(1 + o(1)).$$

The nonzero rational function B_0, if present, has integral degree; $A\sqrt{n}$ has half-integral degree $a + 1/2$. Their leading terms cannot cancel. The rational function C is negligible relative to the factorial growth of E_n. This proves the theorem. □

The scope is essential: the theorem does not bound the common divisor after integerization, and it does not cover a shift order increasing with n. The exact identity $(n+2)RE = -n$ is not a small gamma form, because simultaneously $(n+2)Rq = 0$.

The primitive arithmetic budget. Write $p_n = A_n/b_n$ in lowest terms, $b_n > 0$, and set $g_n = \gcd(A_n, q_n)$. Then

$$P_n = A_n/g_n, \qquad Q_n = b_n q_n/g_n$$

are coprime integers and

$$P_n - \gamma Q_n = \frac{b_n}{g_n}E_n.$$

Hence their nonzero forms tend to zero along an infinite subsequence if and only if, along that subsequence,

$$\log g_n - \log b_n - \log(n!) + 2\sqrt{n} + \tfrac{1}{4}\log n \longrightarrow +\infty. \tag{32.3}$$

This identity locates the missing arithmetic saving; it does not establish it.

Proposition 32.2 (Elementary congruences)**.** *For every $n \ge 1$ and prime ℓ,*

$$q_n \equiv 1 \pmod{n}, \qquad q_n \equiv q_{n \bmod \ell} \pmod{\ell}.$$

If $\ell \le n < 2\ell$, then in $\mathbb{Z}_{(\ell)}$,

$$\ell p_n \equiv 2q_{n-\ell} \equiv 2q_n \pmod{\ell}.$$

Proof. For $k \geq 1$, write $\binom{n}{k}^2 k! = \binom{n}{k} n(n-1)\cdots(n-k+1)$ to obtain divisibility by n. For reduction modulo ℓ, discard $k \geq \ell$ and apply Lucas congruence to the remaining binomial coefficients. For the last claim put $n = \ell + r$, $0 \leq r < \ell$. In $\ell(2H_{n-k} - H_k)$, the residue is 2 for $k \leq r$. For $r < k < \ell$, both harmonic numbers are ℓ-integral and the binomial coefficient also vanishes modulo ℓ. For $k \geq \ell$, the factorial supplies a factor ℓ while the scaled harmonic expression is ℓ-integral. Only $k \leq r$ contributes, giving $2q_r$. □

For odd ℓ in this interval, $q_n \not\equiv 0 \pmod{\ell}$ forces the reduced denominator b_n to contain ℓ exactly once. Uniform nonvanishing cannot simply be assumed: at $n = \ell - 1$,

$$q_{\ell-1} \equiv \sum_{k=0}^{\ell-1} k! =! \ell \pmod{\ell}.$$

Its nonvanishing for every odd prime is Kurepa's left-factorial hypothesis; see R. Meštrović, *Variations of Kurepa's left factorial hypothesis*, arXiv:1312.7037. This is not an equivalence between that hypothesis and the gamma problem.

Another tempting shortcut is actually false. Exact recurrence arithmetic gives

$$\boxed{g_{224} = 233 > 224.}$$

For a compact certificate of the common prime,

$$q_{224} \equiv 208 \cdot 233 \pmod{233^2}, \qquad A_{224} \equiv 31 \cdot 233 \pmod{233^2}, \qquad b_{224} \equiv 16 \pmod{233}.$$

The full integer gcd is computed by the accompanying exact script. Thus neither "all common primes are at most n" nor $g_n \mid n!$ is valid. The same script finds only 27 nontrivial g_n through $n = 3000$, with maximum 233; these finite observations prove no asymptotic gcd bound.

Moving order remains an arithmetic problem. Pascal's identity gives, for every $n, m \geq 0$,

$$\Delta^m L_n(-1) = \sum_{j=0}^{n} \frac{\binom{n}{j}}{(j+m)!} = \frac{n!}{(n+m)!} L_n^{(m)}(-1),$$

so $(n+m)!\Delta^m(q_n/n!)$ is integral. This is an exact identity, not a proof that a substantially smaller denominator suffices. No factorial-size cancellation for moving order, and no complete primitive-gcd obstruction, is established here.

33 Positive Binet kernels: a global floor and extremal zeros

33.1 Definitions

Write

$$Q_M(u) = \prod_{m=2}^{M} (u + m^2), \qquad F(u) = \sum_{m=2}^{M} \frac{R_m}{u + m^2} = \frac{P(u)}{Q_M(u)}.$$

The normalization $\sum_m R_m = 1$ makes P monic of degree $d = M - 2$. Positivity of F on $[0, \infty)$ is exactly positivity of P there. Log balance imposes

$$\sum_m R_m v_p(m) = 0 \quad \text{for every prime } p \leq M.$$

The Binet energy is

$$E(F) = \int_0^\infty \frac{F(u)}{e^{2\pi\sqrt{u}} - 1}\, du.$$

For balanced kernels, the supplied Binet identity gives $E(F) = \gamma - \sum_m R_m h_m$, with $h_m = H_{m-1} + 1/(2m)$. The feasible set is the nonempty compact set established by the preceding positive-kernel construction for $M \geq 4$; its minimum is $E_M^* > 0$.

Lemma 33.1 (Reused nonemptiness and compactness input)**.** *The normalized log-balanced nonnegative kernel set is nonempty and compact for every $M \geq 4$.*

Proof. Take $R_2 = 2$, $R_4 = -1$ and all other residues zero. Then $\sum R_m = 1$, $\sum R_m \log m = 0$, and $F(u) = 2/(u+4) - 1/(u+16) > 0$ for $u \geq 0$. The feasible residue set is a closed convex subset of a finite-dimensional space. A nonzero recession direction would give a nonnegative kernel F_R with $\sum R_m = 0$ and $\sum R_m \log m = 0$. It is integrable on the half-line because $F_R(u) = O(u^{-2})$, and

$$\int_0^\infty F_R(u)\, du = -2 \sum_m R_m \log m = 0.$$

Nonnegativity then forces $F_R = 0$, hence every residue is zero, a contradiction. A closed finite-dimensional convex set with no nonzero recession direction is bounded. This proves compactness. □

33.2 A full-cone monic polynomial norm lemma

If P is any monic real polynomial of degree d nonnegative on $[0, \infty)$, then for every $T > 0$,

$$\int_0^T P(u)\, du \geq \frac{T^{d+1}}{(d+1)\binom{d}{\lfloor d/2 \rfloor}^2} \geq \frac{T^{d+1}}{(d+1)4^d}. \tag{PK.1}$$

Proof. Every such P admits a half-line sum-of-squares representation $P = A^2 + uB^2$. This can be proved directly by factoring: a negative real factor $u + c$ is $(\sqrt{c})^2 + u$; a conjugate pair $(u-\alpha)^2 + \beta^2$ equals $(u-r)^2 + 2(r-\alpha)u$ with $r = \sqrt{\alpha^2 + \beta^2}$; positive roots occur to even order; and representations are preserved by

$$(A^2 + uB^2)(C^2 + uD^2) = (AC - uBD)^2 + u(AD + BC)^2.$$

If $d = 2k$, A has degree k with leading coefficient ± 1, while $\deg B \leq k - 1$. Therefore the integral is bounded below by the minimum squared norm of a monic degree-k polynomial for Lebesgue measure on $[0, T]$. If $d = 2k + 1$, B is monic up to sign of degree k and the lower bound is the corresponding monic norm for weight u.

For $\epsilon = 0$ or 1, shifted Jacobi orthogonality gives the monic degree-k norm

$$\min_{A \text{ monic}, \deg A = k} \int_0^T u^\epsilon A(u)^2 du = \frac{T^{2k+\epsilon+1}}{(2k+\epsilon+1)\binom{2k+\epsilon}{k}^2}.$$

For completeness the normalized orthogonal polynomial on $[0, 1]$ is supplied by

$$R_k(x) = \frac{x^{-\epsilon}}{k!} \frac{d^k}{dx^k} \{x^{k+\epsilon}(x-1)^k\}.$$

Integration by parts proves orthogonality to every lower-degree polynomial. Its leading coefficient is $\binom{2k+\epsilon}{k}$; its squared norm is $1/(2k+\epsilon+1)$, by one further beta-integral computation. Scaling gives the displayed formula. This proves (PK.1). The argument allows arbitrary signs of the coefficients of P.

33.3 The global exponential floor

For $y > 0$ define

$$J(y) = \int_0^1 \log(y^2 + t^2)dt = \log(1 + y^2) - 2 + 2y \arctan(1/y),$$

$$C(y) = 2\pi y + J(y) + \log 4 - 2 \log y.$$

Theorem. For every $M \geq 4$ and every normalized nonnegative kernel above,

$$E(F) \geq \frac{16y^2}{(M-1)(1+y^2)} \exp(-C(y)M). \tag{PK.2}$$

This statement does not even require log balance, so it applies uniformly to the full balanced cone.

Proof. Put $T = y^2M^2$. On $0 \leq u \leq T$,

$$\frac{1}{e^{2\pi\sqrt{u}} - 1} > e^{-2\pi yM}, \qquad Q_M(u) \leq Q_M(T).$$

Thus $E(F) \geq e^{-2\pi yM} Q_M(T)^{-1} \int_0^T P(u)\, du$. Apply (PK.1). To estimate the denominator, $f(t) = \log(y^2 + t^2)$ is increasing, and a right-Riemann-sum bound gives

$$\sum_{m=2}^{M} f(m/M) \leq M \int_0^1 f(t)dt + f(1) - 2f(0).$$

Consequently

$$Q_M(T) \leq M^{2(M-1)} e^{MJ(y)} \frac{1+y^2}{y^4}.$$

Since $d = M - 2$, substitution and cancellation of all powers of M yields exactly (PK.2).

Differentiation gives

$$C'(y) = 2\pi + 2 \arctan(1/y) - 2/y.$$

The unique minimizer is $y = 1/v$, where $v > 0$ is the unique solution of

$$v - \arctan v = \pi.$$

At that point $C_{\min} = \log(4(1 + v^2))$, so

$$\boxed{E_M^* \geq \frac{16}{(1+v^2)(M-1)} [4(1+v^2)]^{-M}.} \tag{PK.3}$$

The numerical values, only explanatory evaluations of the exact formulas, are

$$\begin{aligned} v &= 4.493409457909064\ldots, \\ C_{\min} &= 4.439858114896194\ldots, \\ \frac{16}{1+v^2} &= 0.755047187612980\ldots. \end{aligned}$$

Thus $\limsup_{M\to\infty}(-\log E_M^*)/M \le 4.439858114896194\ldots$. The bound is not asserted sharp. A stronger lower bound matching the lcm exponent, or a matching asymptotic of E_M^*, has not been obtained.

33.4 Extremal zero-count theorem for the full log-balanced cone

Theorem. Every extreme point of the feasible cone slice has a numerator P whose zeros on [0,infinity), counted with multiplicity, total at least

$$M - \pi(M) - 2. \tag{PK.4}$$

There exists an energy minimizer satisfying (PK.4).

Equivalently an extreme minimizing numerator can be written

$$P(u) = u^{\epsilon}A(u)^2W(u), \qquad \epsilon \in \{0,1\}, \quad \deg W \le \pi(M), \quad W(u) > 0 \ (u \ge 0),$$

where all zeros of A lie on $[0,\infty)$. Thus the part not accounted for by a square and a possible factor u has degree at most pi(M), rather than M. This is a necessary structure at some global minimizer; it does not determine the zeros or prove an algorithm finds that minimizer.

Proof. Partial fractions identify R with all polynomials of degree at most $d = M-2$. The affine constraints have rank $r = \pi(M)+1$: if $a_0 + \sum_p a_p v_p(m) = 0$ for every m, inserting $m = p$ gives $a_p = -a_0$, while $m = 4$ gives $a_0 = 0$. Thus every a_p is zero.

Factor $P = GW$, where G consists of all roots on $[0,\infty)$, including the full multiplicity at zero; its positive roots have even multiplicity. Put $s = \deg G$. Then $G \ge 0$ on $[0,\infty)$, and W is strictly positive there and has degree $d-s$. For every polynomial H of degree at most $d-s$, $|H|/W$ is bounded on $[0,\infty)$. Hence for sufficiently small $\eta > 0$, both $G(W+\eta H)$ and $G(W-\eta H)$ remain nonnegative.

The space of perturbations GH has dimension $d-s+1$. If it exceeded r, some nonzero such perturbation would preserve all r affine constraints, contradicting extremality. Thus $s \ge d+1-r = M-\pi(M)-2$. The objective is linear and the feasible slice is compact, so a minimizing face contains an extreme point of the slice.

This theorem does not say all roots are at zero, nor that the remaining coefficients are positive. It therefore does not reduce to the old coefficientwise cone. Optimizing a monomial-positive numerator misses the possible movable positive double roots forced by this full-cone geometry.

33.5 What the arithmetic budget does and does not imply

At the stated target denominator $D_M = 2L_M$, $\log D_M = M + o(M)$. Equation (PK.3) implies that the guaranteed positive error cannot have an unbounded exponent measured against this **unreduced prescribed budget**:

$$\limsup \frac{-\log E(F_M)}{\log(2L_M)} \leq 4.439858114896194\ldots$$

for any feasible sequence F_M. Thus a superexponential gain in M or an unbounded-exponent argument using that budget is ruled out. This is not a bound in terms of the reduced denominator of the rational target: large gcd cancellations are not controlled here.

Crucially, the constant $4.4398\ldots$ exceeds 1. The bound is compatible with E_M^* much smaller than $1/(2L_M)$, so it does not close the natural irrationality window. Even an upper bound $E_M^* < 1/(2L_M)$ alone would not prove the target intersection

$$E_M^* \leq \frac{\{2L_M\gamma\}}{2L_M}.$$

The fractional part could vanish in the rational gamma world. No claim that positivity, compactness, or small unconstrained energy forces the needed rational target is valid.

34 Rational-center Gamma identities and the free first cumulant

The one-center translation and finite-jet invariant-ring argument is a reused input. The extension below considers all rational-center reflection, shift, and multiplication identities simultaneously. Its conclusion applies to the listed identities, not to a second global Gamma function.

34.1 Simultaneous rational-center jet deformation

For each positive rational q, let

$$G_q(z) = \Gamma(q+z).$$

These are separately labelled convergent power-series germs. For a real or complex parameter c, define

$$G_q^{[c]}(z) = e^{cz} G_q(z). \tag{ME.2.1}$$

Proposition 34.1 (Simultaneous rational-center deformation)**.** *The deformation* (ME.2.1)*, with the same c at every center, preserves all the following identities and all their finite derivative consequences:*

$$G_{q+1}(z) = (q+z)G_q(z), \tag{ME.2.2}$$

$$G_q(z)G_{1-q}(-z) = \frac{\pi}{\sin \pi(q+z)} \qquad (0 < q < 1), \tag{ME.2.3}$$

$$G_1(z)G_1(-z) = \frac{\pi z}{\sin \pi z}, \tag{ME.2.4}$$

and, for every positive integer m,

$$G_{mq}(mz) = (2\pi)^{(1-m)/2} m^{mq+mz-1/2} \prod_{k=0}^{m-1} G_{q+k/m}(z). \tag{ME.2.5}$$

It fixes every value $\Gamma(q)$ *and every polygamma value* $\psi^{(r)}(q)$, $r \geq 1$, *while replacing* $\psi(q)$ *by* $\psi(q) + c$. *Consequently the designated first cumulant* $-G_1'(0) = \gamma$ *becomes* $\gamma - c$.

Proof. In (ME.2.2) both sides acquire e^{cz}. In (ME.2.3) and (ME.2.4) the two factors cancel. In (ME.2.5) both sides acquire e^{cmz}. Finally

$$\log G_q^{[c]}(z) = cz + \log G_q(z),$$

which proves all coefficient claims. These are identities of germs, so comparison of any finite number of Taylor coefficients is valid. □

This includes the usual digamma distribution identities. Their constant terms do not isolate the common shift: the sum of m digamma shifts agrees with the shifted term $m\psi(mq)$. Higher polygamma distribution identities contain no shift at all.

34.1.1 Lower incomplete-Gamma data can remain fixed

For a positive cutoff α, set

$$L_{q,\alpha}(z) = \int_0^\alpha e^{-t} t^{q+z-1}\, dt, \qquad U_{q,\alpha}(z) = G_q(z) - L_{q,\alpha}(z).$$

Keep every $L_{q,\alpha}$ fixed, and put

$$U_{q,\alpha}^{[c]}(z) = G_q^{[c]}(z) - L_{q,\alpha}(z). \tag{ME.2.6}$$

Then the split identity, cutoff derivative, and shape recurrence remain true:

$$G_q^{[c]} = L_{q,\alpha} + U_{q,\alpha}^{[c]}, \quad \partial_\alpha U_{q,\alpha}^{[c]} = -e^{-\alpha}\alpha^{q+z-1},$$

$$U_{q+1,\alpha}^{[c]} = (q+z)U_{q,\alpha}^{[c]} + \alpha^{q+z}e^{-\alpha}. \tag{ME.2.7}$$

Indeed the lower recurrence is $L_{q+1,\alpha} = (q+z)L_{q,\alpha} - \alpha^{q+z}e^{-\alpha}$, and subtraction from (ME.2.2) gives (ME.2.7). In particular, every actual algebraic relation or independence statement involving **only the unchanged lower-jet constants** is preserved.

34.1.2 Exact elimination consequence

Let B be the subfield of $\mathbb{C}$ generated by the fixed constants occurring above: zero-order Gamma values, higher polygamma values, $\psi(q)+\gamma$, and lower jets, together with the elementary constants in the identities. For rational q, the combinations $\psi(q) + \gamma$ have the classical expressions in π and logarithms of algebraic numbers; see DLMF 5.4.

An explicit one-parameter model over $B[g]$ is

$$\mathcal{G}_q^{(g)}(z) = \Gamma(q) \exp\left((\psi(q) + \gamma - g)z + \sum_{j\geq 2} \frac{\psi^{(j-1)}(q)}{j!} z^j \right). \tag{ME.2.8}$$

At $q = 1$, this is

$$\mathcal{G}_1^{(g)}(z) = \exp\left(-gz + \sum_{j\geq 2} \frac{(-1)^j \zeta(j)}{j} z^j\right).$$

All identities (ME.2.2)–(ME.2.7) hold coefficientwise in this model, with lower jets unchanged. The construction may be specialized to **any** real algebraic $g = a$, as well as to any real transcendental g. Thus polynomial elimination using exactly this listed identity system cannot produce a nonzero polynomial constraint on the free parameter g, nor exclude all algebraic choices for it.

This is a statement about the specified system of identities. It does not assert that all unknown arithmetic relations among actual Gamma and upper incomplete-Gamma values are preserved.

34.1.3 Essential scope restriction: the model is not a second global Gamma function

The deformed centered germs need not glue to a single function. Where the original germs overlap, their actual consistency is $G_q(z + r) = G_{q+r}(z)$; the deformation instead gives

$$G_q^{[c]}(z + r) = e^{cr} G_{q+r}^{[c]}(z). \tag{ME.2.9}$$

For real c, compatibility at any nonzero rational translation already forces $c = 0$. For complex c, compatibility at all sufficiently small rational translations also forces $c = 0$. Likewise the actual upper-tail condition

$$U_{q,\alpha}(z) \longrightarrow 0 \quad (\alpha \to \infty)$$

is lost: (ME.2.6) tends to $(e^{cz} - 1)G_q(z)$.

Accordingly, this is **not** a countermodel to the definition of the actual Gamma function or actual gamma. It proves that the selected local algebraic jet identities alone leave a free mean. Global analytic compatibility removes that freedom, but says nothing by itself about the arithmetic nature of the selected mean.

34.2 A complete positivity characterization of the remaining parameter

The following result identifies exactly where analytic normalization enters the generalized-moment approach.

For $n \geq 0$, write

$$D_n = \int_0^\infty e^{-t} \log^n(1 + t)\, dt, \quad G_n = (-1)^n \Gamma^{(n)}(1),$$

$$\eta_n = n! \sum_{k\geq 1} \frac{(-1)^{k-1}}{k!\, k^n}.$$

Here $D_0 = G_0 = 1$, $\eta_0 = 1 - e^{-1}$, and D_1 is the Euler–Gompertz constant. Splitting the defining Gamma integral at 1 gives

$$G_n + (-1)^{n+1} \frac{D_n}{e} = \eta_n. \tag{ME.3.1}$$

For a real parameter g, define polynomials $G_n(g)$ by

$$M_g(t) := \sum_{n\geq 0} G_n(g)\frac{t^n}{n!} = \exp\left(gt + \sum_{j\geq 2}\frac{\zeta(j)}{j}t^j\right) = e^{(g-\gamma)t}\Gamma(1-t), \tag{ME.3.2}$$

and set

$$D_n(g) = e(-1)^{n+1}(\eta_n - G_n(g)). \tag{ME.3.3}$$

Theorem 34.2 (Positivity selects the actual first cumulant)**.** *For real g, the following conditions are equivalent:*

1. $g = \gamma$.
2. $D_n(g) \geq 0$ *for every* $n \geq 1$.
3. $D_n(g)/n! \to 0$.

If $g \neq \gamma$, one parity of $D_n(g)$ is strictly negative for all sufficiently large indices of that parity.

Proof. The series definition immediately gives

$$\frac{\eta_n}{n!} = 1 + O(2^{-n}). \tag{ME.3.4}$$

The meromorphic function in (ME.3.2) has its nearest pole at $t = 1$, with principal part $e^{g-\gamma}/(1-t)$. After subtracting this principal part, it is holomorphic in $|t| < 2$. Therefore, for each fixed $1 < R < 2$, Cauchy's estimate gives

$$\frac{G_n(g)}{n!} = e^{g-\gamma} + O_{g,R}(R^{-n}). \tag{ME.3.5}$$

Combining (ME.3.3)–(ME.3.5),

$$\frac{D_n(g)}{n!} = e(-1)^{n+1}\left(1 - e^{g-\gamma} + O_{g,R}(R^{-n})\right). \tag{ME.3.6}$$

For real $g \neq \gamma$, the leading constant in parentheses is nonzero. The claimed eventual negative parity follows, and convergence to zero is impossible. At $g = \gamma$, (ME.3.1) identifies $D_n(g)$ with the strictly positive defining integral; (ME.3.6) also gives the normalized limit zero. □

The theorem can be phrased without gamma in the construction: use the first expression in (ME.3.2), the fixed zeta values, and the explicit E-values η_n. Gamma enters only to prove which real parameter the resulting positivity condition selects. No novelty claim is made for this elementary analytic characterization.

34.2.1 Why this does not settle the algebraic world

A proof that every real algebraic a violates condition 2 would prove gamma transcendental. But Theorem 34.2 alone does not show this violation for any unspecified algebraic a: its eventual-sign argument requires first knowing $a \neq \gamma$.

Moreover, **every finite collection of strict moment-positivity tests admits algebraic choices of the parameter**. The actual sequence D_n is the moment sequence of the positive measure

$$d\nu(y) = e^{1-e^y+y}\,dy \qquad (y > 0). \tag{ME.3.7}$$

Its finite Hankel and shifted Hankel matrices are positive definite, because the integral of any nonzero polynomial square, also after multiplication by y, is strictly positive. Their entries $D_n(g)$ are polynomials in g, so any finitely many such matrices remain positive definite on a neighborhood of γ. That neighborhood contains rational, hence algebraic, parameters. The same applies to any finite list of the individual strict inequalities $D_n(g) > 0$.

Full-moment positivity is even less restrictive: (ME.3.2) is the moment-generating function of a standard Gumbel random variable translated by $g - \gamma$. Thus its full Hamburger moment matrices are positive for every real g; they cannot select the mean at all.

34.3 Differential transcendence and dimension counts do not repair the elimination

Hölder's differential-transcendence theorem concerns the Gamma function, not the arithmetic of a selected derivative value. Each individual germ $e^{cz}\Gamma(q+z)$ remains differentially transcendental over $\mathbb{C}(z)$: otherwise multiplying it by the differentially algebraic function e^{-cz} would make $\Gamma(q+z)$ differentially algebraic. Hence this property also survives the free-mean deformation. See the primary discussion in Wang–Yao, Difference independence of Euler gamma function.

Likewise reflection expresses the even zeta cumulants through π, but does not constrain the first cumulant. Under the hypothesis that gamma is algebraic, the identity

$$G_2 = \gamma^2 + \zeta(2)$$

simply makes G_2 transcendental. Higher transcendence can reside in the zeta cumulants and the upper incomplete-Gamma moments without producing a contradiction at the first cumulant.

35 Published approximation benchmarks and the primitive budget

This section records the source review dated 20 September 2026. The published input statements below are source-reviewed benchmarks, not newly proved approximation theorems or newly measured numerical data. They are retained to distinguish real convergence, integerization, and primitive cancellation. The review was selective; it establishes no classification of all possible approaches.

35.1 The Van Assche–Wolfs benchmark

The source review uses version 3 of Van Assche–Wolfs, arXiv:2404.09799 (15 April 2025) [28], Theorem 1.2 and Propositions 2.5–2.6. Its normalization and approximation exponent must not be mixed with those of version 2. Write

$$d_n = \operatorname{lcm}(1, \dots, n), \qquad A_n = F^{(I)}_{n;1}(1), \qquad B_n = F^{(I)}_{n;2}(1) > 0.$$

These letters refer only to the published family in this subsection. The quoted estimates and integrality are

$$\left|\gamma - \frac{A_n}{B_n}\right| = O\Big(e^{-4n^{3/4}+n^{1/2}+(3/8)n^{1/4}}\Big),$$

$$B_n \asymp n^{-9/8} e^{4n^{3/4}-(1/2)n^{1/2}-(3/8)n^{1/4}},$$
$$P_n = n! d_n A_n \in \mathbb{Z}, \qquad Q_n = n! d_n B_n \in \mathbb{Z}.$$

Multiplying the first estimate by the quoted denominator gives

$$|Q_n \gamma - P_n| \le C n! d_n n^{-9/8} e^{\sqrt{n}/2}. \tag{35.1}$$

This guaranteed upper bound does not tend to zero. It does not prove that every actual residual diverges: the residuals may oscillate, and an upper bound alone excludes no exceptional small subsequence. The leading guaranteed quotient-error exponent relative to the unreduced Q_n is $4/(n^{1/4}\log n) \to 0$; in the linear-form convention the corresponding exponent is this quantity minus one. The repeated-differentiation discussion in the cited paper's Section 5 does not supply a uniform theorem for a parameter increasing with n.

Lemma 35.1 (Exact coefficient identity used in the primitive audit)**.** *For integers $n \ge 1$, $k \ge 0$,*

$$\sum_{\ell=1}^{n} \frac{(-1)^\ell}{\ell} \binom{n+k}{n-\ell} = -\binom{n+k}{k}(H_{n+k} - H_k).$$

Consequently the published integer pair can be written

$$T_{n,k} = \binom{n}{k}^2 \binom{n+k}{k} \frac{n!}{k!},$$
$$Q_n = d_n \sum_{k=0}^{n} T_{n,k},$$
$$P_n = d_n \sum_{k=0}^{n} T_{n,k}(4H_k - 2H_{n-k} - H_{n+k}).$$

The latter formulas rearrange the cited definitions; they do not provide a new approximating family.

Proof. Compute $D^n(x^{n+k}\log x)/n!$ in two ways. Differentiating $D^n x^{n+k+s}/n! = \binom{n+k+s}{n} x^{k+s}$ with respect to s at zero gives

$$\binom{n+k}{n} x^k \big(\log x + H_{n+k} - H_k\big).$$

Leibniz's rule gives the logarithmic term plus

$$x^k \sum_{\ell=1}^{n} \frac{(-1)^{\ell-1}}{\ell} \binom{n+k}{n-\ell}.$$

Comparison proves the identity. Substitution of that identity in the published coefficient sums yields the displayed P_n, Q_n formulas. □

For $g_n = \gcd(P_n, Q_n)$, the source's exact-integer computation reports

n	g_n	$\log(Q_n/g_n)$	$\log g_n/(n\log n)$
10	192	33.048606	0.228330
25	90	112.826729	0.055918
50	96	257.116922	0.023335
100	1692	562.817931	0.016142
200	220	1259.532706	0.005090
400	28836	2724.602499	0.004285

The integer pairs and gcds were computed exactly; the logarithms are rounded diagnostics. These six values are not an asymptotic theorem. Neither $\log g_n = o(n \log n)$ nor $\log g_n = O(n)$ is established. Conversely, if an infinite subsequence has certified nonzero residuals and, for some fixed $\varepsilon > 0$,

$$g_n \geq n! d_n e^{(1/2+\varepsilon)\sqrt{n}},$$

then dividing (35.1) by g_n produces nonzero integer linear forms tending to zero. This is a sufficient factorial-scale saving, not a necessary condition and not an obtained lower bound.

35.2 The conditions on external value theorems remain necessary

The source audit distinguishes the Hessami Pilehrood $k!$ sum used above from Aptekarev's original $\widetilde{q}_n = \sum_k \binom{n}{k}^2 (n+k)!$ family. The identity $q_n = n! L_n(-1)$ is already explicit in the cited Hessami Pilehrood paper. It is not a new structural discovery. The source-reviewed 2026 survey by Aptekarev and the geometric-series paper by Burnol likewise supply no quoted irrationality theorem for γ. In particular, Burnol's fixed-parameter geometric expansion approximates a combination $\gamma + (\ell - 1)\log 2$; geometric convergence alone supplies no nonzero integer form in γ alone.

Fischler–Rivoal's *Rational approximations to values of E-functions*, Theorem 1 in the manuscript dated 3 July 2025 [3], is quoted for irrational values at rational points of E-functions with *rational* Taylor coefficients. Under the stronger hypothesis $\gamma \in \mathbb{Q}$, one has

$$F_\gamma(z) = \sum_{n \geq 0} (H_n - \gamma) \frac{z^n}{n!} \in \mathbb{Q}[[z]], \qquad F_\gamma(1) = \delta.$$

The value is transcendental under that hypothesis by the E-function specialization input, and the quoted approximation result then gives $\mu(\delta) = 2$. This is a conditional consequence, not a contradiction. The rational-coefficient theorem cannot simply be applied under only $\gamma \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$.

The source-reviewed 8 September 2026 Fischler–Rivoal measure theorem [32] for the logarithmic derivative of J_0 requires a permitted value representation. None expressing γ as such a value is given. In particular, $z(F_r' - F_r) = e^z - 1$ does not have the polynomial right-hand side required by the quoted first-order inhomogeneous alternative. Similarly, the source-reviewed 17 September 2026 Rivoal arithmetic-difference-operator theorem [33] requires its Mellin-associated operator to be a G-operator. The theorem's input conditions have not been verified for the gamma approximants at issue. Neither result can be applied solely because a differential or difference equation has been written down.

Finally, the source review of Powers, arXiv:2511.01849v7 [34], reports family-level transcendence results for Gamma derivatives, generalized eta constants, and generalized Gompertz constants. Its stated proportion bound for transcendental Gamma derivatives is of order $N^{-1/2}$, not a positive-density assertion. Those quoted results are not used in the elementary rational-center deformation or positivity proofs above. For the generalized constants discussed in that source, $\eta_0 = 1 - e^{-1}$, so independence over algebraic numbers must not be upgraded to independence of the entire indexed family over a field already containing e.

36 Arithmetic reassessment: primitive targets and limits

36.1 A Dirichlet existence theorem is not an upper ceiling

The claim that every triple of numbers satisfies an approximation ceiling $\tau \leq 2$ is false. Dirichlet's principle supplies good approximations; it does not prohibit much better approximations.

An elementary explicit counterexample is enough. Let

$$\ell = \sum_{k\geq 1} 10^{-k!}, \qquad (x_0, x_1, x_2) = (1, \sqrt{2}, \sqrt{2} + \ell).$$

This triple is linearly independent over $\mathbb{Q}$, since a nontrivial relation would make the Liouville number ℓ algebraic. Put $q_m = 10^{m!}$ and let p_m/q_m be its m-th partial sum. Then

$$0 < | - p_m - q_m\sqrt{2} + q_m(\sqrt{2} + \ell)| = |q_m\ell - p_m| < 2q_m^{-m}.$$

All three integer coefficients are nonzero. Thus no universal exponent-2 ceiling exists even with linear independence and all coordinates active. This does not establish any superior exponent for the actual gamma/Gompertz triple; it invalidates an asserted universal ban.

This correction was already acknowledged in the older integrated archive; it is retained here to prevent reuse of the invalid universal ceiling, without a novelty claim.

36.2 Primitive elimination has two distinct height effects

Suppose

$$R_i = a_i + b_i\gamma + c_i\delta, \quad |a_i|, |b_i|, |c_i| \leq H, \quad |R_i| \leq H^{-\tau}.$$

Eliminating delta gives

$$c_2R_1 - c_1R_2 = A + B\gamma, \quad |A|, |B| \leq 2H^2, \quad |A + B\gamma| \leq 2H^{1-\tau}.$$

If (A, B) is nonzero and $g = \gcd(A, B)$, the primitive output satisfies

$$h \leq 2H^2/g, \qquad |A/g + (B/g)\gamma| \leq 2H^{1-\tau}/g.$$

Consequently the frequently quoted exponent $(\tau - 1)/2$ describes only a raw generic budget. If $g = H^{\sigma+o(1)}$, with $\sigma < 2$ and $\tau + \sigma - 1 > 0$, the displayed estimates certify, up to constants and $o(1)$, an exponent at least

$$\frac{\tau + \sigma - 1}{2 - \sigma}$$

against the primitive height. Actual height drops or better error cancellation can improve this further. Neither such gcd growth nor nonvanishing is automatic.

This also does not defeat the separate fact that a restricted collection of E-function expressions might involve gamma and delta only through one fixed combination: eliminating that combination could eliminate gamma at the same time. These are different obstructions and require different proofs.

The archived Sondow calculation retains both circuit saturation and final target gcd. Its exact quotient norm is invariant under positive rational rescaling of columns. Therefore a large unnormalized quotient index, a large common denominator, or a huge raw coefficient vector is not by itself a primitive-height impossibility theorem.

The source audit reports one archived finite, exactly certified inequality

$$0 < p - q\gamma < q^{-101/100}$$

at a 37-digit denominator. This is compatible with gamma being algebraic. It neither produces an infinite family nor proves eventual denominator growth.

36.3 Gevrey and window-width estimates need their hypotheses

Subexponential convergence for fixed-pole approximants, or factorial costs for one moving-pole construction, does not exclude all varying-degree, mixed, nonlinear, or differently normalized constructions. A rigorous obstruction must specify its input ring, coefficient normalization, degree growth, denominator support, and the permitted transformations.

The earlier common-lcm window budget giving $w < 0.386$ is presentation dependent. In the integerized Sondow family one may instead use each column's own multiplier d_{2n}. On $[N, 2N]$, the old counting argument already yields exponentially small upper bounds with exponent $\log 16 - 2 > 0$. The difficulty is not the absolute impossibility of such a wide window. It is that all the short vectors supplied by counting can map to exact zero.

Likewise, a statement that only two families in all the literature qualify is not established by inspecting two families. It should be replaced by an explicitly delimited list of families actually analyzed.

36.4 The exact Sondow quotient, including both algebraic worlds

Use the established identity

$$e_n = d_{2n} I_n = a_n \gamma - b_n + \sum_p \beta_{p,n} \log p, \qquad a_n, b_n, \beta_{p,n} \in \mathbb{Z},$$

where I_n is Sondow's positive double integral and $d_m = \operatorname{lcm}(1, \dots, m)$. On a finite block define

$$\Lambda = \ker_{\mathbb{Z}} B, \quad \Phi(u) = (a \cdot u, b \cdot u), \quad K_0 = \ker_{\mathbb{Z}}[B; a; b], \quad \Omega = \Phi(\Lambda) \subseteq \mathbb{Z}^2.$$

The first isomorphism theorem gives, unconditionally,

$$\Lambda / K_0 \simeq \Omega, \qquad \operatorname{rank} \Omega \leq 2.$$

This quotient is torsion free. The evaluation of a target (Q, P) is $Q\gamma - P$.

If gamma is irrational, whether algebraic or transcendental, its evaluation kernel on Λ is exactly K_0. Indeed, $Q\gamma = P$ with integers Q, P forces $Q = P = 0$.

If $\gamma = a/b$ is rational in lowest terms, the additional evaluation kernel is

$$\ker(u \mapsto \textstyle\sum u_n e_n)/K_0 \simeq \Omega \cap \mathbb{Z}(b, a).$$

If $\operatorname{rank} \Omega = 2$ this is a rank-one lattice. For every vector whose certified error is less than $1/b$, the image must lie on that rational line. It need not be the zero pair.

Thus the statement "algebraic gamma forces all sufficiently small linear forms to vanish" is wrong without a quantitative, degree-dependent height threshold. Rationality has a fixed spacing $1/b$. Algebraic irrationality has nonzero linear forms and height-dependent lower bounds. The previously established polynomial-collapse result concerns a different family P_D with very strong degree/height/error inequalities; it must not be transferred silently to these linear forms.

36.4.1 Why the existing counting theorem does not survive the quotient

On $[N, 2N]$ there are $N+1$ columns and $O(N/\log N)$ prime rows, all entries $\exp(O(N))$. The elementary counting proof of Siegel's lemma yields nonzero $u \in \Lambda$ with

$$\log \|u\|_\infty = O(N/\log N), \qquad \left|\sum u_n e_n\right| \le \exp[-(\log 16 - 2)N + o(N)].$$

Appending the two rows a and b changes neither asymptotic. The same theorem therefore yields nonzero $u \in K_0$ at exactly this scale. This is a proven mechanism for producing short useless vectors, not merely a hypothetical failure.

One can record the rank of the span of the images of short vectors; it is 0, 1, or 2. Naming that statistic is not an advance. Proving an all-level lower bound on it with the required representative error would be an advance. In particular, proving rank 2 for two errors tending to zero would prove irrationality. A rank-1 family can also suffice if its primitive denominators provably tend to infinity while errors tend to zero.

The stronger assertion that short first-coordinate-dominant vectors exist is a sufficient condition. The converse "gamma irrational implies this particular dominant family exists" has not been proved, so the condition should not be called equivalent to irrationality.

36.4.2 Rank two by itself is inexpensive and insufficient

The source audit reports that the archived exact $n = 1, \dots, 8$ calculation gives two rational-kernel vectors with target determinant

$$\frac{25884800070992513775689 5171}{3003} \neq 0$$

before column integerization. Clearing denominators transfers them to the integerized lattice without destroying independence. Padding these vectors by zeros proves $\operatorname{rank} \Omega = 2$ for every prefix $1, \dots, M$ with $M \ge 8$. This elementary observation is not a tail estimate: both witnesses can remain supported in the first eight columns, so their errors need not decrease.

It does not prove rank two with small representatives in moving windows $[N, 2N]$. Nor does a finite moving-window rank computation prove an asymptotic estimate.

36.5 A successful super-Roth sequence must allow sparse scales

For any real x, let p/q and p'/q' be distinct reduced fractions, with $q' \ge q > 0$, both satisfying

$$|qx - p| \le q^{-1-\eta}, \qquad |q'x - p'| \le (q')^{-1-\eta}, \qquad \eta > 0.$$

Their nonzero integer determinant gives

$$1 \le |qp' - q'p| \le q'q^{-1-\eta} + q(q')^{-1-\eta} \le 2q'q^{-1-\eta}.$$

Hence

$$q' \ge \tfrac{1}{2} q^{1+\eta}.$$

This is an unconditional restriction, including for transcendental x. A proposed construction cannot produce distinct such approximants at every consecutive level with $\log q_N = cN + o(N)$,

$c > 0$. It must allow repeated outputs or sufficiently sparse new levels. Requiring a uniform "new super-Roth target at every nearby height" would wrongly reject every real number, not just an algebraic gamma world.

For two independent target pairs with both denominators <=H and both errors <=E, the simpler restriction is $1 \leq 2HE$. Thus no proof should demand two independent super-Roth approximants at one common comparable height.

The resulting status of the central arithmetic budget. For fixed algebraic r, the moving-degree theorem proves $\log(d_N H(P_N)) \geq N - o(N)$ when $D_N^6 \log(D_N + 2) = o(N)$ in the z-independent case. Its leading constant one is attained by $P_N = Y$ for algebraic-integral r. This is a coefficient denominator-and-height theorem. It is compatible with F_r being an E-function, permits no automatic inference about cancellation after evaluation at a point, and leaves fast-degree, rational-function, and different-representation constructions outside its stated scope. Substitution $r = \gamma$ under an algebraicity hypothesis supplies no contradiction.

The other completed branches constrain fixed shifts, full positive Binet kernels, and a specified local Gamma identity system. None constructs an infinite family with both certified nonzero arithmetic output and sufficiently small error relative to primitive height. Historical priority is unestablished; internal audits and finite exact checks do not constitute external peer review. These results do not prove irrationality or transcendence of γ and do not establish the later rank-four target theorem.

Part V
The four-value problem and cross-field tests

37 The four-value rank and its exceptional points

This continuation separates unconditional deductions from the remaining specialization problem. Throughout this part, put $k = \overline{\mathbb{Q}}$ and use the principal logarithm on $\mathbb{C} \setminus (-\infty, 0]$. Our normalization is

$$\mathrm{Ein}(z) = \sum_{n\ge 1} \frac{(-1)^{n-1} z^n}{n\, n!}, \quad E(z) = e^z, \qquad F(z) = e^z \,\mathrm{Ein}(z),$$
$$E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\, dt, \qquad D(x) = e^x E_1(x) = \int_0^\infty \frac{e^{-t}}{x+t}\, dt \quad (x > 0). \tag{37.1}$$

The connection formula and the distinguished values are

$$\mathrm{Ein}(x) = \gamma + \log x + E_1(x), \qquad \eta = \mathrm{Ein}(1), \qquad \delta = D(1) = e(\eta - \gamma). \tag{37.2}$$

For positive algebraic x, define

$$\rho(x) = \dim_k \mathrm{span}_k\{1, e^x, D(x), F(x)\}. \tag{37.3}$$

The designated assertion $\rho(1) = 4$ is not proved here. It would imply the transcendence of both γ and δ: algebraicity of the former would make $\delta - e\eta + \gamma e = 0$ a nonzero algebraic-coefficient relation, and algebraicity of the latter would already relate 1 and δ. The converse implication is unavailable, as the exact classification below makes particularly clear.

37.1 A finite E-function input with two functional proofs

Lemma 37.1. *If $x_1, \dots, x_m$ are distinct positive algebraic numbers, then*

$$1,\ e^{-x_1}, \dots, e^{-x_m},\ \mathrm{Ein}(x_1), \dots, \mathrm{Ein}(x_m)$$

are linearly independent over k.

Proof. Set $X_i(z) = e^{-x_i z}$ and $U_i(z) = \mathrm{Ein}(x_i z)$. These are E-functions and satisfy the rational differential system

$$X_i' = -x_i X_i, \qquad U_i' = \frac{1 - X_i}{z}, \qquad 1' = 0. \tag{37.4}$$

For clarity, the normalized Taylor coefficients of U_i are $(-1)^{n-1} x_i^n / n$. In a fixed number field all their conjugates have at most exponential growth, and clearing the coefficients through n requires, in addition to an exponential denominator for powers of x_i, only $\mathrm{lcm}(1, \dots, n)$, which is exponentially bounded. The differential equations give holonomicity. Thus the arithmetic E-function hypotheses hold, rather than merely entire-function growth.

First $1, X_1, \dots, X_m$ are independent over $\mathbb{C}(z)$. Clear rational denominators in a relation, order the positive x_i, and let z tend to negative real infinity. The largest exponential with a nonzero polynomial coefficient dominates all smaller ones, a contradiction; after removing all exponentials the polynomial constant term also vanishes.

Suppose a rational relation involving U_i exists, and choose one with the smallest positive number of nonzero U_i coefficients. Normalize one of them to 1 and write

$$\sum_{i\in I} R_i U_i = R_0 + \sum_{j=1}^{m} S_j X_j, \qquad R_i, R_0, S_j \in \mathbb{C}(z).$$

Differentiation, followed by (37.4), produces a relation with fewer U_i terms unless every R_i' vanishes. Minimality forces $R_i = c_i \in \mathbb{C}$. Independence of the exponentials now gives

$$S_i' - x_i S_i = -c_i/z.$$

For $c_i \neq 0$ this has no rational solution: a pole of S_i at zero would yield a pole of strictly higher order in S_i', uncancelled by $-x_i S_i$, whereas if S_i is regular there the left side cannot have a simple pole. Thus every c_i is zero, contradicting the normalization. The $2m+1$ functions are consequently independent over $\mathbb{C}(z)$.

The system has no finite singularity outside zero. Apply the linear ordinary-point case of Beukers' refinement of the Siegel–Shidlovskii theorem at $z = 1$ [4, Theorem 1.3]. It gives the asserted independence of values. This step uses an established theorem for E-functions; it does not use a mixed Gevrey specialization conjecture. □

The following second functional proof records the coefficient audit of the same input. Suppose, after clearing denominators, that

$$P_0(z) + \sum_i P_i(z) e^{-x_i z} + \sum_i Q_i(z)\,\mathrm{Ein}(x_i z) = 0,$$
$$P_i = \sum_j p_{ij} z^j, \qquad Q_i = \sum_j q_{ij} z^j.$$

For n larger than all degrees, the normalized Taylor coefficients give

$$\sum_i (-x_i)^n R_i(n) = 0.$$

Here, with $(T)_j = T(T-1)\cdots(T-j+1)$ and $(T)_0 = 1$, we put

$$R_i(T) = \sum_j p_{ij}(-x_i)^{-j}(T)_j - \sum_j q_{ij}(-x_i)^{-j}\frac{(T)_j}{T-j}. \tag{37.5}$$

If a nonzero R_i with largest x_i were present, division by its exponential term would leave a nonzero rational function, of polynomial asymptotic size, equal to exponentially small terms. Hence every R_i vanishes. Its residue at $T = j$ is exactly $-q_{ij}(-x_i)^{-j} j!$, because all other summands are regular there. Thus all q_{ij} vanish; independence of the falling-factorial basis then gives all $p_{ij} = 0$, followed by $P_0 = 0$. Beukers' theorem completes the specialization as before.

37.2 The unique real triple and the complete rank-three alternatives

Proposition 37.2. *For positive algebraic x one has $\rho(x) \in \{3, 4\}$. Moreover,*

$$\rho(x) = 3 \quad\Longleftrightarrow\quad \gamma + \log x = a_x + b_x e^{-x} + c_x\,\mathrm{Ein}(x) \quad (a_x, b_x, c_x \in k). \tag{37.6}$$

When it exists, this triple is unique and belongs to $(k \cap \mathbb{R})^3$.

Proof. Multiplication of the $m = 1$ case of Lemma 37.1 by e^x proves independence of $1, e^x, F(x)$, so the rank is at least three. Any relation on the four values must therefore have a nonzero coefficient of $D(x)$. Solving for it, dividing by e^x, and using (37.2) gives (37.6), and the steps are reversible. Subtraction of two triples gives a relation on $1, e^{-x}, \mathrm{Ein}(x)$, proving uniqueness. Complex conjugation fixes all three values and the left side, so uniqueness also forces every coefficient to be real. □

We use, for the next stronger conclusion, the established algebraic independence of e and η over k. One primary source is Rivoal [37, §5(ii)]: the series denoted there by $E(-1)$ is $-\mathrm{Ein}(1)$, and replacing e^{-1} by its inverse preserves algebraic independence. This input is stronger than Lemma 37.1.

Theorem 37.3 (Complete classification at rank three)**.** *Assume $\rho(1) = 3$, and let $(a, b, c) \in (k \cap \mathbb{R})^3$ be its unique triple. Then*

$$\gamma = a + b/e + c\eta, \qquad \delta = (1 - c)e\eta - ae - b. \tag{37.7}$$

Exactly the following disjoint alternatives occur:

1. *If $b = c = 0$, then $\gamma = a \in k$ and $k(e, \delta) = k(e, \eta)$; hence e, δ are algebraically independent.*
2. *If $a = 0$ and $c = 1$, then $\delta = -b \in k$ and $k(e, \gamma) = k(e, \eta)$; hence e, γ are algebraically independent.*
3. *In all other cases, $k(\gamma, \delta) = k(e, \eta)$, and γ, δ are algebraically independent.*

Furthermore, under the rank-three assumption, e, γ are algebraically independent exactly when $c \neq 0$, and e, δ are algebraically independent exactly when $c \neq 1$.

Proof. The pair of equations is Proposition 37.2 at $x = 1$ and (37.2). If $\gamma \in k$, subtract it from the first equation and use the independence of $1, e^{-1}, \eta$; this forces $b = c = 0$. The converse is immediate. In this case $\delta = e(\eta - a)$, proving the first field equality. Similarly, the independence of $1, e, e\eta$ applied to the second equation gives $\delta \in k$ if and only if $a = 0, c = 1$. Then $\gamma = \eta + b/e$, proving the second field equality. The two cases are disjoint.

For the remaining cases eliminate η using $\eta = \gamma + \delta/e$ to obtain

$$e\big((1 - c)\gamma - a\big) = c\delta + b. \tag{37.8}$$

The denominator $(1 - c)\gamma - a$ can vanish only in the two cases already listed. Indeed, if $c = 1$ its vanishing means $a = 0$; if $c \neq 1$ it makes $\gamma = a/(1 - c)$ algebraic, and the first case applies. Outside these cases the exact inverse transformation is

$$e = \frac{c\delta + b}{(1 - c)\gamma - a}, \qquad \eta = \gamma + \delta/e.$$

Together with (37.7), it proves equality of the two fields. Their transcendence degree is two, proving the third alternative. Finally, the first equation of (37.7) solves for η from (e, γ) when $c \neq 0$, and puts γ in $k(e)$ when $c = 0$. The second equation has the analogous properties for (e, δ) according as $c \neq 1$ or $c = 1$. □

No triple has been excluded for the actual constants by this classification. In particular, obtaining transcendence or even algebraic independence of γ, δ under a generic rank-three hypothesis is not a contradiction. To isolate the logical issue, the artificial numbers $g = \eta/2$ and $d = e\eta/2$ satisfy $\eta = g + d/e$ and are algebraically independent, but $\{1, e, d, e\eta\}$ has rank three. They are not the Euler and Gompertz constants. The classification also implies the conditional statement that if the actual γ, δ were both transcendental and algebraically dependent, then $\rho(1) = 4$; no such dependence is established here.

37.3 Affine multiplicative restrictions on all deficient points

Theorem 37.4. *Let $B = \{x \in k \cap \mathbb{R}_{>0} : \rho(x) = 3\}$. For distinct $x_1, \dots, x_s \in B$, no nonzero integer vector $(m_1, \dots, m_s)$ satisfies both*

$$\sum_i m_i = 0, \qquad \prod_i x_i^{m_i} = 1. \tag{37.9}$$

Consequently, if the $\mathbb{Q}$-span of $\{\log x : x \in G\}$ has finite dimension d for a multiplicative group $G \subset k \cap \mathbb{R}_{>0}$, then $\#(B \cap G) \le d + 1$.

Proof. Multiply the representation (37.6) for x_i by m_i and add. Since all logarithms are real, (37.9) cancels both γ and the logarithms, leaving

$$0 = \sum_i m_i a_i + \sum_i m_i b_i e^{-x_i} + \sum_i m_i c_i \operatorname{Ein}(x_i).$$

Lemma 37.1 gives $b_i = c_i = 0$ whenever $m_i \neq 0$. Thus $\gamma + \log x_i = a_i \in k$ on the support. The support contains two distinct points, since a single nonzero coefficient cannot have sum zero. Subtracting their equations makes $\log(x_i/x_j)$ a nonzero algebraic number. The Hermite–Lindemann theorem would then make x_i/x_j transcendental, contrary to its algebraicity.

For the cardinality bound, $d + 2$ vectors $(1, \log x_i)$ in a rational space of dimension $d + 1$ have a nontrivial rational linear relation. Clear denominators and exponentiate its logarithmic component to obtain (37.9). Finite generation of G is not required. □

Corollary 37.5. *For every positive algebraic $r \neq 1$,*

$$\#\{n \in \mathbb{Z} : \rho(r^n) = 3\} \le 2.$$

In particular $\max\{\rho(1), \rho(2), \rho(4)\} = 4$, and among $1, \dots, N$ at most $\pi(N) + 1$ points have rank three. The positive integer points of rank four therefore have natural density one.

Proof. The logarithms of r^n span a one-dimensional rational space. More explicitly, for distinct n_1, n_2, n_3 the nonzero coefficients $(n_2 - n_3, n_3 - n_1, n_1 - n_2)$ annihilate both 1 and n, so three such points contradict Theorem 37.4. For $1, 2, 4$ the coefficients are $(1, -2, 1)$. The integers up to N lie in the positive rational group generated by primes up to N, whose rank is $\pi(N)$. The final conclusion uses the classical fact $\pi(N) = o(N)$. □

At each full-rank point both $\gamma + \log x$ and $D(x)$ are transcendental, since algebraicity of either would give a dependence in (37.3). None of these bounds singles out $x = 1$. They also do not constitute an independent new tail-transcendence mechanism: the earlier finite-rank tail-defect arguments already yield related cofinite conclusions for E_1 values. The precise object here is the set of failures of the four-value rank, with its affine multiplicative restriction. No priority claim is made for this short consequence of established E-function theory.

38 Functional rank, mixed division, and the exact missing implication

38.1 A differential system does not determine value rank

Proposition 38.1. *The functions $1, E, F, D$ have rank four over $\mathbb{C}(z)$ and satisfy*

$$1' = 0, \qquad E' = E, \qquad F' = F + (E - 1)/z, \qquad D' = D - 1/z. \tag{38.1}$$

The only finite singularity of this displayed system is zero.

Proof. The equations follow by differentiation. If $A + BE + CF + JD = 0$ with rational functions A, B, C, J, continue once counterclockwise around zero. Only D changes, by $-2\pi iE$. Subtracting the two relations gives $J = 0$. If $C \neq 0$, divide by CE to write $\mathrm{Ein} = R + Se^{-z}$ with rational R, S. Differentiation and independence of $1, e^{-z}$ imply $R' = 1/z$ and $S' - S = -1/z$. A rational derivative has zero residue at every pole, so $R' = 1/z$ is impossible. Hence $C = 0$, and $A + BE = 0$ forces $A = B = 0$. □

For any $s \in \mathbb{C}$, replacement of D by $D_s = D + sE$ preserves the system, the functional rank, and the monodromy. Taking $s = \gamma$ gives

$$D_\gamma = F - E \log z, \qquad D_\gamma(1) = F(1), \tag{38.2}$$

so the values have rank exactly three although 1 is still an ordinary point. This is a countermodel to an inference from these differential facts alone. It does not retain the prescribed positive-ray asymptotic normalization: $D(z) \sim 1/z$, whereas $D_s(z)$ has the additional term se^z for $s \neq 0$. Nor is the arithmetic admissibility of the shift by γ known. The missing information concerns the arithmetic of the canonical solution, not regularity of the evaluation point.

38.2 Formal arithmetic nondivisibility

Use the formal series

$$\widehat{D}(z) = \sum_{n \geq 0} (-1)^n n! z^{-n-1}.$$

Its positive-direction Borel sum is D. We employ the normalized mixed arithmetic space $\mathcal{M}$ consisting of expressions $H(z) + h(1/z)$, where H is an E-function, $h(w) = \sum_{n \geq 1} h_n w^n$ has algebraic coefficients, and its coefficientwise Borel transform $\mathcal{B}h(t) = \sum_{n \geq 1} h_n t^n / n!$ is a G-function. Thus the negative part has no constant term. Arithmetic Gevrey order one here includes holonomicity, exponential conjugate bounds and exponential common-denominator bounds after Borel transformation; analytic factorial growth alone is insufficient. This is the mixed setting of [5, Definition 1 and §2].

Theorem 38.2. *For $a, b, c, d \in k$, put $\Psi = a + bE + cF + d\widehat{D}$. Then*

$$\Psi \in (z-1)\mathcal{M} \quad \Longrightarrow \quad a = b = c = d = 0. \tag{38.3}$$

Proof. Suppose $\Psi = (z-1)(H + \sum_{n \geq 1} h_n z^{-n})$ in the stated arithmetic class. Comparison of nonnegative powers gives

$$a + bE + cF = (z-1)H + h_1.$$

Since H is entire, substitution of $z = 1$ gives $a + be + ce\eta = h_1 \in k$. The independence of $1, e, e\eta$ gives $b = c = 0$ and $h_1 = a$, and then $H = 0$. Comparison of the negative powers now yields

$$h_{n+1} - h_n = d(-1)^{n-1}(n-1)! \quad (n \geq 1).$$

With $S_n = \sum_{j=0}^{n} (-1)^j j!$ this means $h_{n+2} = a + dS_n$. If $\mathcal{B}h$ is a G-function, so is its second derivative

$$B_{a,d}(t) = \sum_{n \geq 0} \frac{a + dS_n}{n!} t^n, \qquad B'_{a,d} - B_{a,d} = -\frac{d}{(1+t)^2}, \qquad B_{a,d}(0) = a + d. \tag{38.4}$$

The coefficient recurrence verifies the equation directly. If $d \neq 0$, the integrating-factor equation $(e^{-t} B_{a,d})' = -de^{-t}/(1+t)^2$ has residue de at $t = -1$. Continuation once around -1 consequently changes $B_{a,d}$ by $2\pi i d e^{t+1}$.

A G-function has a minimal homogeneous differential equation regular singular at every point of the projective line, including infinity [7, §1]. Analytically continued branches solve the same equation, as does their difference. A nonzero multiple of e^t cannot be such a solution: solutions of a regular singular equation at infinity have power-logarithmic, rather than exponential, growth on fixed sectors. This contradiction excludes $d \neq 0$. If $d = 0$, equation (38.4) gives $B_{a,0} = ae^t$, which violates the same condition unless $a = 0$. All coefficients vanish. □

One useful special case makes the index conventions explicit. For every algebraic α,

$$\frac{\widehat{D}(z) - \alpha}{z - 1} = \sum_{m \geq 1} b_m z^{-m}, \qquad b_m = -\alpha + \sum_{j=0}^{m-2} (-1)^j j!, \tag{38.5}$$
$$B(t) = \sum_{m \geq 1} \frac{b_m}{m!} t^m, \qquad B' - B = -\alpha + \log(1+t), \qquad B(0) = 0.$$

The empty sum is zero. For $Y = B'$ one obtains $Y' - Y = 1/(1+t)$ and $Y(0) = -\alpha$. The residue of $e^{-t}/(1+t)$ at -1 is e, so continuation adds $2\pi i e^{t+1}$ to Y. The same regular-singularity argument proves that the quotient in (38.5) is never arithmetic Gevrey one. The statement holds independently of whether its Borel-summed numerator vanishes at one.

Proposition 38.3 (The restricted zero-division gate)**.** *For the four-dimensional algebraic-coefficient family above, the assertion*

$$(Z_4): \qquad \Psi_0(1) = 0 \Longrightarrow \Psi \in (z-1)\mathcal{M} \tag{38.6}$$

is equivalent to $\rho(1) = 4$. The subscript 0 denotes summation in the positive real direction.

Proof. If (Z_4) holds, Theorem 38.2 makes every algebraic-coefficient relation $a + be + ce\eta + d\delta = 0$ trivial. Conversely, full value rank makes the hypothesis in (38.6) possible only for the zero expression, whose quotient belongs to $\mathcal{M}$. □

Fischler–Rivoal's general mixed zero-removal statement is Conjecture 3, and their Theorem 4 assumes that conjecture before deducing ordinary-point linear specialization [5]. The present system meets its differential and summation conditions at 1 in direction zero, so the conjecture would imply the target. Its conditional hypothesis cannot be dropped. Formula (38.3) is an unconditional formal obstruction; (38.6) is the unresolved passage from an analytic zero to membership in the arithmetic quotient class. In particular, formal independence in $\mathcal{M}/(z-1)\mathcal{M}$ does not establish injectivity of analytic evaluation on these classes.

38.3 Why analytic division does not supply algebraic coefficients

Let $A(z) = a + bE(z) + cF(z)$ with $a, b, c \in k$. Entire analytic division does give $H_A(z) = (A(z) - A(1))/(z-1)$. If $H_n = \sum_{j=1}^{n} 1/j$ and $H_0 = 0$, then $F(z) = \sum_{n \geq 0} H_n z^n/n!$, and coefficient comparison gives

$$[z^n]H_A = b\left(e - \sum_{j=0}^{n} \frac{1}{j!}\right) + c\left(e\eta - \sum_{j=0}^{n} \frac{H_j}{j!}\right). \tag{38.7}$$

Indeed the coefficient of a quotient $(A - A(1))/(z-1)$ is the tail of the Taylor series of A beyond degree n, which proves the formula. If $(b, c) \neq (0, 0)$, each displayed coefficient is transcendental: its algebraicity would give a forbidden nontrivial relation on $1, e, e\eta$. Analytic

tail estimates therefore cannot be substituted for E-function arithmetic conditions. Under a hypothetical relation $A(1) + d\delta = 0$, replacing $A(1)$ by $-d\delta$ only moves the same unknown connection constant; it does not place these coefficients in a number field. When $b = c = 0$ the coefficients in the negative quotient are algebraic, but Theorem 38.2 still excludes the required G-function denominator bounds.

This also prevents an application of a fresh-prime theorem whose parameter is required to be algebraic to the expression $a + b/e + c\eta$ solely because it is hypothesized to equal γ. That expression need not be algebraic outside the exceptional strata of Theorem 37.3. The differential equation

$$\Psi_0' - \Psi_0 = -a + \frac{cE - c - d}{z}$$

is itself compatible with a zero at the ordinary point one; it merely specifies an initial condition. Finally, the target $\delta \notin \operatorname{span}_k\{1, e, e\eta\}$ has two separate arithmetic aspects: excluding algebraic δ, and excluding a transcendental δ in that E-value span. An intersection assertion that common E- and divergent values are algebraic would address the second aspect; pure divergent zero removal would address the first. Neither is supplied here by the differential system alone.

39 Spectral reformulation and its precise limitations

For a nonnegative self-adjoint Jacobi operator J on $\ell^2(\mathbb{Z}_{\geq 0})$, use the convention

$$m_J(z) = \langle e_0, (J + zI)^{-1} e_0 \rangle \qquad (z > 0).$$

Here a_n is the entry between coordinates n and $n+1$, and b_n is the diagonal entry at n. This fixes the resolvent sign throughout.

39.1 Laguerre determinants and an exact raw integer error

For the probability measure $e^{-t}\,dt$ on $[0, \infty)$, the orthonormal polynomials $p_n(t) = (-1)^n L_n(t)$ satisfy

$$tp_n = (n+1)p_{n+1} + (2n+1)p_n + np_{n-1}. \tag{39.1}$$

These standard Laguerre formulas can be found in [6, §§18.2,18.9]. Thus the Jacobi coefficients are $a_n = n + 1$, $b_n = 2n + 1$. The Jacobi Carleman criterion $\sum_n a_n^{-1} = \infty$ makes the operator on finite sequences essentially self-adjoint; its spectral representation identifies its closure with multiplication by t for the displayed measure [38, Chapter 2]. In particular,

$$m_J(z) = D(z), \qquad m_J(1) = \delta. \tag{39.2}$$

Schur complementation gives the classical continued fraction

$$\delta = \cfrac{1}{2 - \cfrac{1^2}{4 - \cfrac{2^2}{6 - \cfrac{3^2}{8 - \cdots}}}}.$$

The minus signs agree with the $J + zI$ convention; compare [6, §6.9].

Proposition 39.1. *Let J_N be the leading $N \times N$ truncation, and set*

$$Q_N = \det(J_N + I) = N! L_N(-1), \qquad \frac{P_N}{Q_N} = \langle e_0, (J_N + I)^{-1} e_0 \rangle.$$

For $N = 0$ use $Q_0 = 1, P_0 = 0$. Then P_N, Q_N are integers,

$$X_{N+1} = 2(N+1)X_N - N^2 X_{N-1}, \quad (Q_0, Q_1) = (1, 2), \quad (P_0, P_1) = (0, 1), \tag{39.3}$$

and

$$Q_N \delta - P_N = N! \int_0^\infty \frac{e^{-s} s^N}{(1+s)^{N+1}}\, ds > 0. \tag{39.4}$$

For $N \geq 1$,

$$Q_N \delta - P_N \geq \frac{N!}{\sqrt{N}+2} e^{-2\sqrt{N}-1} \longrightarrow \infty. \tag{39.5}$$

Proof. The monic characteristic polynomial is $(-1)^N N! L_N(t)$, by (39.1). Substitution of $t = -1$ gives Q_N; the cofactor formula gives integer P_N. Determinant or continuant expansion gives (39.3) and its initial values. For a direct second-kind remainder calculation define

$$I_N = N! \int_0^\infty \frac{e^{-t} L_N(t)}{1+t}\, dt, \qquad \widetilde{P}_N = Q_N \delta - I_N.$$

The Laguerre recurrence, and $\int_0^\infty e^{-t} L_N(t)\, dt = 0$ for $N \geq 1$, show that I_N obeys the same recurrence for $N \geq 1$: indeed, writing $2N + 1 - t = 2N + 2 - (1 + t)$ produces exactly the vanishing orthogonality integral. The initial values are $I_0 = \delta$ and $I_1 = 2\delta - 1$. Hence $\widetilde{P}_0 = 0$, $\widetilde{P}_1 = 1$, and uniqueness for the recurrence identifies $\widetilde{P}_N = P_N$.

Insert $(1+t)^{-1} = \int_0^\infty e^{-(1+t)s}\, ds$. Interchanging the integrals is justified by absolute integrability, since L_N is a polynomial. Its Laplace transform is

$$\int_0^\infty e^{-(1+s)t} L_N(t)\, dt = \frac{s^N}{(1+s)^{N+1}}.$$

For completeness, expansion $L_N(t) = \sum_{j=0}^N \binom{N}{j} (-t)^j / j!$ makes the integral equal to $(1 + s)^{-1} \sum_j \binom{N}{j} [-(1 + s)^{-1}]^j$, proving the formula. It follows that I_N is the right side of (39.4), which is positive.

Restrict that integral to $s \in [\sqrt{N}, \sqrt{N} + 1]$. On this interval $e^{-s} \geq e^{-\sqrt{N}-1}$, $(1 + s)^{-1} \geq (\sqrt{N} + 2)^{-1}$, and

$$\left(\frac{s}{1+s}\right)^N \geq \exp\{-N \log(1 + 1/\sqrt{N})\} \geq e^{-\sqrt{N}}.$$

Integration over an interval of length one proves (39.5); its divergence follows from factorial growth, for example from Stirling's formula. □

The initial convergents are 1/2, 4/7, and 20/34. This is the classical Laguerre denominator with its homogeneous Gompertz companion, also related to the constructions in [27]. Positivity certifies nonvanishing but does not make these raw integer linear forms small. Dividing P_N, Q_N by their common divisor changes the arithmetic estimate; no uniform gain from that primitive normalization is established by the lower bound above.

39.2 An integer Jacobi family with rational designated values

Theorem 39.2. *For each integer $B \geq 2$, let K_B be the self-adjoint closure of the Jacobi matrix*

$$a_n = 2(n+1), \qquad b_0 = B, \qquad b_n = 5n \quad (n \geq 1). \tag{39.6}$$

It is nonnegative, has compact resolvent and infinite spectrum, and all its moments $\langle e_0, K_B^n e_0\rangle$ are integers. Its canonical Weyl function is meromorphic on $\mathbb{C}$, has infinitely many poles, and satisfies

$$m_{K_B}(1) = 1/B, \qquad \lim_{z\to+\infty} z m_{K_B}(z) = 1. \tag{39.7}$$

In particular it is neither algebraic over $\mathbb{C}(z)$ nor D-finite.

Proof. Since $\sum_n 1/[2(n+1)] = \infty$, the Jacobi Carleman criterion gives essential self-adjointness. It suffices first to prove positivity for $K = K_2$, since $K_B = K + (B-2)P_0$, where $P_0 v = \langle e_0, v\rangle e_0$, is a nonnegative bounded rank-one perturbation. Conjugate K by the diagonal unitary $Ue_n = (-1)^n e_n$ to make its off-diagonal entries negative. Choose

$$h_0 = h_1 = 1, \qquad h_n = (3/4)^{n-1} \quad (n \geq 1).$$

Expansion of the squares gives, for finitely supported v,

$$\langle v, UKU^{-1}v\rangle = \sum_{n\geq 0} a_n h_n h_{n+1} \left|\frac{v_n}{h_n} - \frac{v_{n+1}}{h_{n+1}}\right|^2 + \sum_{n\geq 0} R_n |v_n|^2,$$
$$R_n = b_n - a_{n-1}\frac{h_{n-1}}{h_n} - a_n \frac{h_{n+1}}{h_n},$$

where the first subtracted term is omitted for $n = 0$. Direct calculation gives $R_0 = R_1 = 0$ and $R_n = 5n/6 - 3/2 > 0$ for $n \geq 2$. This proves nonnegativity on a core, and hence for the closure.

Set $w_n = (-1/2)^n$. For $n \geq 1$, division of $((K_B+I)w)_n$ by w_n gives $-4n+(5n+1)-(n+1) = 0$. Its zeroth component is $(B+1) - 1 = B$. Thus

$$(K_B + I)w = Be_0.$$

The vector w is square summable, and its formal image under K_B is $Be_0 - w \in \ell^2$. It belongs to the maximal Jacobi domain, which essential self-adjointness identifies with the self-adjoint domain. Since $K_B + I$ is invertible, its $(0,0)$ resolvent entry is $w_0/B = 1/B$.

The elementary inequality $2|u\overline{v}| \leq |u|^2 + |v|^2$ applied to adjacent coordinates gives

$$K_B \geq \operatorname{diag}(n-2)_{n\geq 0}$$

as a quadratic-form inequality. Consequently a bounded subset of the form domain, with norm coming from $K_B + 3I$, has uniformly small ℓ^2 tails: its tail from N onward is bounded by a constant divided by $N+1$. Finite-coordinate projections then prove compactness of the form-domain embedding, and hence compactness of the resolvent. Its spectrum is an infinite discrete sequence tending to infinity.

All a_n are nonzero. The recurrence shows inductively that every e_n belongs to the span of $e_0, K_B e_0, \ldots, K_B^n e_0$, so e_0 is cyclic. Every eigenvalue therefore has a nonzero spectral weight in its probability spectral measure. The spectral representation of m_{K_B} gives a meromorphic function with a pole at the negative of each eigenvalue, and thus infinitely many poles. An algebraic function over $\mathbb{C}(z)$ has only finitely many possible finite singularities; a solution of a nonzero linear differential equation with polynomial coefficients likewise has singularities only

among the finitely many zeros of its leading coefficient. The infinite-pole property excludes both algebraicity and D-finiteness.

Finally $z/(z+t) \to 1$, bounded between zero and one for $z > 0, t \geq 0$, so dominated convergence for the probability spectral measure proves the normalization in (39.7). Since $K_B^n e_0$ has finite support and integer coordinates, its zeroth coordinate, the nth moment, is an integer. □

This counterexample keeps positivity, integer coefficients, essential self-adjointness, unbounded spectrum, integer moments, canonical normalization and functional transcendence. These general properties therefore do not imply irrationality of a designated resolvent value. The example is not the Laguerre operator, and it makes no contrary claim about the actual arithmetic nature of δ.

39.3 Meixner identification and full differential transcendence

Non-D-finiteness by itself does not rule out nonlinear algebraic differential equations. We now retain the full argument establishing the stronger property for $K = K_2$.

Lemma 39.3 (Direct Meixner spectral representation)**.** *The canonical Weyl function of K in Theorem 39.2 is*

$$m_K(z) = \frac{R(z)}{1+2R(z)}, \qquad R(z) = \frac{3}{4}\sum_{j\geq 0} \frac{4^{-j}}{3j+z-1}. \tag{39.8}$$

Proof. Fix $0 < c < 1$ and put $\mu_j = (1-c)c^j$. Define polynomials $M_n(j)$ by the convergent generating function near $u = 0$,

$$G_j(u) = \sum_{n\geq 0} M_n(j)u^n = \frac{1}{1-u}\left(\frac{1-u/c}{1-u}\right)^j.$$

A geometric sum, absolutely convergent for sufficiently small u, v, gives

$$\begin{aligned}\sum_{j\geq 0} \mu_j G_j(u)G_j(v) &= \frac{1-c}{(1-u)(1-v)-c(1-u/c)(1-v/c)} \\ &= \frac{1}{1-uv/c}.\end{aligned}$$

Coefficient comparison proves $\sum_j \mu_j M_n(j)M_m(j) = c^{-n}\delta_{nm}$. Thus $p_n(j) = (-\sqrt{c})^n M_n(j)$ are orthonormal and have positive leading coefficients. Differentiation of G_j gives

$$(1-c)jG_j = (c-u)G_j - (c-u)(1-u)G_j'.$$

Comparing coefficients, with $M_{-1} = p_{-1} = 0$, yields

$$\begin{aligned} jM_n &= -\frac{c(n+1)}{1-c}M_{n+1} + \frac{(1+c)n+c}{1-c}M_n - \frac{n}{1-c}M_{n-1}, \\ jp_n &= a_n p_{n+1} + b_n p_n + a_{n-1}p_{n-1}, \\ a_n &= \frac{\sqrt{c}(n+1)}{1-c}, \qquad b_n = \frac{(1+c)n+c}{1-c}. \end{aligned}$$

We also need completeness, so that this is an identification of the actual spectral measure. If $f \in L^2(\mu)$ is orthogonal to every polynomial, the series $H(s) = \sum_{j\geq 0} \mu_j f(j)s^j$ is analytic

for $|s| < c^{-1/2}$ by Cauchy–Schwarz. Its derivatives at $s = 1$ vanish by orthogonality to the falling factorials; termwise differentiation is justified on compact subdisks. Since 1 is interior to the disk, $H \equiv 0$, and hence $f = 0$. Complex conjugation, if required by the inner-product convention, does not change the argument.

The unitary map $e_n \mapsto p_n$ consequently identifies the closure of the Jacobi matrix with multiplication by j on $L^2(\mu)$. Indeed multiplication extends the displayed recurrence, and Carleman's criterion ensures that this self-adjoint extension is unique. At $c = 1/4$ the coefficients are $a_n = 2(n+1)/3$ and $b_n = (5n+1)/3$. Therefore $L = 3J - I$ has off-diagonal entries $2(n+1)$, diagonal entries $5n$, and spectral measure $\mu_j = (3/4)4^{-j}$ at $3j-1$. Its Weyl function is precisely R in (39.8).

Since $K = L + 2P_0$, the rank-one resolvent identity gives, wherever the resolvents exist,

$$(K + zI)^{-1} = (L + zI)^{-1} - \frac{2(L + zI)^{-1}P_0(L + zI)^{-1}}{1 + 2R(z)}.$$

Taking the $(0, 0)$ entry yields $m_K = R - 2R^2/(1+2R) = R/(1+2R)$. Meromorphic continuation proves the identity throughout its domain. □

Lemma 39.4. *For any field C_0 of characteristic zero, the rational difference equation*

$$f(t + 1) - 4f(t) = -1/t \tag{39.9}$$

has no solution in $C_0(t)$, where the shift fixes C_0.

Proof. Work over an algebraic closure of C_0. If f has a finite pole, partition its finite pole set into orbits under translation by integers. Within any occupied orbit choose a representative α and the smallest and largest integer indices $p \le q$ for which $\alpha+p, \alpha+q$ are poles. The difference in (39.9) has a noncancelling pole at $\alpha+p-1$, contributed only by $f(t+1)$, and another at $\alpha+q$, contributed only by $-4f(t)$. They are distinct, and poles from other orbits cannot cancel them. The right side has exactly one pole, a contradiction. If f has no finite poles it is polynomial, which also cannot produce the pole on the right side. □

Theorem 39.5. *The canonical Weyl function m_K is differentially transcendental over $\mathbb{C}(z)$: it satisfies no nonzero polynomial differential equation with coefficients in $\mathbb{C}(z)$. Nevertheless $m_K(1) = 1/2$.*

Proof. The series defining R converges normally on compact sets avoiding its simple poles, so R is meromorphic on $\mathbb{C}$. Shifting the summation index gives

$$R(z + 3) = 4R(z) - \frac{3}{z - 1}.$$

Thus $y(t) = R(3t + 1)$ is meromorphic and satisfies $y(t + 1) = 4y(t) - 1/t$.

Let $\mathcal{P}$ be the field of one-periodic meromorphic functions of t. Hardouin–Singer's Proposition 3.9(2) states, in the form needed here, that a meromorphic solution $y \notin \mathcal{P}(t)$ of $y(t + 1) = ay(t) + b(t)$, with constant $a \neq 0$ and $b \in \mathbb{C}(t)$, is differentially algebraic over $\mathcal{P}(t)$ if and only if $b = f(t + 1) - af(t)$ for some $f \in \mathbb{C}(t)$ [39, Proposition 3.9(2)]. A nonzero constant a is in the standard form required by that proposition.

We check the nonmembership hypothesis explicitly. The coordinate t is transcendental over $\mathcal{P}$: a nonzero polynomial relation with periodic coefficients, after every integer translation, would make the same polynomial have infinitely many distinct roots $t + n$ in the field of meromorphic functions. Thus $\mathcal{P}(t)$ is an abstract rational function field on which translation fixes the

coefficient field. If y belonged to it, Lemma 39.4 with $C_0 = \mathcal{P}$ would be contradicted. The same lemma with $C_0 = \mathbb{C}$ excludes the rational solution required by Hardouin–Singer. The proposition therefore proves differential transcendence of y over $\mathcal{P}(t)$, and hence over $\mathbb{C}(t)$. An affine change of the independent variable preserves differential algebraicity, so R is differentially transcendental over $\mathbb{C}(z)$.

Finally $R = m_K/(1-2m_K)$, and the denominator is not identically zero since $m_K(z) \to 0$ on the positive real ray. Differentially algebraic functions form a field, so differential algebraicity of m_K would imply that of R, a contradiction. At $z = 1$, the $j = 0$ summand of R has a simple pole of residue $3/4$ and the remaining summands are regular. The quotient $R/(1+2R)$ has a removable singularity with value $1/2$, consistently with (39.7). □

This conclusion uses the cited nonlinear differential-algebraic theorem, not an inference from infinitely many poles. It is an application of existing theory to an explicit comparison operator, and supplies no new arithmetic statement about the Euler resolvent.

39.4 A nearby arithmetic Gevrey example and a degenerate change of basis

The probability density $(1+t)e^{-t}/2$ gives another useful comparison:

$$\widetilde{D}(z) = \int_0^\infty \frac{(1+t)e^{-t}}{2(z+t)}\,dt = \frac{1+(1-z)D(z)}{2}, \qquad \widetilde{D}(1) = \frac{1}{2}. \tag{39.10}$$

The identity follows from $1+t = (z+t)+(1-z)$. Its moments are $(n+2)n!/2$, and the shifted Borel kernel of its inverse-power expansion is

$$\sum_{n\ge 0} \frac{(-1)^n}{n!}\frac{(n+2)n!}{2}u^n = \frac{u+2}{2(1+u)^2}.$$

Thus it retains a probability measure with unbounded support, the normalization $1/z$, and an arithmetic Gevrey-one expansion. It is not rational, since (39.10) would then make D rational; the nonzero logarithmic monodromy of D excludes that possibility. However, direct differentiation gives

$$\widetilde{D}' = \frac{z}{z-1}\widetilde{D} + \frac{1-2z}{2z(z-1)}, \qquad \frac{\widetilde{D}-1/2}{z-1} = -D/2.$$

The corresponding formal quotient is already arithmetic Gevrey one. The change of basis from $(1, D)$ to $(1, \widetilde{D})$ degenerates at $z = 1$, where the displayed scalar equation is singular. This example therefore does not contradict an ordinary-point mixed lifting statement for the original, nondegenerate four-function system.

40 Exact local radii, denominator growth, and the foliation test

40.1 A number-field initial-value family

For $\alpha \in k$ consider the solution at zero of

$$Y_\alpha' = Y_\alpha - \frac{1}{1+t}, \qquad Y_\alpha(0) = \alpha. \tag{40.1}$$

Write $S_n = \sum_{j=0}^n (-1)^j j!$, with $S_{-1} = 0$. Coefficient comparison, or induction in the differential equation, gives

$$Y_\alpha(t) = \sum_{n\ge 0} \frac{\alpha - S_{n-1}}{n!}t^n. \tag{40.2}$$

For example the recurrence for $c_n = [t^n]Y_\alpha$ is $(n+1)c_{n+1} = c_n - (-1)^n$, and multiplication by $n!$ telescopes to the formula. In the complex realization uniqueness gives

$$Y_\alpha(t) = D(1+t) + (\alpha - \delta)e^t. \tag{40.3}$$

The local complex radius is exactly one: the equation gives analyticity on $|t| < 1$, and a radius greater than one would make its left side regular at $t = -1$, contradicting the inhomogeneous pole there. If δ were algebraic, $\alpha = \delta$ would be the initial value selected by the real positive-ray normalization. This statement alone does not impose any nonarchimedean initial-value matching.

Theorem 40.1 (Exact nonarchimedean radius)**.** *Fix a prime p, a finite extension $L/\mathbb{Q}_p$, and an image of α in L. Normalize the absolute value by $|p|_p = p^{-1}$, and put*

$$\delta_p = \sum_{j\geq 0} (-1)^j j! \in \mathbb{Z}_p.$$

The radius of convergence of (40.2) is

$$R_p(\alpha) = \begin{cases} 1, & \alpha = \delta_p, \\ p^{-1/(p-1)}, & \alpha \neq \delta_p. \end{cases} \tag{40.4}$$

Proof. The factorial series converges because $v_p(j!) \to \infty$. If $\alpha \neq \delta_p$, its coefficient numerators $\alpha - S_{n-1}$ converge to a nonzero element. Their valuations are eventually constant, since an ultrametric-convergent sequence with nonzero limit eventually has the absolute value of that limit. Legendre's formula gives $v_p(n!)/n \to 1/(p-1)$. Cauchy–Hadamard then gives the second radius in (40.4).

If $\alpha = \delta_p$, then $\alpha - S_{n-1} = \sum_{j\geq n}(-1)^j j!$ belongs to $n!\mathbb{Z}_p$, so all Taylor coefficients are integral and the radius is at least one. If it were greater than one, the series and its derivative would be analytic at $t = -1$. Multiplying (40.1) by $1 + t$ and substituting $t = -1$ would give $0 = -1$, a contradiction. The radius is therefore exactly one. □

Equality modulo p or any fixed power of p does not imply the first case: the improved radius requires equality with the entire p-adic sum. This distinction will also appear in the curvature calculation.

Theorem 40.2 (Zero finite-place radius product)**.** *Let $K = \mathbb{Q}(\alpha)$, $d = [K : \mathbb{Q}]$, and for every finite place $v \mid p$ set $n_v = [K_v : \mathbb{Q}_p]/d$. Interpret the radius in Theorem 40.1 in K_v. Then*

$$\prod_{v \text{ finite}} R_v(\alpha)^{n_v} = 0. \tag{40.5}$$

If $d > 1$, the following stronger estimate holds prime by prime:

$$\prod_{v\mid p} R_v(\alpha)^{n_v} \leq p^{-(1-1/d)/(p-1)}. \tag{40.6}$$

If $\alpha \in \mathbb{Q}$, the mismatching primes instead satisfy

$$\sum_{p:\alpha\neq\delta_p} \frac{\log p}{p-1} = \infty. \tag{40.7}$$

Proof. Suppose first that $d > 1$. At a matching place $\alpha = \delta_p$ lies in $\mathbb{Q}_p$. Since $K = \mathbb{Q}(\alpha)$, its completion K_v is generated topologically over $\mathbb{Q}_p$ by the image of α; therefore $K_v = \mathbb{Q}_p$ and $n_v = 1/d$. There is at most one matching place over a given p: two embeddings of $\mathbb{Q}(\alpha)$ into $\mathbb{Q}_p$ sending its generator to the same prescribed element δ_p coincide and induce the same place. Thus the total matching weight is at most $1/d$. The weights above p sum to one, and Theorem 40.1 gives (40.6). The classical divergence of $\sum_p \log p/(p-1)$ proves the zero product.

Now write $\alpha = A/B \in \mathbb{Q}$ in lowest terms with $B > 0$, and let $M = \{p : \alpha \neq \delta_p\}$. Suppose contrary to (40.7) that $C = \sum_{p\in M} \log p/(p-1) < \infty$. At matching primes all coefficients of (40.2) are integral. At the other primes the factorial denominators show that a common denominator through degree N is bounded by

$$B \prod_{p\in M} p^{v_p(N!)} \leq B \exp(CN),$$

where only finitely many factors are nontrivial and $v_p(N!) \leq N/(p-1)$. The complex analytic radius is positive, so the rational coefficients have exponential absolute-value bounds. They are holonomic: applying $(1+t)\partial_t + 1$ to $(\partial_t - 1)Y_\alpha = -1/(1+t)$ gives a homogeneous linear differential equation with rational coefficients. These are precisely the G-function requirements, so Y_α would be a G-function.

But $(e^{-t}Y_\alpha)' = -e^{-t}/(1+t)$ has residue $-e$ at -1. Continuation around -1 changes Y_α by $-2\pi i e^{t+1}$. As in Theorem 38.2, this is incompatible with the global regular-singularity theorem for G-functions. Thus (40.7) holds, and the radius dichotomy proves (40.5) also for rational α. □

The rational case is the denominator obstruction expressed in local radius language, rather than a new distribution theorem for factorial residues. For primary background on real and p-adic factorial sums and their approximation, see [40]. The proof above specifies its own use of the global G-function theorem.

Proposition 40.3 (Common denominator consequences)**.** *Let q_N be the least positive rational integer for which all coefficients of (40.2) through degree N become algebraic integers after multiplication by q_N. If $\alpha \in \mathbb{Q}$, then*

$$\frac{\log q_N}{N} \longrightarrow \infty.$$

If α is algebraic irrational, then there are constants $c_\alpha, C_\alpha > 0$ such that

$$c_\alpha N! \leq q_N \leq C_\alpha N! \qquad (N \geq 1).$$

Proof. For rational α, at each fixed mismatching prime p the valuations of $\alpha - S_{N-1}$ eventually stabilize at a finite constant. The degree-N coefficient alone then forces $v_p(q_N) \geq v_p(N!) - O_{\alpha,p}(1)$. For each finite set T of mismatching primes this gives

$$\liminf_{N\to\infty} \frac{\log q_N}{N} \geq \sum_{p\in T} \frac{\log p}{p-1}.$$

The right side is unbounded as T increases by (40.7), proving the first assertion.

For irrational α, choose an integer $A_0 > 0$ making $A_0\alpha$ integral. Then $A_0 N!$ clears every coefficient through N, proving the upper bound. In a fixed normal closure L choose a conjugate $\alpha^\sigma \neq \alpha$. Since $q_N(\alpha - S_{N-1})/N!$ and its conjugate are algebraic integers, so is their nonzero difference

$$\frac{q_N}{N!}(\alpha - \alpha^\sigma).$$

Its absolute field norm from L to $\mathbb{Q}$ is at least one. Writing $h = [L : \mathbb{Q}]$ gives $(q_N/N!)^h |N_{L/\mathbb{Q}}(\alpha - \alpha^\sigma)| \geq 1$, which proves the lower bound with a fixed positive constant. □

40.2 Factorial truncations in characteristic p

Proposition 40.4. *On $\mathbb{G}_m \times \mathbb{A}^1$ over $\mathbb{F}_p$, let*

$$V = \partial_z + (y - 1/z)\partial_y, \qquad T_p(z) = \sum_{j=0}^{p-1} (-1)^j j! z^{-j-1}.$$

As derivations of the rational function field,

$$V^p = (y - T_p(z))\partial_y, \qquad T_p' - T_p = -1/z. \tag{40.8}$$

The rank-one foliation generated by V is not stable under pth powers at any prime. At a good reduction of $(1, \alpha)$, its curvature class vanishes at that point exactly when

$$\alpha \equiv T_p(1) = S_{p-1} \equiv \delta_p \pmod{\mathfrak{p}}. \tag{40.9}$$

Proof. For $m \geq 1$, induction gives $V^m(y) = y - \sum_{j=0}^{m-1} \partial_z^j(1/z)$, since V acts as ∂_z on rational functions of z. At $m = p$ this is $y - T_p(z)$, while $V^p(z) = 0$. In characteristic p the $p$fold iterate of a derivation is a derivation, because the interior binomial coefficients in the iterated Leibniz rule vanish. Its values on z, y therefore prove the first identity of (40.8). Termwise differentiation of T_p telescopes against T_p; the last derivative term has factor $p!$ and is zero, giving the second identity.

If $V^p = fV$ for a rational function f, comparison of the z components would give $f = 0$, whereas $y - T_p(z)$ is not the zero rational function. Thus the generic foliation is not p-closed. The class of V^p modulo the line generated by V vanishes at $(1, \alpha)$ if and only if its vertical coefficient is zero, since V has horizontal component one. This is the first congruence in (40.9). The equality with δ_p modulo $\mathfrak{p}$ follows because every $j!$ with $j \geq p$ is divisible by p. □

The full formal differential system also has an explicit $p$fold iteration. In characteristic p, with E, F, D treated as differential module coordinates satisfying (38.1), one has

$$E^{(p)} = E, \qquad D^{(p)} = D - T_p, \qquad F^{(p)} = F - z^{-p}E - T_p. \tag{40.10}$$

Here is a direct check of the less immediate last formula. Iterating $F' = F + (E - 1)/z$ gives $F^{(p)} - F = \sum_{j=0}^{p-1} \partial_z^j((E-1)/z)$. The non-exponential part is $-T_p$, while its exponential coefficient is

$$\sum_{j=0}^{p-1} (\partial_z + 1)^j (1/z) = \partial_z^{p-1}(1/z) = -z^{-p}.$$

The polynomial identity $\sum_{j=0}^{p-1}(X+1)^j = X^{p-1}$ holds in $\mathbb{F}_p[X]$; the last equality uses $(p-1)! = -1$ in $\mathbb{F}_p$ and is also valid for $p = 2$. Neither γ nor δ enters these module identities.

40.3 Checking the algebraization hypothesis rather than assuming it

Bost's algebraicity criterion for leaves concerns a smooth variety over a number field, an involutive subbundle of its tangent bundle, and a rational base point. Besides a complex Liouville condition on the leaf, Theorem 2.1 requires the reduced foliation to be closed under pth powers at almost every finite prime [41, Theorem 2.1]. Proposition 40.4 shows that this required global

closure fails at every prime for the displayed Euler foliation. The pointwise congruence (40.9) is not a substitute for that hypothesis. It is also weaker than the exact equality $\alpha = \delta_p$ that produces radius one.

Likewise Theorem 40.2 leaves no positive product of these particular Taylor radii. This is a coordinate-specific calculation for the displayed parameter and local series. It neither identifies these radii with all intrinsic formal-germ sizes or capacities nor rules out other arithmetic geometries, reparametrizations, or arithmetic Gevrey estimates. Those distinctions matter in the broader formal-germ and capacity theory of [42]. The concrete missing implication is that the real canonical boundary condition might force compatible nonarchimedean conditions; no such implication has been proved. A general rank-three relation is still further from a number-field initial-value problem, since it need not make δ algebraic at all.

41 Galois transport and exact joint-span bookkeeping

41.1 What an automorphism actually transports

If $\gamma = a + b/e + c\eta$ with $a, b, c \in k$, then for an abstract automorphism σ of $\mathbb{C}$ the valid identity is

$$\sigma(\gamma) = \sigma(a) + \frac{\sigma(b)}{\sigma(e)} + \sigma(c)\sigma(\eta). \tag{41.1}$$

Conjugating only the algebraic coefficients while holding all three transcendental values fixed would require a separately justified automorphism with those fixed points. The numerical relation itself may obstruct that prescription. Nor does an arbitrary field automorphism commute with the limits defining exp or Ein: there is no general identity $\sigma(e^{\alpha}) = e^{\sigma(\alpha)}$ or its Ein analogue. The continuous automorphisms of $\mathbb{C}$ are the identity and complex conjugation; the latter fixes the real constants here and gives only the realness already proved in Proposition 37.2. Also, the argument 1 itself is fixed by every field automorphism, so its algebraic orbit cannot produce new evaluation points.

If γ were algebraic, its algebraic conjugates would be roots of its minimal polynomial, but their identification with the analytically normalized Euler constant would still require proof. Coefficient traces and norms have the same issue: all constants in the relation must be tracked under the embeddings. Without additional field-intersection information they do not turn an algebraic-coefficient relation on fixed transcendental numbers into a rational-coefficient relation on those same numbers. Genuine E-function lifting can sometimes justify descent by first producing functional relations; D is not an E-function, and extending that step to this mixed relation is precisely missing arithmetic input.

41.2 Circuits constrain the exceptional set but do not propagate it

If $1 \in B$, Theorem 37.4 implies that every finite subset of $B \setminus \{1\}$ is multiplicatively independent. Indeed, given an integer multiplicative relation on that subset, add at the point 1 the negative sum of its coefficients. This produces the forbidden affine relation. It does not, however, assert that another bad point exists. The configuration $B = \{1\}$ satisfies every circuit restriction and all finite-rank bounds in this part. Changing the geometric base leaves $1 = r^0$ in every progression, so intersecting such bounds cannot eliminate this configuration.

The exact scaling formula also shows the unremoved terms:

$$E_1(rx) = E_1(x) - \int_x^{rx} \frac{e^{-t}}{t}\, dt.$$

By (37.2) the difference introduces $\log r$ and $\mathrm{Ein}(rx) - \mathrm{Ein}(x)$. Without an additional relation eliminating these new quantities it does not transport rank failure from 1 to 2 or 4. Density-one full rank, or even sequences of full-rank points approaching one, likewise gives no continuity principle for arithmetic independence of values.

Proposition 41.1 (Joint span on a geometric progression)**.** *Fix positive algebraic $r \neq 1$, let $L = \log r$, and let $S \subset \mathbb{Z}$ contain $m \geq 2$ distinct integers. Put $x_n = r^n$ and*

$$V_S = \mathrm{span}_k\{1, e^{-x_n}, \mathrm{Ein}(x_n) : n \in S\},$$
$$J_S = \mathrm{span}_k\{1, e^{-x_n}, \mathrm{Ein}(x_n), E_1(x_n) : n \in S\}.$$

Then

$$\dim_k V_S = 2m + 1, \qquad J_S = V_S + \mathrm{span}_k\{\gamma, L\}, \qquad \dim_k J_S = 2m + 1 + s_S, \tag{41.2}$$

where

$$s_S = \dim_k \frac{V_S + \mathrm{span}_k\{\gamma, L\}}{V_S} \in \{0, 1, 2\}.$$

For nested sets $S \subset S'$ the parameter $s_{S'} \leq s_S$. If $n \notin S$, then

$$\dim_k J_{S\cup\{n\}} - \dim_k J_S = 2 + s_{S\cup\{n\}} - s_S. \tag{41.3}$$

Along any nested sequence the total loss relative to increments of two is at most two, and the parameters eventually stabilize.

Proof. The dimension of V_S is Lemma 37.1. For each $n \in S$ the connection formula gives

$$\mathrm{Ein}(x_n) - E_1(x_n) = \gamma + nL.$$

This places every $E_1(x_n)$ in the right side of (41.2). Conversely, choose distinct $n, n' \in S$; the difference of these two identities, divided by the nonzero integer $n - n'$, expresses L as an element of J_S, and either identity then expresses γ. The vector-space equality and its dimension formula follow. If S is enlarged, V_S increases while $\mathrm{span}_k\{\gamma, L\}$ is fixed. The dimension of its image in the quotient therefore cannot increase. Subtraction of the dimension formulas gives (41.3). A nonincreasing sequence in $\{0, 1, 2\}$ stabilizes, and the total decrease is at most two. □

This is an exact formula with an undetermined arithmetic parameter; the finite E-value input alone does not determine s_S, its eventual value, or a stabilization index. The use of unscaled E_1 values is essential to the stated formula. Multiplication of one local block by the common nonzero scalar e^{-x} preserves its k-rank. Multiplication of different blocks by different transcendental scalars does not generally preserve the rank of the combined vector, so (41.2) cannot simply be rewritten with all $E_1(x_n)$ replaced by $D(x_n)$.

For distinct points the local E-value spaces $\mathrm{span}_k\{1, e^{-x}, \mathrm{Ein}(x)\}$ intersect only in k, as follows by subtracting two representations and applying Lemma 37.1. At a bad point r^n, the value $\gamma + nL$ lies in that local three-dimensional space. This explains the cancellation geometry behind Theorem 37.4, while leaving the actual position of γ at $n = 0$ undetermined.

41.3 Scope of the continuation

The unconditional conclusions are the exact rank-three alternatives, the exceptional-set restrictions, the formal arithmetic nondivision theorem, the spectral error identity and counterexamples, the local radius and denominator calculations, the curvature obstruction, and the joint-span

identities. Their proofs use identified classical or published inputs. They do not prove the designated rank-four assertion, the irrationality of either constant, or a new transcendence conclusion for γ or δ. A successful completion would need an unconditional arithmetic argument excluding an isolated rank defect at one, respecting the canonical analytic normalization. Neither a renamed zero-removal gate nor a general spectral or differential property supplies that argument.

42 What the integrated argument decides

The construction and exclusion results in this manuscript have different logical outputs. The binomial and prime results characterize arithmetic behaviour under explicit hypotheses. The moment and Fourier sections solve the auxiliary reconstruction problems that they state. The functional results exclude differential-algebraic descriptions of specified transforms. The rank-four investigation restricts where a rank failure can occur, and the final spectral and p-adic calculations test proposed transfer mechanisms. None of these deductions alone identifies the arithmetic nature of the designated constant γ.

Three gaps deserve separate names only for clarity, not as claims of progress from renaming the problem.

1. **Nonzero arithmetic output.** A short vector in a logarithm elimination lattice may lie in the kernel of the rational output map. A useful infinite family must retain nonzero output while satisfying the needed height and error estimates.
2. **Arithmetic specialization.** Analytic division by $z - 1$ after a zero at 1 does not automatically preserve the arithmetic Gevrey class. In the restricted four-dimensional family, the requested lifting statement is equivalent to the designated rank-four target, once the formal nondivisibility result is used.
3. **Compatibility between places.** A real asymptotic normalization does not by itself identify the initial value with the p-adic factorial sum. The exact Taylor-radius calculation measures this missing equality; it does not establish it.

The counterexamples locate the scope of such gaps. A deformation that preserves a restricted list of identities may lose the canonical normalization of the original analytic function. Conversely, the positive integer Jacobi example preserves positivity, integer moments, infinite spectrum and canonical resolvent normalization, and its Weyl function is differentially transcendental, yet its value at 1 is rational. A proof for the original Euler system must therefore use additional information distinguishing that system from the example.

The final status is

$$\boxed{\rho(1) = 4 \text{ remains unproved in this manuscript.}}$$

Neither an explicit contradiction in the algebraic-gamma alternative nor one in the transcendental-gamma alternative has been obtained. This statement concerns the results established here; it is not an impossibility theorem for future methods.

43 Source record, corrections and reproducibility

The requested starting point is the evaluation question quoted in the Korean reader's guide. The following source families cover the mathematical developments from that evaluation through the last cross-field investigation. They are internal research manuscripts and proof archives. The earlier September 11 Two-World manuscript is the object of the initial review, not a second copy inserted into this edition. Manuscripts 001 and 002 are historical antecedents; the reused foundations needed in this range are reproduced where invoked.

Stage	Recovered source	Content retained in the integrated text
R	`KEIO_TwoWorld_Review_Addendum_20260920.tex`	Review corrections; sharp interval exponent; rational and algebraic floors; collision-ring information; periodic deformation.
A1	`KEIO_Gamma_AB_Construction_20260920.tex`	Binomial closure; Borel division; multiplication and Fourier constraints; coefficient-rank criteria.
A2	`KEIO_Gamma_AB_Extension_II_20260920.tex`	Minimal prime certificates; denominator and density refinements; finite Fourier ranges; multiple poles; categorical comparison.
M	`KEIO_Gamma_Moneyball_20260920.tex`	Prime-based independence; denominator defects; Hankel construction; countermodels; imported zero theorems; parity-moment reconstruction.
S	`KEIO_Gamma_Snowball_20260920.tex`	Functional differential transcendence; recurrence restrictions; nonlinear descent; two-trace inverse data; local denominators; residue-profile transfer and supporting foundations.
C	`KEIO_Gamma_Core_Research_20260920.zip`	Moving-degree arithmetic cost and its proof; fixed-transform, positive-kernel and moment-elimination limits; arithmetic reassessment.
R4	`KEIO_Rank4_Proofs.zip`	Differential system; rank-three classification; geometric-point exception bounds; exact formal quotient obstruction.
X	`KEIO_Cross_Field_Proofs.zip`	Laguerre resolvent; integer Jacobi and Meixner counterexample; difference-Galois proof; adelic radii and p-curvature; Galois and multipoint elimination audit.

43.1 Corrections that affect interpretation

The initial review corrects several claims in its predecessor: uniform framing statements need a common interval; finite approximation ratios are not asymptotic irrationality or transcendence exponents; and restricted countermodels do not establish independence of the actual analytically defined constant. The later independence-selection statements are existential: a suitable density-zero set can be deleted, not an arbitrary one. The functional results do not identify the arithmetic nature of the value at zero; for the entire transform $H(z) = \int_0^1 e^{zx}(1 - \log x)^{-1}\,dx$, that value is exactly $H(0) = \delta$.

The generic rank-three case must also be read correctly. Deriving transcendence or even algebraic independence of γ, δ from that case is not a contradiction: the rank condition is

stronger than those assertions. Finally, positivity or differential transcendence of an auxiliary function cannot be treated as an unproved value-transcendence principle.

43.2 Rebuilding the document

The delivered TeX is a single source, including all mathematical text, macros and bibliography; it does not use external `\input`, `\include`, graphics or a separate BibTeX database. Compile it twice with XeLaTeX. Standard TeX packages, `kotex`, and a Korean font (Noto Serif CJK KR, with a fallback to UnBatang) are required. These are typesetting dependencies, not additional manuscript source files.

The historical computational archives remain separate research sources. Exact finite checks supplement the proofs and are identified as such; finite decimal observations have not been converted into claims about unbounded sequences. The present compilation validates the TeX build, cross-references and PDF rendering and incorporates the mathematical audits described in the text. It does not claim formal verification or an external assessment of originality.

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